

of the economy. This is the conditional forward probability of survival, $\text{Prob}_t[\tau > U | \tau > T]$. By the Bayes rule, we can write¹³

$$\text{Prob}_t[\tau > U | \tau > T] = \frac{\text{Prob}_t[\tau > U | \tau > t]}{\text{Prob}_t[\tau > T | \tau > t]} = \frac{Q(t, U)}{Q(t, T)} \quad (17.28)$$

and hence the forward conditional probability of default regarding the future time period $[T, U]$ is given by

$$\begin{aligned} \text{Prob}_t[\tau \leq U | \tau > T] &= 1 - \text{Prob}_t[\tau > U | \tau > T] \\ &= -\frac{Q(t, U) - Q(t, T)}{Q(t, T)} \end{aligned} \quad (17.29)$$

17.2.1.3 Forward Default Rates

We can now introduce a key concept in the context of reduced-form models, the default rate $H(t, T)$, defined as the risk-neutral default probability associated with a given horizon divided by the length of the horizon¹⁴

$$H(t, T) \equiv \frac{\text{Prob}_t[\tau \leq T | \tau > t]}{T - t} = \frac{1 - Q(t, T)}{T - t} \quad (17.30)$$

The forward default rate, as seen at time t , that corresponds to the future period $[T, U]$ is analogously defined as

$$H(t, T, U) \equiv \frac{\text{Prob}_t[\tau \leq U | \tau > T]}{U - T} = -\frac{Q(t, U) - Q(t, T)}{U - T} \frac{1}{Q(t, T)} \quad (17.31)$$

If we now let $U \equiv T + \Delta T$, we can define the time- t instantaneous forward default rate as:

$$h(t, T) \equiv \lim_{\Delta T \rightarrow 0} H(t, T, T + \Delta T) \quad (17.32)$$

¹³The Bayes rule simply says that, given two events A and B , and denoting $\text{Prob}[A \& B]$ as the probability that both A and B occur, one can write

$$\text{Prob}[A \& B] = \text{Prob}[A] \text{Prob}[B | A]$$

where $\text{Prob}[A]$ is the unconditional probability that A will take place and $\text{Prob}[B | A]$ is the probability of B occurring given that A has taken place.

In the context of (17.28), we can think of event A as the event defined as $\tau > T$, B as $\tau > U$, and, given $U > T$, $A \& B$ corresponds to $\tau > U$. (To be sure, all of the corresponding probabilities would have to be defined as being conditional on information available at time t , as well as on survival through that time.)

¹⁴Unless otherwise indicated, all probabilities discussed in this chapter are risk-neutral probabilities.

and it is not difficult to see that

$$h(t, T) = -\frac{\partial Q(t, T)}{\partial T} \frac{1}{Q(t, T)} \quad (17.33)$$

To see how we arrived at (17.33), recall that (17.31) implies that

$$h(t, T) \equiv \left[\lim_{\Delta T \rightarrow 0} -\frac{[Q(t, T + \Delta T) - Q(t, T)]}{\Delta T} \right] \frac{1}{Q(t, T)} \quad (17.34)$$

but, assuming that certain technical conditions on $Q(\cdot)$ are satisfied, the expression within the square brackets is nothing more than the definition of the negative of the derivative of $Q(t, T)$ with respect to T .

Equation (17.33) is probably the single most important preliminary result derived so far in Section 17.2. In particular, integrating both sides of (17.33) from T to U , we obtain

$$e^{-\int_T^U h(t, v) dv} = \frac{Q(t, U)}{Q(t, T)} = \text{Prob}_t[\tau > U | \tau > T] \quad (17.35)$$

where the last equality results from (17.28).

Equation (17.35) is the forward survival probability associated with the future period $[T, U]$, given survival through time T and conditional on all available time- t information. Indeed, we can now express default and survival probabilities over any horizon for a given entity simply on the basis of the instantaneous forward default rate $h(t, s)$.¹⁵ For instance, the time- t probability that the firm will survive through time U is

$$\text{Prob}_t[\tau > U | \tau > t] = Q(t, U) = e^{-\int_t^U h(t, v) dv} \quad (17.36)$$

For $t = T$, we can rewrite (17.34) as

$$h(t, t) \equiv \lim_{\Delta t \rightarrow 0} -\frac{[Q(t, t + \Delta t) - Q(t, t)]}{\Delta t} \frac{1}{Q(t, t)} \quad (17.37)$$

but, by definition, $Q(t, t) = 1$, and thus, for infinitesimally small dt ,

$$\text{Prob}_t[\tau \leq t + dt | \tau > t] = h(t, t)dt \quad (17.38)$$

¹⁵ Readers familiar with yield curve models will notice the analogy between the concept of instantaneous forward interest rates and that of instantaneous forward default rates, especially regarding their relationship to zero-coupon bond prices and survival probabilities, respectively. (Appendix A provides a brief overview of instantaneous forward interest rates.)

which, in words, is the time- t instantaneous probability of default, assuming no default at t . Thus, for small time intervals Δt , the probability of default between t and $t + \Delta t$, given no default by t , is approximately equal to $h(t, t)\Delta t$.¹⁶

17.2.2 Default Intensity

In the context of most reduced-form models, the random nature of defaults is typically characterized in terms of the first “arrival” of a Poisson process. In particular, for a given reference entity, if we assume that defaults arrive (occur) randomly at the mean risk-neutral rate of λ per year, the time- t risk-neutral probability of no default by time T can be written as¹⁷

$$Q(t, T) = e^{-\lambda(T-t)} \quad (17.39)$$

provided λ is time invariant and, of course, given no default by time t . In the credit risk literature λ is commonly called the intensity of default or the hazard rate.

If λ is not constant over time, but varies deterministically, we can follow the same logic discussed in Section 17.2.1 to show that if λ changes continuously over infinitesimally small time intervals we can write

$$Q(t, T) = e^{-\int_t^T \lambda(v)dv} \quad (17.40)$$

and thus, as suggested by a comparison between (17.40) and (17.36), with deterministic λ , the terms default intensity and forward default rate can essentially be used interchangeably.

¹⁶ Given the definition of derivative in classic calculus:

$$\frac{\partial Q(t, T)}{\partial T} \equiv \lim_{dt \rightarrow 0} -\frac{[Q(t, T+dt) - Q(t, T)]}{dt}$$

we can write

$$Q(t, T+dt) \approx Q(t, T) + \frac{\partial Q(t, T)}{\partial T} dt = Q(t, T)[1 - h(t, T)dt]$$

For $T = t$, and given that $Q(t, t) = 1$, we can write:

$$1 - Q(t, t+dt) \approx h(t, t)dt$$

which becomes an equality if $Q(t, T)$ has a derivative at $T = t$ and we let $dt \rightarrow 0$. Thus, given that $1 - Q(t, t+dt) = \text{Prob}_t[\tau \leq t+dt | \tau > t]$, we arrive at (17.38)—Neftci (2002)[62] discusses basic notions from classic calculus that are relevant for finance.

¹⁷ For a Poisson process with a constant mean arrival rate of λ , a basic result from statistics states that the time until the “first arrival,” τ , is a random variable with probability density function $\lambda e^{-\lambda\tau}$, which characterizes the exponential distribution. From this, equation (17.39) easily follows. See Appendix B for further details on the Poisson and exponential distributions.

But default intensity can vary stochastically over time in response to, say, unanticipated developments regarding the economy or the financial condition of the firm. With stochastic default intensity, it can be shown that, under some technical conditions that go beyond the scope of this book—see, for instance, Lando (1998)[51]—the expression for the time- t survival probability of a given firm can be written in a way that is entirely analogous to (17.25):

$$Q(t, T) = \tilde{E}_t \left[e^{-\int_t^T \lambda(v) dv} \right] \quad (17.41)$$

Models where (17.41) holds are typically called doubly stochastic models of default because they assume that not only is the time of default a random variable, but so is the mean arrival rate of default at any given point in time.

A comparison between equations (17.41) and (17.36) suggests that the equality between time- t forward default rates and future values of intensity generally breaks down when λ is assumed to be stochastic. For instance, consider the value of default intensity at some future time v . That value, $\lambda(v)$, incorporates all information available at time v , including the values of any stochastic factors that may affect default intensity at time v , such as the prevailing states of the economy and of the firm at time v . In contrast, the forward default rate $h(t, v)$ can be thought of as the intensity rate for the future time v , as seen on the basis of currently available (time- t) information, and conditional on the firm surviving through time v .

Thus, whereas $\lambda(v)$ is based on all information available at time v , the only time- v information on which $h(t, v)$ is conditional is the survival of the firm through time v . Still, from the perspective of time t , it can be shown the following relationship between forward default rates and default intensity holds:

$$h(t, t) = \lambda(t) \quad (17.42)$$

which says that when defaults occur according to a Poisson process, today's (time- t) instantaneous default rate associated with the infinitesimally small time interval $[t, t + dt]$ is simply the default intensity of the reference entity, which is assumed to be known at time t . Moreover, we can write the conditional instantaneous time- t probability of default as:

$$\text{Prob}_t[\tau \leq t + dt | \tau > t] = \lambda(t)dt \quad (17.43)$$

Reduced-form models can essentially be characterized in terms of the particular assumptions that they make regarding how λ changes over time. Indeed, the case of constant default intensity constitutes the simplest of all reduced-form models.

A simple model that admits random variation in default intensity involves the following specification for the evolution of λ :

$$d\lambda(t) = k[\bar{\lambda} - \lambda(t)]dt + \sigma\sqrt{\lambda(t)}dW(t) \quad (17.44)$$

where $dW(t)$ is an infinitesimal increment in a standard Brownian motion.¹⁸

Equation (17.44) closely parallels the yield curve model proposed by Cox, Ingersoll, and Ross (1985)[15]. In words, it says that, although the default intensity associated with a given reference entity varies stochastically over time, it has a tendency to revert to its long-run mean of $\bar{\lambda}$, where k indicates the degree of mean reversion.

17.2.3 Uncertain Time of Default

The use of reduced-form models in the valuation of defaultable securities and related credit derivatives often requires the derivation of a probability distribution function for τ , the time of default. This is the function $G(s)$ such that

$$\text{Prob}[s < \tau \leq s + ds] = G(s + ds) - G(s) \approx g(s)ds$$

where $g(s)$ is the probability density function (p.d.f.) of τ and the approximation error in the above equation is negligible for very small values of ds —see Grimmett and Stirzaker (1998)[36], p. 90.¹⁹

The probability density of default time can be thought of as the product involving the probability of no default by time s and the probability of default in the interval from s to $s + ds$ conditional on no default by time s :

$$\text{Prob}[s < \tau \leq s + ds] = \text{Prob}[\tau > s] \text{ Prob}[\tau \leq s + ds | \tau > s]$$

which is an implication of the Bayes rule.

Conditional on all information available at time t , and given no default by time t , we can use equations (17.41) and (17.43) to write the conditional

¹⁸ A discrete-time approximation helps convey some of the intuition behind equation (17.44):

$$\lambda_{t+\delta t} \approx \lambda_t + k[\bar{\lambda} - \lambda_t]\delta t + \sigma\sqrt{\lambda_t} \epsilon_{t+\delta t}$$

where ϵ is a normally distributed random variable with mean zero and variance δt . Thus, we can think of $\lambda_{t+\delta t}$ as a mean-reverting random variable with conditional variance $\text{var}[\lambda_{t+\delta t} | \lambda_t]$ equal to $\sigma^2 \lambda_t \delta t$.

¹⁹ $G(\cdot)$ is the so-called cumulative distribution function of τ . Appendix B reviews key concepts surrounding cumulative distribution and probability density functions.

p.d.f. of τ as the function $g_t(s)$ such that:

$$g_t(s)ds = \text{Prob}_t[s < \tau \leq s + ds] = \tilde{E}_t \left[e^{-\int_t^s \lambda(u)du} \lambda(s)ds \right] \quad (17.45)$$

where we are assuming that ds is infinitesimally small and that $s \geq t$.

For nonrandom intensity, we can make use of (17.40), and equation (17.45) simplifies to

$$g_t(s)ds = \text{Prob}_t[s < \tau \leq s + ds] = Q(t, s)\lambda(s)ds$$

17.2.4 Valuing Defaultable Bonds

Suppose we want to value a zero-coupon bond that pays \$1 at time T if its issuer has not defaulted by then. Otherwise the bond becomes worthless, a zero recovery rate. As in previous chapters, we let $Z_0^d(t, T)$ denote today's (time- t) price of this bond.

In Section 15.2.1, where survival probabilities and future values of the riskless rate were assumed to be known with certainty, we argued that one can write the price of a defaultable bond as the product of the bond issuer's risk-neutral survival probability and the price of a riskless zero-coupon bond with the same maturity date and face value as the defaultable bond. That result still holds when interest rates are stochastic but independent from the default intensity process, an assumption that we will maintain throughout this chapter.²⁰ Thus, given (17.25) and (17.41), we can write:

$$Z_0^d(t, T) = \tilde{E}_t \left[e^{-\int_t^T [r(v) + \lambda(v)]dv} \right] = Z(t, T)Q(t, T) \quad (17.46)$$

which gives the logical result that, in the absence of default risk, the expression for $Z_0^d(t, T)$ reduces to $Z(t, T)$.

Given the above equation, it is relatively straightforward to derive the yield spread associated with this risky bond. In particular, recall from Chapter 15 that the yield to maturity of this bond is the annualized rate $R^d(t, T)$ that discounts the face value of the bond back to its market price. Thus, $R^d(t, T)$ is such that $Z_0^d(t, T) = e^{-R^d(t, T)(T-t)}$ for a bond with a face

²⁰ Two events are said to be independent if the knowledge that one has occurred does not affect one's assessment of the probability that the other will occur. Whenever we refer to independent random variables in this book, we mean independence with respect to risk-neutral probabilities. Independence is discussed further in Appendix B. We briefly discussed some empirical results regarding the relationship between defaults and riskless interest rates in Chapter 15.

value of \$1. As a result,

$$R^d(t, T) = \frac{-\log(Z_0^d(t, T))}{T - t} = \frac{\int_t^T [f(t, v) + h(t, v)] dv}{T - t} \quad (17.47)$$

where the second equality stems from equation (17.36) and from the basic result, reviewed in Appendix A, that the price of a riskless zero coupon bond can be expressed in terms of the time- t instantaneous forward riskless interest rates, $f(t, v)$, that span the remaining maturity of the bond:

$$Z(t, T) = e^{-\int_t^T f(t, v) dv}$$

The risky bond spread $S(t, T)$, defined as $R^d(t, T) - R(t, T)$, is given by

$$S(t, T) = \frac{\int_t^T h(t, v) dv}{T - t}$$

In light of (17.36), we can write

$$S(t, T) = \frac{-\log(Q(t, T))}{T - t}$$

which has the intuitive implication that the higher the risk-neutral survival probability $Q(t, T)$ of the bond issuer, the lower the corresponding risk spread.

If we assume that both r and λ are deterministic and time invariant, equation (17.46) reduces to

$$Z_0^d(t, T) = e^{-(r+\lambda)(T-t)} \quad (17.48)$$

and it is easy to see that, with zero recovery, the spread between the yield to maturity on a defaultable bond, $r + \lambda$, over that on a riskless bond, r , is equal to the default intensity of the issuer of the defaultable bond:

$$S(t, T) = \lambda$$

17.2.4.1 Non zero Recovery

Armed with the probability distribution of default times, we can now discuss the valuation of defaultable bonds with a nonzero recovery rate. In particular, we will take advantage of the fact that such bonds can be thought of as a portfolio involving two simpler securities: an otherwise comparable zero-recovery defaultable bond and a contingent claim that pays X at the time of default, if a default occurs before the maturity date

of the bond, and zero otherwise, where X is the recovery value of the original bond.

Let $\Phi(t, T)$ be the time- t price of the contingent claim just described. In order to value such a claim we rely on the continuous-time analog of the argument developed in Section 15.4 in the valuation of defaultable bonds with a nonzero recovery value. In particular $\Phi(t, T)$ is equal to the probability-weighted average of all possible recovery payment scenarios involving the bond, where the weights are given by the risk-neutral probability density function of the time of default. Thus,

$$\Phi(t, T) = X \int_t^T Z(t, v) g_t(v) dv \quad (17.49)$$

where $g_t(v) \equiv \tilde{E}_t[e^{-\int_t^v \lambda(u) du} \lambda(v)]$ is the probability density function of the default time, which we discussed in Section 17.2.3.

We are now ready to derive the valuation formula for a zero-coupon bond that has a value of X in the event of default. Given the above discussion, we can write

$$Z^d(t, T) = Z^d(t, T) + \Phi(t, T) \quad (17.50)$$

where the two right-hand side terms of (17.50) are given by (17.46) and (17.49), respectively.

For the simple reduced-form model that assumes that both r and λ are time invariant, the valuation formula for the defaultable zero-coupon bond with recovery value X can be shown to be:

$$Z^d(t, T) = e^{-(r+\lambda)(T-t)} + \frac{X\lambda}{r+\lambda}(1 - e^{-(r+\lambda)(T-t)}) \quad (17.51)$$

17.2.4.2 Alternative Recovery Assumptions

Thus far in the context of reduced-form models we have essentially assumed that investors recover, at the time of default, a fraction of the defaulted instrument's original face value. It should be noted, however, that this is only an assumption, and that there are alternative ways for reduced-form models to handle the valuation of bonds with nonzero recovery values.

One alternative to the framework based on immediate recovery of face value is the so-called equivalent recovery assumption. This was actually the recovery assumption made in the seminal work of Jarrow and Turnbull (1995) [46]. In the original Jarrow-Turnbull model, a defaulted security with face value \$1 is immediately replaced by X otherwise equivalent riskfree zero-coupon bonds, with $0 \leq X \leq 1$. (Obviously, $X = 1$ would constitute the case of a riskfree bond to begin with.) By “otherwise equivalent,” we

mean that the newly issued riskfree zero-coupon bonds will have the same maturity date and face value as the defaulted bond.

How does the valuation of defaultable bonds under the equivalent recovery assumption differ from (17.50)? The assumption that, in the event of default, the recovery payment only takes place at the original maturity of the risky bond makes this valuation exercise simpler because we no longer need to explicitly derive a p.d.f. for the time of default. To see this, note that, at the maturity date T , the bond holder will either receive the full face value of the bond (\$1) or the recovery value X , and the risk-neutral probabilities associated with these events are $Q(t, T)$ and $1 - Q(t, T)$, respectively. Thus the risk-adjusted expected value of the bondholder's payout, based on information available at time t , is

$$Q(t, T) + [1 - Q(t, T)]X$$

and, following the logic set out in Chapter 15, the time- t value of this expected payout is:

$$Z^d(t, T) = Z(t, T)\{Q(t, T) + [1 - Q(t, T)]X\}$$

Recall that $Q(t, T)Z(t, T)$ is simply $Z_0^d(t, T)$. Thus, the bond valuation formula becomes:

$$Z^d(t, T) = Z_0^d(t, T)(1 - X) + Z(t, T)X \quad (17.52)$$

and a comparison of (17.50) and (17.52) makes it clear that one obtains different values for a defaultable bond depending on the assumed recovery scheme.

A third common recovery assumption is the fractional recovery of market value framework, proposed by Duffie and Singleton (1999)[24]. Duffie and Singleton essentially assume that upon a default at time τ the bond loses a fraction L of its market value. (This is equivalent to saying that upon default the bondholder recovers a fraction $1 - L$ of the no-default market value of the bond.) Duffie and Singleton show that, under such circumstances, the value of a defaultable zero-coupon bond would satisfy

$$Z^d(t, T) = \tilde{E}_t \left[e^{-\int_t^T [r(s) + L\lambda(s)] ds} \right] \quad (17.53)$$

which simplifies to

$$Z^d(t, T) = Z(t, T)\tilde{E}_t \left[e^{-\int_t^T L\lambda(s) ds} \right] \quad (17.54)$$

when the riskless rate process is independent of the default process.

TABLE 17.1
Effect of Alternative Recovery Assumptions on the Valuation of a Five-year
Zero-coupon Bond^a

Recovery Assumption	Price (\$)	Yield (percent)	Risk Spread (basis points)
Fractional recovery of face value	.699	7.18	218
Equivalent recovery	.676	7.83	283
Fractional recovery of market value	.664	8.20	320
No recovery	.552	13.00	800
<i>Memo: Riskless five-year bond</i>	.779	5.00	

^aBased on the assumption of flat riskless curve. The riskless rate r is set at 5 percent, and the default intensity of the bond issuer is assumed to be constant at 8 percent. For the nonzero recovery rate cases, X is set at 60 percent ($L = 40$ percent). Face value of the bonds = \$1.

To illustrate the effect of the recovery assumption on the valuation of a zero-coupon bond, we compare the prices of a five-year zero-coupon bond implied by equations (17.50), (17.52), and (17.54) derived from an otherwise identical reduced-form model and based on the same parameter values. Suppose, for instance, that both the riskless interest rate and default intensity are constant at 5 percent and 8 percent, respectively. Let the relevant recovery rates be 60 percent under each of the recovery rate assumptions examined.²¹

Table 17.1 shows that the recovery assumption can have a non-trivial effect on the model-implied prices and yield-to-maturities of this bond, with the latter ranging from 7.18 percent when the bond is priced under the assumption of fractional recovery of the face value of the bond to 8.20 percent under the assumption of partial recovery of the market value of the bond. The case of recovery of face value has the lowest yield given that it involves an immediate payment to the bond holder upon default, as well as a payment that represents 60 percent of the full par value of the bond. Under the equivalent recovery case, even though the recovery value still corresponds to 60 percent of the par value, that payment is effectively received in full only at the bond's original maturity date so that the investor effectively has to be compensated for this "delay" by receiving a higher yield.

The lowest bond price (highest yield) corresponds to the case involving the fractional recovery of market value. This occurs because the market

²¹ For the recovery-of-market-value framework, we assume that L is equal to $1 - X$ or 40 percent.

value of the bond just before default can be substantially below par for a zero-coupon bond. The table also shows the price of the defaultable bond under the assumption of a zero recovery value, which corresponds to a yield of 13 percent (a spread of 800 basis points over the yield on a comparable riskfree bond, shown as a memo item).

The differences in model-implied bond prices can be significantly less dramatic than those shown in Table 17.1 for zero-coupon bonds of shorter maturity. For instance, for a one-year bond, the theoretical yields would range from 8 percent to 8.2 percent given the assumptions listed in the table's footnote. Differences in bond prices across different recovery assumptions can also be much smaller than those in the table for coupon-bearing defaultable bonds, as shown by Duffie and Singleton (1999)[24] in the context of a reduced-form model with stochastic riskfree rates and intensity.

Which recovery assumption should one favor? Each has its pros and cons, and neither has gained complete acceptance among either academics or practitioners. The assumption of fractional recovery of face value is closest to the market convention for defaulted bonds, where the obligations of a liquidated debtor tend to have the same value, assuming the same level of seniority, regardless of their maturity date. Nonetheless, as discussed by O'Kane and Schlogl (2001)[64], the recovery of face value assumption imposes upper bounds on the yields of defaultable bonds, as does the equivalent recovery assumption. The latter has the advantage of being easier to deal with analytically than the recovery of face value assumption. (This can be seen by examining how much simpler it was to arrive at (17.52) than at (17.50), which involved, for instance, the computation of the probability distribution of default times.)

From an analytical perspective, the most tractable of the recovery assumptions is likely the formulation involving fractional recovery of the market value of the bond. Indeed, this assumption has the advantage of making the valuation of defaultable bonds almost entirely analogous to that of default-free bonds. One drawback of the recovery-of-market-value assumption, however, is that one can no longer separately infer default probabilities from observed market quotes using the simple steps outlined in Chapter 16.

17.2.5 *Extensions and Uses of Reduced-form Models*

Most of the model-specific results derived thus far, such as the valuation formulae for defaultable zero-coupon bonds with zero recovery and with fractional recovery of face value—equations (17.48) and (17.51), respectively—came from the simple reduced-form model based on constant intensity and riskless rate of interest and nonrandom recovery rates. To be

sure, these are over-simplistic assumptions, which were made solely for the sake of analytical tractability and pedagogical convenience.

Richer models do exist with various degrees of complexity, ranging from specifications with stochastic riskless rates but deterministically time-varying intensity to fully stochastic models with uncertain recovery. Examples of work in this strand of the literature include Lando (1998)[51], Duffie and Singleton (1999)[24], Jarrow and Yu (2001)[47], Madan and Unal (2000)[58], Schonbucher (1998)[70], and many others.

Reduced-form models are commonly used in practice to extract default probability information from the prices of actively traded instruments and use those probabilities to value, for instance, less liquid or more complex credit derivatives. Nonetheless, given that we essentially did this in Chapter 16 without ever having explicitly to resort to a model, one might wonder why reduced-form models would be used in this context. The answer partly resides in the fact that an important motivation for using such models relates to an assumption proposed in the simple exercise carried out in Chapter 16. There we made the assertion that default intensities were independent of the riskless interest rate process. But one might well suspect that the level of market interest rates and default probabilities are correlated, in which case a credit risk model—such as the reduced-form and structural frameworks described in the current chapter—is needed.²² Other instances where a particular model for the forward default rate are needed include some valuation problems for derivatives involving spread optionality, some of which are discussed in Chapter 18.

17.2.6 Credit Default Swap Valuation

Models in the reduced-form tradition can be used to price both single- and multi-issuer credit derivatives. As an illustration, we will use it to value a vanilla credit default swap. As in Section 17.1.5, we will let S denote the credit default swap premium for a contract that matures at time T . For simplicity, we will continue to assume that the premium is paid continuously and that the notional amount of the contract is \$1. In addition, we will take both the riskless interest rate and default intensity to be time invariant.

In the context of the basic reduced-form model, the present value of the premium leg of the credit default swap can be written as:

$$\eta(t) = \int_t^T S e^{-(r+\lambda)(s-t)} ds = \frac{S}{r+\lambda} (1 - e^{-(r+\lambda)(T-t)}) \quad (17.55)$$

²²We discussed some empirical results on the relationship between defaults and interest rates in Chapter 15.

Notice above that, as in equation (17.48), the premium stream is discounted at the risky rate $r + \lambda$, reflecting the uncertainty surrounding the default event. (λ is the default intensity associated with the reference entity.)

To value the protection leg, we note that it is equivalent to a contingent claim that pays $(1 - X)$ in the event of default before T , where X is the recovery rate of the reference entity's defaulted liabilities. The value of such a claim is given by equation (17.49) with X replaced with $1 - X$. Letting $\theta(t)$ denote the value of the protection leg, we can write

$$\theta(t) = \frac{\lambda(1 - X)}{r + \lambda} (1 - e^{-(r+\lambda)(T-t)}) \quad (17.56)$$

A default swap typically has zero market value when it is set up, and thus pricing such a contract is equivalent to finding the value of S that makes the two legs of the swap have equal value. This is given by

$$S = \lambda(1 - X) \quad (17.57)$$

which gives us the result that, for a given recovery rate and constant r and λ , the credit default swap premium tells us about the default intensity of the reference entity. In other words, by assuming a value for the recovery rate, we can use the above expression and observable default swap spreads to infer the default intensity associated with the reference entity. Such a CDS-implied default intensity can then be used in the valuation of other credit derivatives instruments.

This simple CDS pricing exercise has made several restrictive assumptions, such as continuously paid premiums and constant interest rates and default intensity. As discussed in Section 17.2.4, many of these assumptions can be relaxed in order to bring the valuation exercise closer to reality.

17.3 Comparing the Two Main Approaches

There is no clear consensus in either the practitioner or academic literatures about which of the two credit risk modeling approaches—structural and reduced-form—is the most appropriate one. For instance, while the structural form approach might be said to have the advantage of relying explicitly on the fundamentals underlying a given firm—as these are reflected in the firm's balance sheet—such reliance could also be seen as a drawback. Indeed, balance sheet information tends to become available only on a quarterly basis, which could be a limitation if one is interested, for instance, in accurately marking positions to market. (Moreover, balance-sheet information can be quite noisy as the financial reporting scandals of the early 2000s painfully reminded the markets.) At the same time,

reduced-form models might be of more limited value if one's interest is, for example, in assessing how a change in the capital structure of a given firm may affect its financing costs, in which case careful use of a structural model might be the most appropriate approach.

The two approaches can also be compared from a methodological perspective. For instance, those familiar with models of the term structure of interest rates tend to feel more at ease, from a purely technical standpoint, in the world of reduced-form models. Take, for instance, the relationship between the survival probability, $Q(t, T)$, and the forward default rate, $h(t, s)$,

$$Q(t, T) = e^{-\int_t^T h(t, s) ds}$$

As noted, this is similar to the mathematical relationship between zero-coupon bond prices $Z(t, T)$ and time- t instantaneous forward interest rates. Likewise, the mathematical treatment of default intensity, λ , as well as its relationship with $h()$ and $Q()$, is analogous to the links among the spot short-term interest rate, instantaneous forward interest rates, and zero-coupon bond prices. Moreover, when λ is assumed to be stochastic, many of the basic models for describing its evolution mirror common specifications developed for interest rate models, such as the well-known yield curve model of Cox, Ingersoll, and Ross (1985)[15].

While fixed-income modelers might find substantial commonality between techniques used in interest rate models and those used in the reduced-form approach, equity-minded analysts will no doubt see familiar ground in the structural approach. In addition to focusing on balance-sheet information, to which many equity analysts are already used, a centerpiece of the structural approach is the use of equity-based option theoretic results to price defaultable debt instruments.

To sum up, from a methodological perspective, the pros and cons of each approach have to be examined in the context in which the models will be used, with certain models being more naturally suited for certain applications. In addition, some analysts might be more attracted to one approach vs. the other based on how comfortable they feel with the underlying methodological framework behind each class of models.

Empirically, while structural models are appealing in that they attempt to link explicitly the likelihood of default of a given firm to its economic and financial condition, traditional forms of such models tend not to fit the data as well as reduced-form models, especially, as noted in Section 17.1.4, in relation to short-term credit spreads. Indeed, in the intensity-based framework, defaults can happen suddenly and unexpectedly without having to be presaged by observable phenomenon, such as the value of the firm approaching the default barrier in the typical BSM-style model. In this regard, what may be characterized as another class of credit risk models has emerged.

These models attempt to combine the economic/intuitive appeal of structural models with the empirical plausibility of the intensity-based approach. The model of Duffie and Lando (2001)[23] is one of the better-known works in this strand of the literature.²³

The Duffie-Lando model can be thought of as a hybrid structural/intensity-based model because it is essentially a first-passage model that, contrary to standard structural models, also has an intensity-based interpretation. In particular, rather than being given exogenously as in the pure reduced-form approach, default intensity in the Duffie-Lando model can be calculated in terms of observable variables related to the balance-sheet fundamentals of the firm. Moreover, Duffie and Lando argue that many estimation methods used in the context of pure intensity-based models are also applicable to the hybrid framework.

Essentially, a main thrust of this hybrid approach to credit risk modeling is to assume that investors only have imperfect information about the true financial condition of the firm. For instance, investors may not know with certainty just how far the value of the firm is from its default boundary. As a result, the possibility of a default in the very near term cannot be fully discarded, and thus the pattern of short-term credit spreads generated by these models tends to be more realistic than that implied by traditional BSM-type models.²⁴

17.4 Ratings-based Models

Instead of allowing for the firm to be only in one of two states—default and survival—ratings-based models allow for a variety of states, where each non-default state might correspond, for instance, to a given credit rating—such as AAA, A, BB⁺, etc.—assigned to the firm by a major credit rating agency. Such models are not widely used for the direct pricing of vanilla credit derivatives as most such instruments do not have payouts that are ratings-dependent. Nonetheless, ratings-based models can be useful in the context of credit derivatives that involve collateral requirements that are linked to the credit rating of the counterparties.

We will not describe ratings-based models in detail in this book, but simply highlight their main features and how they relate to the models examined in previous sections. A well-known ratings-based model is that

²³Other related work includes Giesecke (2001)[32] and Giesecke and Goldberg (2004)[33].

²⁴Within the structural framework, Zhou's (1997)[76] jump-diffusion model, discussed in Section 17.2, constitutes an alternative approach to address the empirical implausibility of the short-term credit spreads implied by traditional structural models.

of Jarrow, Lando, and Turnbull (1997)[45]. In what follows, we limit ourselves mostly to providing a basic description of a discrete-time version of the Jarrow-Lando-Turnbull (JLT) model. In so doing, we follow JLT themselves, who also used a discrete-time setting to introduce their model.

Suppose that a given bond issuer can have one of $J - 1$ credit ratings, with 1 representing the highest credit quality and $J - 1$ representing the rating just prior to default. We will also allow for a J th “rating,” which will correspond to default. Let $\omega_{i,j}$ represent the actual (not necessarily risk-neutral) probability, based on all information available at time t , of the firm migrating from a rating of i at time t to one of j at time $t + 1$. For simplicity we shall assume that these probabilities are time-invariant over the horizon of interest, which we assume to span from time 0 to time U . Let us define the $J \times J$ transition matrix Ω such that its (i, j) th element is $\omega_{i,j}$:

$$\Omega = \begin{pmatrix} \omega_{1,1} & \omega_{1,2} & \dots & \omega_{1,J} \\ \omega_{2,1} & \omega_{2,2} & \dots & \omega_{2,J} \\ \dots & \dots & \dots & \dots \\ \omega_{J-1,1} & \omega_{J-1,2} & \dots & \omega_{J-1,J} \\ 0 & 0 & \dots & 1 \end{pmatrix} \quad (17.58)$$

We further assume that:

$$\omega_{i,j} \geq 0 \quad \text{for all } i, j, i \neq j \quad (17.59)$$

$$\omega_{i,i} \equiv 1 - \sum_{j=1, j \neq i}^J \omega_{i,j} \quad \text{for all } i \quad (17.60)$$

Equation (17.60) essentially acknowledges the fact that for any given firm rated i at time t , its time- $t + 1$ rating will have to be one of the J ratings. In particular, the probability that the firm will retain its i rating at $t + 1$ must be one minus the sum of the probabilities associated with migration to any one of the remaining $J - 1$ ratings.

In technical terms, the last row of Ω says that default—the J th “rating”—is an “absorbing state,” meaning that once the firm enters into a state of default, we assume that it will stay there with probability $\omega_{J,J} = 1$. In corporate finance terms, the model assumes that there is no reorganization after default.

Let $\omega_{i,j}(t, t + n)$ be the probability, conditional on information available at time t , that the firm’s rating will change from i at time t to j at time $t + n$. If $\Omega(t, t + n)$ is the matrix such that its (i, j) th element is $\omega_{i,j}(t, t + n)$,

it can be shown that:

$$\Omega(t, t + n) = \Omega^n \quad (17.61)$$

i.e., ratings transitions are said to follow a Markovian process in that the current ratings transition matrix is assumed to contain all currently available relevant information regarding future ratings transitions.

Empirical estimates of Ω are published regularly by some of the major credit-rating agencies based on actual rating changes in the universe of firms covered by these agencies. For pricing purposes, of course, what matter are the risk-neutral transition probabilities $\tilde{\omega}_{i,j}$, rather than the empirical probabilities in Ω . JLT propose the following mapping between empirical and risk-neutral probabilities:

$$\tilde{\omega}_{i,j}(t, t + 1) = \pi_i(t)\omega_{i,j} \quad \text{for all } i, j, i \neq j \quad (17.62)$$

for $\pi_i(t) \geq 0$ for all i and t . Consistent with the discussion in Chapter 15, $\pi_i(t)$ can be thought of as a risk-premium-induced adjustment to the actual transition probabilities.

The risk-neutral transition probabilities are assumed to satisfy conditions analogous to those in equations (17.59) and (17.60). In addition, JLT imposed the technical condition that $\tilde{\omega}_{i,j}(t, t + 1) > 0$ if and only if $\omega_{i,j} > 0$, for $0 \leq t \leq U - 1$.

Thus, we can also define the risk-neutral transition matrix $\tilde{\Omega}(t, t + n)$, and it should by now be clear that its (i, j) th entry is $\tilde{\omega}_{i,j}(t, t + n)$, which is the risk-neutral probability that the entity will migrate from a rating of i at time t to one of j at time $t + n$. If we make the simplifying assumption that both Ω and π_i are time invariant:

$$\tilde{\Omega}(t, t + n) = \tilde{\Omega}(t, t + 1)^n \quad (17.63)$$

Suppose one is interested in the risk-neutral probability, conditional on all information available at time t , that an i -rated firm will survive through some future date T . Given all the assumptions discussed thus far, this probability is simply

$$Q_i(t, T) = 1 - \tilde{\omega}_{i,J}(t, T) \quad (17.64)$$

Thus, as discussed throughout this part of the book, and assuming that riskless interest rates are independent of the stochastic process underlying the ratings transitions of the firm, the time- t price of a zero-recovery, zero-coupon bond that will mature at time T with a face value of \$1 is:

$$Z_0^d(t, T; i) = Z(t, T)[1 - \tilde{\omega}_{i,J}(t, T)] \quad (17.65)$$

where we added the argument i to the zero-coupon bond price to indicate that this bond was issued by a firm that is currently rated i .

If $R^d(t, T; i)$ denotes the yield to maturity on this bond, we can use results discussed in earlier sections in this chapter to derive the credit risk spread associated with this firm. In particular,

$$\begin{aligned} R^d(t, T; i) - R(t, T) &= \frac{-\log(Z_0^d(t, T; i)) + \log(Z(t, T))}{T - t} \\ &= \frac{-\log(1 - \tilde{\omega}_{i,J}(t, T))}{T - t} \end{aligned} \quad (17.66)$$

where we continue to assume that $R(t, T) \equiv \frac{-\log(Z(t, T))}{T - t}$ is the yield to maturity on a riskless zero-coupon bond with the same maturity date and face value as the risky bond.

Thus far we have been using the model essentially to derive expressions for prices and spreads that we were also able to examine with the modeling approaches summarized in Sections 17.1 and 17.2. As their name suggests, however, ratings-based models are particularly suitable for analyses involving yield spreads across different ratings. For instance, the model-implied yield spread between two bonds rated i and j is

$$R^d(t, T; i) - R^d(t, T; j) = \log \left[\frac{1 - \tilde{\omega}_{j,J}(t, T)}{1 - \tilde{\omega}_{i,J}(t, T)} \right] \frac{1}{T - t} \quad (17.67)$$

Equations (17.66) and (17.67) can be used to identify potentially profitable opportunities across different issuers with various credit ratings, by, for instance, comparing model-implied spreads to the ones observed in the market place. Alternatively, one may be interested in using equations like (17.66) and (17.67) to calibrate the model to the data in order to use the resulting risk-neutral transition probabilities to value financial instruments and contracts with ratings-dependent payoffs, such as bonds with ratings-dependent coupons and credit derivatives contracts with ratings-linked collateral requirements.

A full continuous-time version of the model described in this section is provided in the original JLT paper—Jarrow, Lando, and Turnbull (1997)[45]—which also addresses calibration-related issues. Discussing the technical details behind that version of the model, as well as model calibration and other implementation topics, is outside the scope of this book. Instead we limit ourselves to providing a very brief overview of some basic concepts that are germane to the continuous-time specification of the JLT model. In particular, in the simplest case of time-invariant risk-neutral transition probabilities, the $J \times J$ transition matrix for the continuous-time

version of the model can be written as:

$$\Omega(t, T) = e^{\Lambda(T-t)} \quad (17.68)$$

where the $J \times J$ matrix Λ is typically called the generator matrix. The i th diagonal element of Λ , $\lambda_{i,i}$, can be thought of as the exit rate from the i th rating, and, for $i \neq j$, the (i, j) th element of Λ , $\lambda_{i,j}$, is the transition rate between ratings i and j . The concepts of exit and transition rates are analogous to that of default intensity, examined in Section 17.2. In this sense, one can think of the JLT model as a generalized intensity-based model, and, indeed, the JLT model is essentially an extension of the Jarrow and Turnbull (1995)[46] model.

The literature on ratings-based models is a vast one, and the uses and implications of these models go well beyond the analysis and valuation of credit derivatives. Other contributions to the literature include the work of Kijima and Komoribayashi (1998)[50], Lando and Skodeberg (2002)[52], Das and Tufano (1996)[19], and Arvanitis, Gregory, and Laurent (1999)[3]. Some structural credit risk models that are commercially available also incorporate the analysis of ratings transitions. We summarize the main features of a few well-known commercial models in Chapter 22.

18

Valuing Credit Options

Chapter 8 contained a basic discussion of the main features of spread and bond options. In this chapter we describe a relatively simple framework for valuing these instruments. We start Section 18.1 with a discussion of forward-starting credit default swaps, introducing some concepts that will come in handy in the valuation of credit default swaptions, the subject of Section 18.2. Section 18.3 generalizes the valuation approach for credit default swaptions so it can be used with other spread options. Extensions and alternatives to the simple framework described in Sections 18.2 and 18.3 are briefly discussed in Section 18.4. The valuation of bond options is sketched out in Section 18.5.

18.1 Forward-starting Contracts

At the end of Chapter 7 we briefly mentioned the forward-starting total return swap, which is a contractual commitment to enter into a total return swap at a fixed future date and at a predetermined spread. In this discussion of valuation methods for credit default swaptions and other credit options, we will meet two additional types of forward-starting contracts, the forward-starting credit default swap and forward contracts involving floaters. The aims of such contracts are self-evident; they are agreements to enter into a credit default swap and to buy and sell floaters, respectively, at future dates and at predetermined premiums (in the case of a CDS) and spreads (the forward floater contract).

As we shall see below, forward-starting credit default swaps can be thought of as the underlying “asset” in a credit default swaption, and thus it will be instructive to have a basic understanding of how they are valued before proceeding to examine the valuation of credit default swaptions. The same idea applies to forward contracts written on floaters, and so we shall examine them in some detail.

18.1.1 Valuing a Forward-starting CDS

Consider a forward-starting CDS agreement entered into at time t where one party agrees to buy protection in a CDS that will start at the future date T with a corresponding CDS premium of K , and premium payment dates $T_1, T_2, \dots, T_n$. For simplicity, we assume that the notional amount of the CDS is \$1.

From Section 16.2, we know that the time- t market value of such an agreement to the protection buyer can be written as

$$W(t) = \sum_{j=1}^n Z(t, T_j) \{ [Q(t, T_{j-1}) - Q(t, T_j)](1 - X) - Q(t, T_j) \delta_j K \} \quad (18.1)$$

where $Z(t, T_j)$ corresponds to the proxy for a riskfree discount factor—which, as discussed in Chapter 16, tends to be derived in practice from the LIBOR/swap curve to reflect the funding costs of the large banks that tend to be most active in the CDS market—and X and $Q(t, T_j)$ relate to, respectively, the recovery rate of the reference entity ($0 \leq X < 1$) and the risk-neutral probability that the reference entity will survive through T_j , conditional on all information available at time t . δ_j is the accrual factor for the j th premium payment (the number of days between the $(j - 1)$ th and j th premium payment dates divided by the number of days in the year, based on the appropriate day-count convention).

We can now introduce the notion of the forward CDS premium, which can be thought of as the value of K in (18.1) such that the forward-starting credit default swap has zero market value at time t . We shall let $S^F(t, T, T_n)$ denote the forward CDS premium, as seen at time t , for a CDS contract that will start at time T and have premium payment dates at $T_1, T_2, \dots, T_n$. Solving (18.1) for K while requiring $W(t)$ to be zero, we can write

$$S^F(t, T, T_n) = \frac{\sum_{j=1}^n Z(t, T_j) [Q(t, T_{j-1}) - Q(t, T_j)] (1 - X)}{\sum_{j=1}^n Z(t, T_j) Q(t, T_j) \delta_j} \quad (18.2)$$

and substituting this last expression into (18.1) we arrive at a convenient formula for the market value of a protection-buying position in a

forward-starting credit default swap:

$$W(t) = \sum_{j=1}^n Z(t, T_j) Q(t, T_j) \delta_j [S^F(t, T, T_n) - K] \quad (18.3)$$

which has the intuitive implication that the market value of a forward-starting CDS depends crucially on the difference between the corresponding forward CDS premium and the predetermined premium written into the contract.¹

18.1.2 Other Forward-starting Structures

The valuation of other credit-related forward-starting structures, such as forward-starting asset swaps and forward contracts involving floating-rate notes, can be carried out using similar methods to the one just described for forward-starting credit default swaps. Consider, for instance, a forward contract to receive par for a floating-rate note at a future date with a prespecified spread over LIBOR. Assume, for simplicity, that the floater has a zero recovery rate. (This is a forward contract to sell a given floater for its par value at a future date at a predetermined spread.)

Recall, from Chapter 4, that the time- T market value of a just-issued par floater with a face value of \$1 and coupon payment dates $T_1, T_2, \dots, T_n$ can be written as

$$1 = \sum_{j=1}^n Z_0^d(T, T_j) \delta_j [F^*(T, T_{j-1}, T_j) + s(T, T_n)] + Z_0^d(T, T_n) \quad (18.4)$$

where $F^*(T, T_{j-1}, T_j)$ is the point on the forward LIBOR curve, as seen at time T , that corresponds to a loan lasting from the future date T_{j-1} to T_j ; $s(T, T_n)$ is the par floater spread, and, to simplify the notation, $Z_0^d(T, T_j) \equiv Z(T, T_j) Q(t, T_j)$.

Likewise, for a par floater that pays LIBOR flat:

$$1 = \sum_{j=1}^n Z(T, T_j) \delta_j F^*(T, T_{j-1}, T_j) + Z(T, T_n) \quad (18.5)$$

which differs from the previous equation only because of the zero spread and the choice of discount factors.

¹ Note the similarity between (18.3) and the expression for marking to market a CDS position in Chapter 16.

The time- t value of the latter par floater, for $t < T$, can be shown to be

$$Z(t, T) = \sum_{j=1}^n Z(t, T_j) \delta_j F^*(t, T_{j-1}, T_j) + Z(t, T_n)$$

which can be verified given the definition of forward LIBOR (see Chapter 4):

$$F^*(t, T_{j-1}, T_j) \equiv \delta_j^{-1} \left[\frac{Z(t, T_{j-1})}{Z(t, T_j)} - 1 \right]$$

As for the riskier par floater, its time- t value becomes

$$Z_0^d(t, T) = \sum_{j=1}^n Z^d(t, T_j) [F^*(t, T_{j-1}, T_j) + s^F(t, T, T_n)] \delta_j + Z_0^d(t, T_n)$$

where $s^F(t, T, T_n)$ is defined as the forward par floater spread associated with this particular issuer, as seen at time t , for future borrowing between times T and T_n .

Given the above, the task of valuing an arbitrary forward contract involving a floater that will pay a spread of say K , which is not necessarily the par spread, is relatively straightforward, and the reader can easily verify that, from the perspective of the party committed to selling the floater, the time- t value of such a contract can be written as

$$W^{FL}(t) = \sum_{j=1}^n Z^d(t, T_j) \delta_j [s^F(t, T, T_n) - K] + Z^d(t, T) \quad (18.6)$$

which, again, has the simple intuition that a contract to sell a floater at a future date for par—in essence, a contract to pay a given spread starting at some future date—will have positive market value whenever the spread K written into the contract is below the corresponding forward spread associated with the issuer.

We carried out this discussion with a forward contract to sell a floater. The results would be entirely analogous for a forward-starting asset swap, and we leave this exercise to the reader.

18.2 Valuing Credit Default Swaptions

Let $W(t)$ be the time- t value, to a protection buyer, of a forward-starting credit default swap. Continuing with the same setup introduced in the

previous section, the CDS will start at a future time T , with payment dates at $T_1, T_2, \dots, T_n$, and the premium is set at K . As a result:

$$W(t) = \sum_{j=1}^n Z(t, T_j) Q(t, T_j) \delta_j [S^F(t, T, T_n) - K]$$

which is simply (18.3).

Consider now a European option, written at time t , to buy protection in the contract underlying the forward-starting CDS described in the previous section. At time T , the exercise date of the option, $S^F(T, T, T_n) = S(T, T_n)$, i.e., the forward premium converges to the spot premium, and the value of the default swaption will be:

$$V(T) = \text{Max}(W(T), 0) = \text{Max} \left\{ \sum_{j=1}^n Z_0^d(T, T_j) \delta_j [S(T, T_n) - K], 0 \right\} \quad (18.7)$$

Equation (18.7) tells us that the holder of this credit default swaption will exercise it only if the underlying CDS has positive market value at T , which is the case whenever the then-prevailing par CDS premium exceeds the premium written into the option (otherwise the holder would be better off paying the prevailing CDS premium, $S(T, T_n)$, in a par CDS, which has zero market value).

One can think of the term $\sum_{j=1}^n Z_0^d(T, T_j) \delta_j$ in equation (18.7) as an annuity factor that gives the time- T value of the entire stream of differences between the premium payments in a par CDS contract and the one specified in the default swaption. If we let $A(T, T_n)$ denote this factor, we can write the time- T value of the default swaption as

$$V(T) = \text{Max}[A(T, T_n)(S^F(T, T, T_n) - K), 0] \quad (18.8)$$

To find the time- t value of the default swaption, it is convenient to rewrite (18.8) as

$$V(T) = A(T, T_n) \text{Max}[(S^F(T, T, T_n) - K), 0] \quad (18.9)$$

which tells us that the time- T value of the option is simply a function of the present value of the difference between the premium payments of the two credit default swaps.

If we now recall that the time- t value of any financial asset is simply the risk-adjusted expected present value of its cash flow, we can write

$$V(t) = A(t, T) E_t^* [\text{Max}(S^F(T, T, T_n) - K, 0)] \quad (18.10)$$

where $E_t^*[\cdot]$ denotes the expected value of “.” conditional on information available at time t , computed on the basis of probabilities that are appropriately adjusted for risk in a way that follows the spirit of the risk-neutral probabilities discussed in Chapter 15.²

In order to derive a pricing formula for this default swaption, we need to have an explicit assumption (a model) that describes the evolution of the forward CDS premium over time. A common assumption is to assert that $S^F(t, T, T_n)$ is lognormally distributed, which allows one to use the option pricing formula derived by Black (1976)[8].³ If we let $\sigma(t, T, T_n)$ denote the volatility of percentage changes in $S^F(t, T, T_n)$, we can write the Black formula for a credit default swaption as:

$$V(t) = A(t, T_n)[S^F(t, T, T_n)N(d_1) - KN(d_2)] \quad (18.11)$$

with

$$d_1 \equiv \frac{\log(S^F(t, T, T_n)/K)}{\sqrt{\sigma(T, T_n)^2(T-t)}} + .5\sigma(T, T_n)\sqrt{T-t}$$

$$d_2 \equiv \frac{\log(S^F(t, T, T_n)/K)}{\sqrt{\sigma(T, T_n)^2(T-t)}} - .5\sigma(T, T_n)\sqrt{T-t}$$

$N(\cdot)$ is the cumulative standard normal distribution, and we made the simplifying assumption that $\sigma(t, T, T_n)$ is time-invariant.

As with vanilla calls and puts, credit default swaption prices are strictly increasing in the volatility of the relevant forward CDS premium. Other basic features of call and put options, such as put-call parity, also hold.

18.3 Valuing Other Credit Options

The valuation of other credit options, such as an option to sell the floater underlying the forward contract discussed in Section 18.2, can be carried out following essentially the same steps outlined for credit default swaptions.

²The reader with some familiarity with continuous-time finance methods may recognize the probability measure embedded in $E_t^*[\cdot]$ as that corresponding to the so-called “annuity measure” (Hunt and Kennedy, 2000[43]). Under this probability measure, both the relative price $\frac{V(t)}{A(t, T_n)}$ and the forward par CDS premium $S^F(t, T, T_n)$ follow a random walk.

³The Black pricing formula is a variant of the well-known Black-Scholes formula. Black originally derived it for the pricing of options on futures contracts, but it can be shown that it applies directly to the pricing of credit default swaptions and many other related options. See Hull (2003)[41] for a textbook discussion of the Black formula.

Consider, for instance, a put option on a floater with a face value of \$1. The option expires at date T and has a strike price of \$1. As discussed in Chapter 8, this is basically a call option on the floater spread.

At time T , the value of the put option will be

$$V^{FL}(T) = \text{Max}(1 - W^{FL}(T), 0) \quad (18.12)$$

i.e., the option holder will choose to sell the floater for its face value only if the market value of the floater, $W^{FL}(T)$, is less than \$1.

We can use the results of Section 18.1.2 to rewrite equation (18.12) as

$$V^{FL}(T) = A(T, T_n) \text{Max}[s^F(T, T, T_n) - K, 0] \quad (18.13)$$

and from here on one would proceed as in Section 18.2 to obtain the Black pricing formula for this call spread option.

18.4 Alternative Valuation Approaches

An implicit (but fundamental) assumption made in this chapter is that credit spreads and, thus, CDS premiums are stochastic variables. Indeed, it would make no sense to write an option on something that behaves deterministically. Going back to results derived in previous chapters, however, which suggested a close relationship between CDS premiums and the default probabilities of the reference entity, the assumption of stochastic forward CDS premiums is tantamount to admitting that default probabilities are themselves stochastic.

Indeed, an alternative approach to valuing spread and bond options is to model the stochastic behavior of default intensity directly, often jointly with the behavior of short-term interest rates, and use the resulting framework to derive the option prices of interest. As discussed by Arvanitis and Gregory (2001)[2] and Schonbucher (1999)[71], stochastic default probability models can also be used to value more complex credit options than the ones examined in this chapter, such as Bermudan options and other structures with more than one exercise date.

18.5 Valuing Bond Options

We discussed above the valuation of options written on floating-rate bonds and on credit default swaps. Similar methods apply to the valuation of options written on fixed-rate bonds. For instance, one may assume that the forward price of the bond is lognormally distributed and then use the Black formula as in Sections 18.2 and 18.3. Hull (2003)[41] provides a useful textbook discussion of the valuation of fixed-rate bond options.

This Page Intentionally Left Blank

Part IV

Introduction to Credit Modeling II: Portfolio Credit Risk

This Page Intentionally Left Blank

19

The Basics of Portfolio Credit Risk

The credit risk models we have examined thus far in this book have all focused on single default events, or on the likelihood that a given firm will default on its financial obligations within a given period of time. We shall now shift gears, so to speak, and take a quick tour of approaches and techniques that are useful for modeling credit risk in a portfolio setting. As we saw in Chapters 9, 10, and 14, two key concepts in the modeling of portfolio credit risk are default correlation and the loss distribution function. The basic model discussed in this chapter allows us to take a closer quantitative look at these concepts.

Before we proceed, however, one caveat is in order. The discussion in this part of the book only scratches the surface of what has become a large and growing technical literature on the modeling of portfolio credit risk. Our goal is only to introduce the reader to some of the main issues and challenges facing both academics and practitioners in the real world when it comes to modeling the default risk embedded in a portfolio of credit-related instruments.

19.1 Default Correlation

Intuitively, default correlation is a measure related to the likelihood that two or more reference entities will default together within a given horizon. The higher the default correlation between two firms, the higher the chances that default by one of them may be accompanied by a default by the other. Given that corporate defaults are relatively rare events,

empirical measures of default correlation are not easy to come by. The available evidence suggests that default correlation tends to be higher for lower quality credits for higher borrowers, presumably because less-credit worthy firms are more sensitive to the ups and downs of the economy, and that the extent of default correlation depends on the time horizon in question (Lucas, 1995)[57]. In addition, it seems plausible that default correlation tends to be higher among firms in the same industry than among firms in different lines of business altogether, although one can imagine situations where a default by one firm strengthens the position of its main competitor, making it less likely to default.

19.1.1 Pairwise Default Correlation

From elementary statistics, we know that the correlation coefficient, ρ_{Y_1, Y_2} , involving two random variables Y_1 and Y_2 is defined as

$$\rho_{Y_1, Y_2} \equiv \frac{\text{Cov}[Y_1, Y_2]}{\sqrt{\text{Var}[Y_1]} \sqrt{\text{Var}[Y_2]}} \quad (19.1)$$

where $\text{Cov}[Y_1, Y_2]$ denotes the covariance between Y_1 and Y_2 , and $\text{Var}[Y_1]$ and $\text{Var}[Y_2]$ stand for the variances of Y_1 and Y_2 , respectively, all of which are defined below.

Mathematically, one can write an expression for the coefficient of default correlation for any two entities—called the pairwise default correlation—in terms of their respective default probabilities. To see this, we shall take another look at the two hypothetical entities we have been analyzing throughout this book: XYZ Corp. and AZZ Bank.

Using notation introduced earlier, we shall let ω_A and ω_X denote the (risk-neutral) default probabilities of AZZ and XYZ, respectively, over a given time horizon.¹ As for the probability that both AZZ Bank and XYZ Corp. will default together, we shall denote it as $\omega_{A\&X}$. These probabilities are represented in the diagram in Figure 19.1. The area encompassed by the rectangle in the figure represents all possible survival and default outcomes associated with AZZ and XYZ over a given time horizon. Those outcomes that involve a default by AZZ are shown within area A and those involving defaults by XYZ are represented by area X. For simplicity, we shall assume that the area of the rectangle is equal to 1 and thus we can think of the areas A and X as the probabilities that AZZ Bank and XYZ Corp. will default, respectively, within the prescribed time horizon. Furthermore, the region of overlap between areas A and X corresponds to the probability $\omega_{A\&X}$ that both AZZ Bank and XYZ Corp. will default.

¹In terms of the notation used in Part III, ω_A and ω_X can be written as $1 - Q_A(t, T)$ and $1 - Q_X(t, T)$, respectively, where, for instance, $Q_A(t, T)$ is the risk-neutral probability, conditional on all information available at time t , that AZZ will survive through some future time T . As in previous chapters, we continue to denote the current time as t .

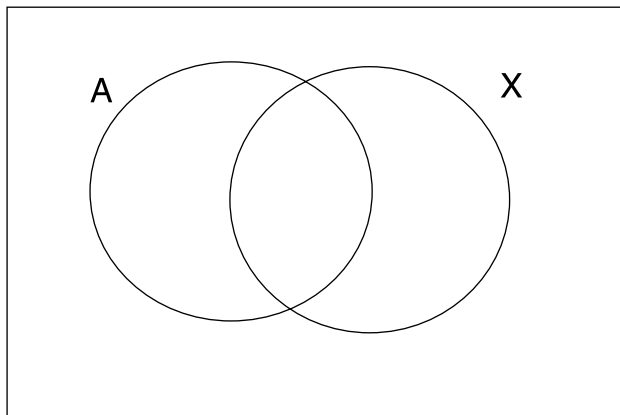


FIGURE 19.1. Diagrammatic Representation of Default Probabilities

Let us now define I_X as a random variable that takes the value of 1 in the event of default by XYZ Corp. over a given horizon and 0 otherwise. I_A is likewise defined for AZZ Bank. In technical terms, I_X and I_A are called indicator functions regarding default events by XYZ and AZZ. It is straightforward to see that the expected values of I_X and I_A are simply the associated default probabilities over the horizon of interest. For instance, given that I_A can only take two values, and relying on risk-neutral probabilities, its expected value is:

$$\tilde{E}[I_A] = \omega_A \times 1 + (1 - \omega_A) \times 0 = \omega_A \quad (19.2)$$

where $\tilde{E}[I_A]$ is the expectation of I_A , computed on the basis of risk-neutral probabilities.

Again using basic results from elementary statistics, we can also compute the variance of I_A and the covariance between I_A and I_X . The variance is

$$\text{Var}[I_A] \equiv \tilde{E} \left[(I_A - \tilde{E}[I_A])^2 \right] = \omega_A(1 - \omega_A) \quad (19.3)$$

and the covariance can be written as

$$\text{Cov}[I_A, I_X] \equiv \tilde{E} \left[(I_A - \tilde{E}[I_A])(I_X - \tilde{E}[I_X]) \right] = \tilde{E}[I_A I_X] - \tilde{E}[I_A] \tilde{E}[I_X]$$

But notice that $I_A I_X$ is only nonzero when both I_A and I_X are 1, and that happens with probability $\omega_{A \& X}$. Thus

$$\tilde{E}[I_A I_X] = \omega_{A \& X}$$

and we can write

$$\text{Cov}[I_A, I_X] = \omega_{A\&X} - \omega_A\omega_X \quad (19.4)$$

We can now derive an expression for the pairwise default correlation for AZZ Bank and XYZ Corp. entirely in terms of their respective default probabilities:

$$\rho_{A,X} \equiv \frac{\text{Cov}[I_A, I_X]}{\sqrt{\text{Var}[I_A]}\sqrt{\text{Var}[I_X]}} = \frac{\omega_{A\&X} - \omega_A\omega_X}{\sqrt{\omega_A(1 - \omega_A)}\sqrt{\omega_X(1 - \omega_X)}} \quad (19.5)$$

Equation (19.5) formalizes an intuitive and key result regarding pairwise default correlations. In particular, other things being equal, the higher the probability that any two entities will default together over the prescribed horizon—the region of overlap between areas A and X in Figure 19.1—the higher their pairwise default correlation coefficient over that horizon.²

It can be shown that (19.5) is such that the pairwise default correlation coefficient lies between 1 (perfect positive correlation) and -1 (perfect negative correlation). In the former case, the two entities either default or survive together; in the latter case, only one of the entities will survive through the end of the relevant horizon and a position in a bond issued by one entity can be used as a hedge against default-related losses in an otherwise comparable bond issued by the other.

The case of perfect positive correlation, $\rho_{A,X} = 1$, corresponds to a scenario where the two entities have identical default probabilities, $\omega_A = \omega_X = \bar{\omega}$, and where $\omega_{A\&X} = \bar{\omega}$. The case of perfect negative correlation corresponds to a scenario where $\omega_{A\&X}$ is zero, its lower bound, and $\omega_X = 1 - \omega_A$, i.e., there is no region of overlap between areas A and X in Figure 19.1 and area X takes up the entire portion of the rectangle that is not encompassed by area A.³

In addition to the two polar cases of perfect positive and negative correlation, another special case is the situation of zero default correlation between any two entities. As can be seen in equation (19.5), this corresponds to the case where $\omega_{A\&X} = \omega_A\omega_X$.

Before we proceed, we can use a variant of Figure 19.1 to verify a result advanced in Chapter 9. We claimed in that chapter that as the default

² Equation (19.5) is only one of several possible ways to define default correlation. For instance, an alternative approach would be to focus on the correlation of default times. Technically oriented readers interested in this topic may wish to consult Li (2000)[55] and Embrechts, McNeil, and Strautman (1999)[26].

³ One way to verify that 1 is the maximum value that $\rho_{A\&X}$ can achieve is to convince yourself that $\omega_{A\&X}$ cannot be larger than $\text{Min}[\omega_A, \omega_X]$. Hint: Look at the diagram in Figure 19.1.

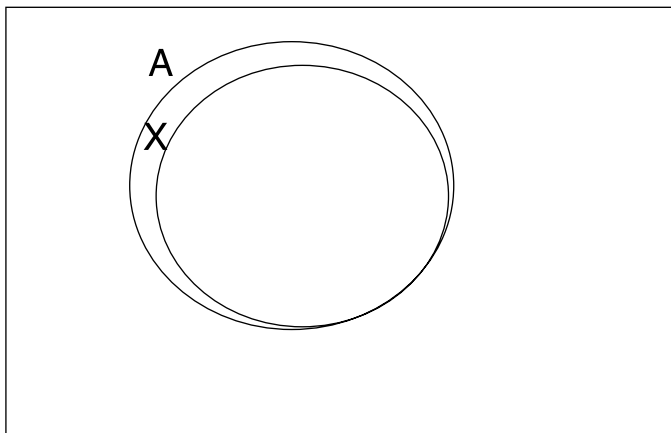


FIGURE 19.2. Diagrammatic Representation of a Case of High Default Correlation

correlation between two entities approaches 1, the probability that both entities will default over a given period approaches the default probability of the entity with the higher credit quality. This is illustrated in Figure 19.2, which shows a near complete overlap between areas A and X—or a default correlation of nearly one according to the results discussed above—but where area A is larger than area X—AZZ bank is of an inferior credit quality (greater default probability) than XYZ Corp. In the case depicted in the figure, all scenarios where XYZ Corp. defaults are also scenarios where AZZ Bank defaults. Hence the region of overlap between areas A and X is the same as area X.

19.1.2 Modeling Default Correlation

Thus far we have essentially limited ourselves to a portfolio with debt instruments issued by only two entities. How do we examine default correlation in more realistic settings, which often involve a large number of entities? We discuss below a very simple modeling framework that allows us to start tackling this issue and that should expose many of the difficulties and challenges associated with modeling portfolio credit risk.⁴

The model builds on the seminal work of Black and Scholes (1973)[9] and Merton (1974)[59], which we discussed in Part III. In particular, for a hypothetical firm i , we define default as a situation where the return R_i on

⁴ Versions of the simple model discussed in this section can be found in Gupton, Finger, and Bhatia (1997)[37], O’Kane and Schlogl (2001)[64], Crosbie (2002)[16], and many others.

owning the firm falls below a given threshold C_i at a given date T .⁵ Thus,

$$\text{Default by firm } i \text{ at time } T \iff R_i \leq C_i \quad (19.6)$$

and the default probability associated with this firm is simply the probability that its return will fall short of C_i over the relevant time horizon.

To simplify the analysis, especially when debt instruments issued by several firms are included in the portfolio, we shall assume that the above returns have already been standardized, i.e., for each firm i in the portfolio, we express returns in terms of deviations from the sample mean and divide the result by the standard deviation of returns:

$$R_{i,t} \equiv \frac{\bar{R}_{i,t} - \mu_i}{\sigma_i}$$

where $\bar{R}_{i,t}$ are the raw (“non-standardized”) returns associated with owning firm i at time t , and μ_i and σ_i are the mean and standard deviation of $\bar{R}_{i,t}$, respectively. Note that, by working with standardized returns, all the firms in the portfolio have the same expected returns and standard deviations of returns—0 and 1, respectively.

Let us now consider a portfolio that contains debt instruments issued by I different firms, where default events for each firm are defined as in equation (19.6). Two key assumptions of the model we are about to introduce are that (i) individual returns R_i are normally distributed, and (ii) returns (and thus defaults) across firms are correlated through dependence on one common factor. A straightforward corollary of the first assumption is that, for a given time horizon, the probability of default by any one single firm is given by

$$\omega_i \equiv \text{Prob}[\text{default by firm } i] = \text{Prob}[R_i \leq C_i] = N(C_i) \quad (19.7)$$

where $N(\cdot)$ is the cumulative distribution function of the standard normal distribution.

⁵ Strictly speaking, R_i could be more properly called a log-return given that we implicitly define it in discrete time as

$$R_{i,t} \equiv \log(A_{i,t}) - \log(A_{i,t-1})$$

where $A_{i,t}$ is the value of firm i at time t . In the remainder of this book, however, we will economize on terminology and simply call $R_{i,t}$ the return associated with firm i .

Readers of earlier chapters will notice that we have shifted from an asset-value-based definition of default—recall, for instance, that the Black-Scholes-Merton framework discussed in Chapter 17 modeled defaults as situations where the *value* of the firm fell below some critical level—to one that is based on (log) returns. These two approaches to specifying defaults are perfectly consistent with one another: In the Black-Scholes-Merton framework, the value, A_i , of the firm was assumed to be lognormally distributed, which has the implication that log-returns are normally distributed, as assumed here.

Here we pause to note that the probabilities involved in equation (19.7)—and, indeed, throughout this part of the book—correspond to the concept of risk-neutral probabilities, which we discussed in Part III. Thus, if they are available, one can use market prices of liquid assets, such as credit default swap premiums for certain reference names, as a means of arriving at a value for ω_i and then rely on (19.7) to back out a market-implied value for C_i :

$$C_i = N^{-1}(\omega_i) \quad (19.8)$$

where $N^{-1}(\cdot)$ is the inverse of $N(\cdot)$, or the function that determines the value of C_i in (19.7) that corresponds to a given value of ω_i .

Going back to the second assumption (correlation through dependence on a common factor), we complete the model by making the following assumption regarding the evolution of returns:

$$R_{i,t} = \beta_i \alpha_t + \sqrt{1 - \beta_i^2} \epsilon_{i,t} \quad (19.9)$$

for all firms in the portfolio, i.e., for $i = 1, 2, \dots, I$.

Equation (19.9) says that individual returns, and thus the likelihood of default, depend on a factor α , which affects all entities represented in the portfolio, and on an entity-specific factor ϵ_i . For instance, the entities represented in the portfolio may all be sensitive to conditions in the overall stock market, or, alternatively, α_t may stand for the current state of the economy.

To be consistent with the definition of $R_{i,t}$ as a random variable with zero mean and unit variance, we further assume that α_t and, for all i , $\epsilon_{i,t}$ have also been standardized—they have zero mean and unit variance—and that they are independently and normally distributed.⁶ We also assume that, for any two firms i and j , $\epsilon_{j,t}$ and $\epsilon_{i,t}$ are independently distributed and that α_t and $\epsilon_{i,t}$, and thus $R_{i,t}$ are serially uncorrelated, i.e.,

$$\tilde{E}[\alpha_t \alpha_{t-s}] = \tilde{E}[\epsilon_{i,t} \epsilon_{i,t-s}] = 0$$

for all nonzero values of s .⁷

⁶ It is easy to verify that (19.9) is consistent with the assumption of zero mean and unit variance of returns. $\tilde{E}[R_{i,t}] = 0$ follows directly from the zero-mean assumption on α and ϵ_i . To see that the variance of $R_{i,t}$ remains unit, simply note that we can write it as

$$\text{Var}[R_{i,t}] = \beta_i^2 \tilde{E}[\alpha_t^2] + (1 - \beta_i^2) \tilde{E}[\epsilon_{i,t}^2] + 2\beta_i \sqrt{1 - \beta_i^2} \tilde{E}[\alpha_t \epsilon_{i,t}]$$

which is equal to one given the variance and independence assumptions regarding α_t and $\epsilon_{i,t}$.

⁷ The concept of serial independence is discussed briefly in Appendix B.

In light of equation (19.9), we can show that β_i plays a crucial role in capturing the extent of return correlation among any two entities in the portfolio. In particular, the covariance between $R_{i,t}$ and $R_{j,t}$ can be written as

$$\text{Cov}(R_{i,t}, R_{j,t}) \equiv \tilde{E}[(R_{i,t} - \tilde{E}[R_{i,t}])(R_{j,t} - \tilde{E}[R_{j,t}])] = \beta_i \beta_j \quad (19.10)$$

which, given the assumption of unit variances, also corresponds to the correlation between $R_{i,t}$ and $R_{j,t}$.

From a modeling standpoint, one advantage of the above framework is that, for a given value of the common factor, α , returns involving any two firms in the portfolio, and thus their respective default events, are uncorrelated.⁸ To verify the conditional independence of defaults, we start by computing the conditional mean and variance of returns under the model:

$$\tilde{E}[R_{i,t}|\alpha_t] = \beta_i \alpha_t \quad (19.11)$$

$$\text{Var}[R_{i,t}|\alpha_t] = 1 - \beta_i^2 \quad (19.12)$$

and it is then straightforward to see that returns are indeed pairwise conditionally uncorrelated:

$$\text{Cov}[R_{i,t}, R_{j,t}|\alpha_t] = \sqrt{1 - \beta_i^2} \sqrt{1 - \beta_j^2} \tilde{E}[\epsilon_{i,t} \epsilon_{j,t}] = 0 \quad (19.13)$$

where $\tilde{E}[R_{i,t}|\alpha_t]$ is the expected value of $R_{i,t}$ conditioned on the time- t value of α , and $\text{Var}[\cdot|\alpha_t]$ and $\text{Cov}[\cdot|\alpha_t]$ are analogously defined. Note that (19.13) follows from the fact that $\epsilon_{i,t}$ and $\epsilon_{j,t}$ are mutually independently distributed, i.e., $\tilde{E}[\epsilon_{i,t} \epsilon_{j,t}] = 0$.

We can also compute conditional default probabilities over given horizons for any firm in the portfolio. For instance, for $\alpha_T = \alpha_t$, the time- t probability that the firm will default at time T can be written as

$$\omega_i(\alpha_t) \equiv \text{Prob}[\text{default by firm } i \text{ at } T|\alpha_t] = \text{Prob}\left[\epsilon_{i,T} \leq \frac{C_i - \beta_i \alpha_t}{\sqrt{1 - \beta_i^2}}\right] \quad (19.14)$$

which, unlike the expression for unconditional default probabilities, equation (19.7), depends importantly on β_i .

⁸Throughout this chapter we assume that C_i is a deterministic variable for all i .

19.1.3 Pairwise Default Correlation and “ β ”

We have shown that the model-implied correlation between returns on any two assets in the portfolio is:

$$\text{Cor}(R_{i,t}, R_{j,t}) = \text{Cov}(R_{i,t}, R_{j,t}) = \beta_i \beta_j \quad (19.15)$$

How does that relate to the concept of pairwise default correlation discussed in Section 19.1.1? There we saw that we can write the default correlation between any two entities i and j in the portfolio as

$$\rho_{i,j} = \frac{\omega_{i\&j} - \omega_i \omega_j}{\sqrt{\omega_i(1 - \omega_i)} \sqrt{\omega_j(1 - \omega_j)}} \quad (19.16)$$

where we have already seen that the model implies that the probabilities ω_i and ω_j can be written as $N(C_i)$ and $N(C_j)$, respectively. Now, $\omega_{i\&j}$ denotes the probability that both i and j will default over the time horizon of interest. In the context of the model,

$$\omega_{i\&j} \equiv \text{Prob}[R_i \leq C_i \text{ and } R_j \leq C_j] \quad (19.17)$$

but, given that R_i and R_j are individually normally distributed with correlation coefficient $\beta_i \beta_j$, they are jointly distributed according to the bivariate normal distribution. Thus,

$$\omega_{i\&j} = N_2(C_i, C_j, \beta_i \beta_j) \quad (19.18)$$

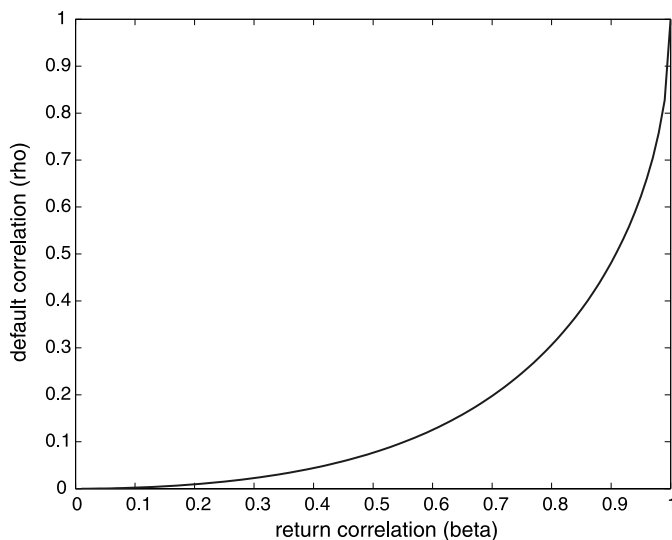
where $N_2()$ is the cumulative distribution function of the bivariate normal distribution.⁹

We can now write out the default correlation between any two entities in the portfolio entirely in terms of model parameters:

$$\rho_{i,j} = \frac{N_2(C_i, C_j, \beta_i \beta_j) - N(C_i)N(C_j)}{\sqrt{N(C_i)(1 - N(C_i))} \sqrt{N(C_j)(1 - N(C_j))}} \quad (19.19)$$

Equation (19.19) shows the explicit link between the return correlation parameters, β_i and β_j , and the degree of default correlation between reference entities i and j . Figure 19.3 illustrates the nature of this link for a particular parameterization of default probabilities— $\omega_i = \omega_j = .05$ —and return correlations— $\beta_i = \beta_j = \beta$, and β varies from 0 to 1. The figure shows that default correlation increases monotonically with return correlation, but the relationship is very nonlinear. Given this one-to-one mapping

⁹ See Appendix B.

FIGURE 19.3. Return Correlation (β) and Pairwise Default Correlation

between β and $\rho_{i,j}$, we shall couch the discussion of default correlation in this part of the book mostly in terms of β . We do this simply for analytical convenience.

19.2 The Loss Distribution Function

We first met the concept of the loss distribution function in this book in Chapter 10, where we discussed portfolio default swaps. To recap, an informal definition of the loss distribution function would say that, for a given portfolio, it is the function that assigns probabilities to default-related losses of various magnitudes over a given time horizon.

We will continue to build on the simple modeling framework introduced in the previous section to take a closer look at the loss distribution function and its relation to default correlation. For convenience, however, we will make a few additional simplifying assumptions. First, we assume that the portfolio is composed of a set of homogeneous debt instruments, i.e., for all i ,

$$\beta_i = \beta$$

$$C_i = C$$

Second, we assume that each entity represented in the portfolio corresponds to an equal share of the portfolio. Henceforth, we will refer to such a portfolio as an equally weighted homogeneous portfolio.

The first assumption ensures that all entities represented in the portfolio have the same default probability over the time period of interest; the second guarantees that there is a one-to-one correspondence between the number of defaults in the portfolio and the size of the percentage default-related loss in the portfolio. In particular, in a portfolio with, say, I reference entities, the probability of k defaults among the entities in the portfolio is equivalent to the probability of a $100\frac{k}{I}$ percent default-related loss in the portfolio.¹⁰

19.2.1 Conditional Loss Distribution Function

Armed with the tools developed thus far we can compute the probability distribution of default-related losses for a given value of α , which corresponds to the concept of the conditional loss distribution function. Given that, conditional on α , individual returns are independently and normally distributed, a basic result from statistics says that the number of defaults in the portfolio is binomially distributed for a given value of α .¹¹ Thus, if we let L denote the percentage (default-related) loss in the portfolio over, say, the next year, we can write the conditional probability of a given loss as

$$\text{Prob} \left[L = \frac{k}{I} | \alpha \right] \equiv \text{Prob} [k \text{ defaults} | \alpha] = \frac{I!}{k!(I-k)!} \omega(\alpha)^k [1 - \omega(\alpha)]^{I-k} \quad (19.20)$$

where

$$\omega(\alpha) \equiv \text{Prob} \left[\epsilon_{i,t} \leq \frac{C - \beta\alpha}{\sqrt{1 - \beta^2}} \right] = N \left(\frac{C - \beta\alpha}{\sqrt{1 - \beta^2}} \right) \quad (19.21)$$

is the conditional probability of default of each reference entity represented in the equally weighted homogeneous portfolio.

Equation (19.20) is essentially the expression for the conditional loss distribution function implied by (19.9), applied to the case of an equally-weighted homogeneous portfolio. To highlight its dependence on the common factor, α , Figure 19.4 shows conditional loss distribution functions corresponding to a period of, say, one year for a portfolio with 20

¹⁰ For added convenience, we are assuming a zero recovery rate for all entities in the portfolio.

¹¹ See Appendix B for an overview of the binomial distribution.

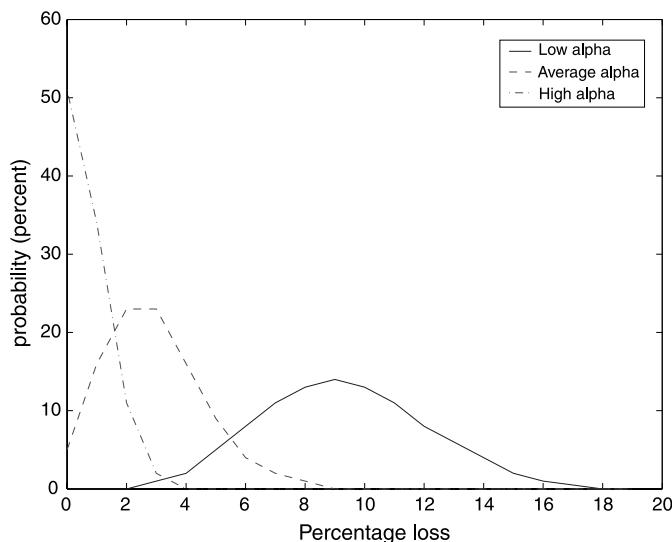


FIGURE 19.4. Conditional Loss Distribution Functions for an Equally Weighted Homogeneous Portfolio

reference entities, each with an individual default probability of 5 percent and a β of 0.5. The figure depicts three cases, one with α set at its average value of zero, and the others corresponding to α set at plus and minus one standard deviation, -1 and 1 , respectively.¹²

As can be seen in Figure 19.4, the conditional loss distribution function flattens out as α declines. This is consistent with the intuition that, with positive dependence of returns on α , conditional default probabilities increase for all entities in the portfolio as α decreases. Lower values of α in this case pull individual returns closer to their default thresholds, C , increasing the likelihood of larger losses in the portfolio and thus allowing for “fatter tails” in the conditional loss distribution function.

19.2.2 Unconditional Loss Distribution Function

When pricing multi-name credit derivatives such as basket swaps and portfolio default swaps it is often the unconditional loss distribution function that will matter most. As its name suggests, this concept is not predicated on any one particular value of the common factor, α , and thus it fully

¹² Figure 19.4 illustrates an additional use of the modeling framework examined in this chapter, which is to assess likely losses in the portfolio under different scenarios involving the common factor, a practice commonly called scenario stress testing.

takes on board the reality that future values of α are themselves generally subject to substantial uncertainty.

In what follows we shall use the model to derive an analytical solution for the unconditional loss distribution function of a homogeneous portfolio. The technical requirements for this derivation are perhaps a bit beyond the scope of this book, but we should note that readers less interested in statistical and mathematical details may skip this subsection entirely without any fear of missing out on what is to come in the remainder of Part IV. Indeed, the main reason for obtaining an expression for the loss distribution here—equation (19.24) below—is so we can use it later on to check the accuracy of the large-portfolio approximation and simulation-based methods described in sections 19.2.3 and 19.4, respectively. It is the latter method, in particular, that will be our tool of choice throughout most of this part of the book, not just for deriving loss distribution functions, but also for valuing multi-name credit derivatives.

To derive an expression for the unconditional loss distribution, we appeal to a basic result from statistics, known as the law of iterated expectations. According to this “law,” the unconditional expectation of having, say, k defaults in the portfolio is given by the probability-weighted average of conditional probabilities of having k defaults. These latter probabilities are computed over all possible values of the common factor α , and the weights are given by the probability density function of α . Thus,

$$\text{Prob}[k \text{ defaults}] = \int_{-\infty}^{\infty} \text{Prob}[k \text{ defaults} | \alpha = y] n(y) dy \quad (19.22)$$

where $n(\cdot)$ is the probability density function of the standard normal distribution.¹³

Substituting equations (19.20) and (19.21) into (19.22) we obtain the following expression for the unconditional probability of k defaults in the portfolio:

$$\text{Prob}[k \text{ defaults}] = \int_{-\infty}^{\infty} \frac{I!}{k!(I-k)!} \omega(y)^k (1 - \omega(y))^{I-k} n(y) dy \quad (19.23)$$

where $\omega(y) \equiv N\left(\frac{C - \beta y}{\sqrt{1 - \beta^2}}\right)$, as we saw earlier.

¹³ See Appendix B for a quick review of key results from statistics.

Given equation (19.23), the analytical solution for the loss distribution function of a homogeneous portfolio is:

$$\text{Prob}[l \leq K] = \sum_{k=0}^K \int_{-\infty}^{\infty} \frac{I!}{k!(I-k)!} \omega(y)^k (1 - \omega(y))^{I-k} n(y) dy \quad (19.24)$$

where $\text{Prob}[l \leq K]$ denotes the probability that the number of defaults in the portfolio will be equal to or less than K . Equation (19.24) is a somewhat cumbersome expression that can be solved, for instance, via numerical integration, provided I is not too large.

19.2.3 Large-Portfolio Approximation

As we saw in the previous section, allowing for both the common and the firm-specific factors to be fully stochastic makes the mathematical analysis of credit risk in portfolios substantially more involving than in the case of fixed values for the common factor, especially for portfolios with a large number of assets. One technique that greatly simplifies the analysis of relatively homogeneous portfolios with many assets is the so-called large-portfolio approximation method—see, for instance, Vasicek (1987)[73].¹⁴ As the name suggests, the main thrust of this approach is to assume that the portfolio has a sufficiently large number of reference entities so that the expected fraction of entities defaulting over a given time horizon can be approximated by the corresponding individual default probabilities of the entities.¹⁵ In terms of the conditional default probabilities derived in Section 19.2.1, this implies

$$\theta \equiv \tilde{E}[L|\alpha] \approx \omega(\alpha) \quad (19.25)$$

where θ and L are the expected and actual percentage loss in the portfolio, respectively.

Note that for any given value of θ , one can back out the implied value of α upon which the conditional expectation in (19.25) is based:

$$\alpha = \omega^{-1}(\theta) \quad (19.26)$$

where $\omega^{-1}(\cdot)$ is the inverse function of $\omega(\alpha)$.¹⁶

¹⁴ Our presentation of the large-portfolio approximation method partly follows O’Kane and Schlogl (2001)[64].

¹⁵ Readers with some familiarity with statistics will recognize the (conditional) law of large numbers at work here.

¹⁶ We will ignore the approximation error embedded in (19.25) from now on. Obviously, that error can be significant, especially for small portfolios, which, as we will see in Section 19.4, can be examined with alternative methods.

Suppose now that the actual value of α turns out to be larger than the one used in (19.25). Other things being equal, the actual percentage loss, L , will be smaller than the expected loss, θ , because, for positive β , individual returns will be farther away from the default boundary C than implicit in (19.25). Mathematically, this can be summed up as:

$$\alpha \geq \omega^{-1}(\theta) \Leftrightarrow L \leq \theta \quad (19.27)$$

Thus, we can make the following probabilistic statement:

$$\text{Prob}[\alpha \geq \omega^{-1}(\theta)] = \text{Prob}[L \leq \theta] \quad (19.28)$$

Given that α was assumed to be normally distributed, and relying on the symmetric nature of the normal probability density function,¹⁷

$$\text{Prob}[\alpha \geq \omega^{-1}(\theta)] = \text{Prob}[\alpha \leq -\omega^{-1}(\theta)] = N(-\omega^{-1}(\theta))$$

Thus we arrive at the result:

$$\text{Prob}[L \leq \theta] = N(-\omega^{-1}(\theta)) \quad (19.29)$$

which is an approximate expression for the unconditional loss distribution of the large homogeneous portfolio.

To write out (19.29) explicitly in terms of the parameters of the model, note that, ignoring any errors introduced by the large portfolio approximation, equations (19.21) and (19.25) imply

$$\theta = N \left(\frac{C - \beta\alpha}{\sqrt{1 - \beta^2}} \right) \quad (19.30)$$

and thus we can write

$$\alpha = \frac{C - N^{-1}(\theta)\sqrt{1 - \beta^2}}{\beta} \quad (19.31)$$

which, together with (19.26), implies

$$\omega^{-1}(\theta) = \frac{C - N^{-1}(\theta)\sqrt{1 - \beta^2}}{\beta} \quad (19.32)$$

¹⁷ By symmetry of the normal distribution we mean the fact that, for any normally distributed random variable α that has an expected value of zero,

$$\text{Prob}[\alpha \geq U] = \text{Prob}[\alpha \leq -U]$$

where U is any arbitrary real number.

and we arrive at our final expression for the (approximate) unconditional loss distribution of a large homogeneous portfolio:

$$\text{Prob}[L \leq \theta] = N\left(-\frac{C - N^{-1}(\theta)\sqrt{1-\beta^2}}{\beta}\right) \quad (19.33)$$

19.3 Default Correlation and Loss Distribution

We are now ready to start tying together some of the different topics discussed in this chapter by examining how the probability distribution of future losses is affected by the degree of default correlation among the issuers in the portfolio. This analysis also expands on an early exercise on correlation and loss distribution, discussed in Chapter 10.

Our goal is to examine the crucial role that default correlation plays in the determination of a portfolio's loss distribution function and, as we shall see in greater detail in Chapters 20 and 21, in the valuation of multi-name credit derivatives. To highlight that role, we shall examine two large homogeneous portfolios—which we shall call portfolios *A* and *B*—that are identical in every respect, except for their extent of return, and, thus, default correlation. In particular, for a given horizon, the individual default probabilities of the entities represented in either portfolio are assumed to be 5 percent, which, following the spirit of equation (19.25), means that both portfolios have expected default-related losses of 5 percent. (For simplicity, we continue to assume zero recovery rates for all entities.) As for their degree of correlation with the common factor, we shall assume that portfolio *A* has a β of 0.2, and portfolio *B*'s β is 0.5.

Figure 19.5 shows the unconditional loss distribution functions for the two portfolios described above. The loss distributions are quite different despite the fact that both portfolios have the same expected loss. In particular, the loss distribution of portfolio *A* (low β) shows a virtually zero probability of default-related losses amounting to more than 15 percent of the portfolio. In contrast, for portfolio *B* (high β), that probability is distinctively positive, although still relatively small. Portfolio *B* also has a higher probability of very small losses than does portfolio *A*, reflecting the fact that its higher correlation results in a greater tendency for its reference entities to either survive or default together. In statistical parlance, the portfolio with the greater default correlation has a loss distribution with “fatter tails,” i.e., the loss distribution of portfolio *B* assigns greater odds to extreme events than does that of portfolio *A*.

From a practical perspective, the basic insights derived from the analysis of portfolios *A* and *B* are very powerful. First, the degree of default correlation in a portfolio can dramatically affect its risk characteristics. For instance, while holding a portfolio that includes a large number of issuers

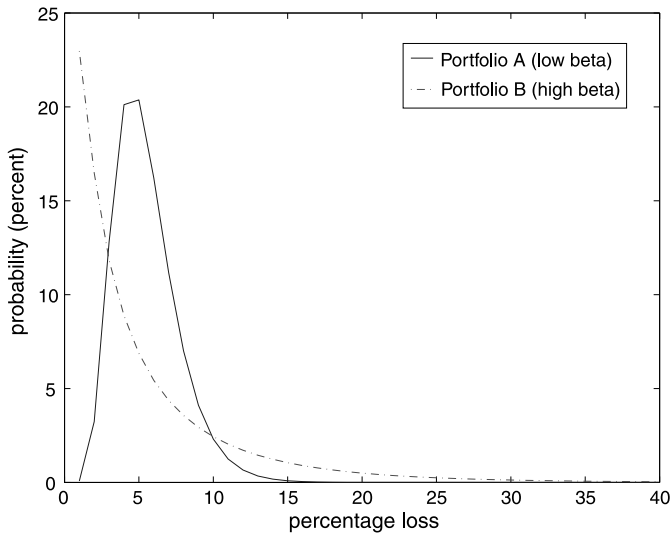


FIGURE 19.5. Unconditional Loss Distribution Functions for Two Large Homogeneous Portfolios

generally contributes to diversification across different types of risks, that diversification may be significantly reduced if the degree of default correlation among the issuers is high. Indeed, higher odds of extreme events can be thought of as reduced diversification. For instance, as we discussed in Chapter 10, in the extreme case of perfect correlation either all of the issuers in the portfolio survive or default together, which is akin to holding only one asset (no diversification).

A second basic insight from Figure 19.5 regards the valuation of credit derivatives that involve some tranching of credit risk, such as baskets and portfolio default swaps, which we will examine more fully in the next two chapters. As we saw in Chapters 9 and 10, the valuation of these derivatives can be importantly affected by the probability distribution of losses in the underlying portfolio, which, as we have just seen, depends crucially on the degree of default correlation among the entities represented in the portfolio.

19.4 Monte Carlo Simulation: Brief Overview

As an alternative to the large portfolio approximation, one can use Monte Carlo simulation methods to compute the (approximate) loss distribution

function of a portfolio. In their simplest form, described below, simulation methods place less mathematical demands on the user, at the cost of longer computer running times than the version of the large portfolio method discussed in Section 19.2.¹⁸ An advantage of simulation-based methods is that they are best applied to smaller portfolios, for which the large portfolio approximation method is less suitable. In addition, they can be easily used in the analysis of heterogeneous portfolios and in versions of the model that allow for more than one common factor.

We can use equations (19.6) and (19.9)—repeated below for convenience—to illustrate the basic principle of the Monte Carlo simulation approach:

$$\text{Default by firm } i \iff R_{i,t} \leq C_i$$

$$R_{i,t} = \beta_i \alpha_t + \left(\sqrt{1 - \beta_i^2} \right) \epsilon_{i,t}$$

where, as assumed before, α and ϵ_i are mutually independent random variables that are normally distributed with zero mean and unit variance.

The basic thrust of simulation-based methods is very straightforward. It consists of generating a large number of draws from the standard normal distribution for α and, for all i in the portfolio, for ϵ_i . For instance, for a portfolio with 20 reference entities, each draw will consist of 21 values randomly selected according to the standard normal distribution, and, for each draw, one compares the resulting return R_i for each entity to its default boundary, C_i , to determine whether or not a default has occurred. After a sufficiently large number of draws, one can count the number of defaults and estimate the probability distribution of losses as

$$\text{Prob}[j \text{ defaults}] = M_j/M$$

where M_j is the number of draws where there were j defaults in the portfolio, and M is the total number of draws.¹⁹

We illustrate the simple simulation-based method described above for an equally weighted homogeneous portfolio with 40 reference entities, each with a one-year individual default probability of 5 percent and a degree of correlation (β) of 0.2. The solid line in Figure 19.6 shows the corresponding loss distribution function for this portfolio, based on 500,000 draws from the standard normal distribution. To keep the analysis comparable

¹⁸ Hull (2003)[41] provides an overview of simulation techniques in finance.

¹⁹ If one is interested in the distribution of percentage losses in the portfolio, one can modify these calculations accordingly. For instance, M_j could stand for the number of draws where percentage losses in the portfolio amounted to j percent.

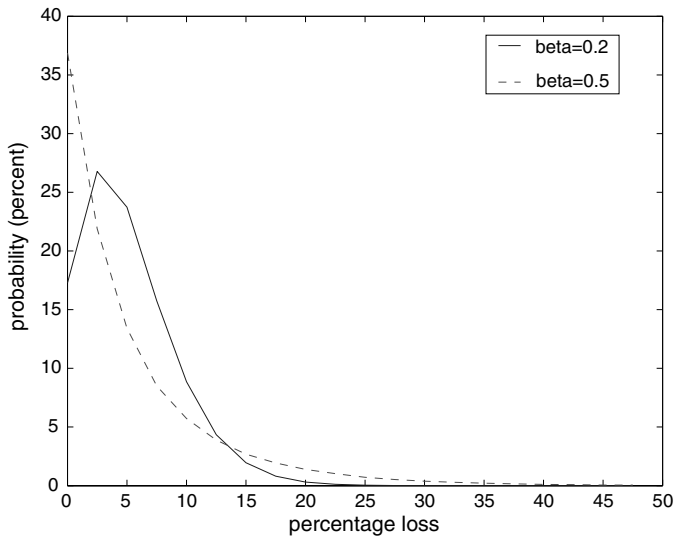


FIGURE 19.6. Unconditional Loss Distribution Functions for Smaller Portfolios (based on 500,000 simulations of the model)

to that of large portfolios, we also computed the probability loss distribution for an otherwise identical portfolio with a β of 0.5. We arrive at some of the same conclusions discussed in Section 19.3. Portfolios with a larger degree of default correlation have more probability mass at tail events than do portfolios with less correlation. Again, we see the benefits of effective diversification (less default correlation) at work.

19.4.1 *How Accurate is the Simulation-Based Method?*

We have mentioned some desirable features of the simulation-based method—such as ease of use and applicability to both small and heterogeneous portfolios, as well as to multifactor models—and a potential drawback—Monte Carlo simulations tend to be computer intensive. Simulation-based methods would be of limited value, however, if the resulting loss distribution function were a poor representation of the true function implied by the model—equation (19.24)—or if it took an unreasonably large number of simulations—and thus substantial computing time—for the simulation method to get it right.

Figure 19.7 takes an informal look at the accuracy of the Monte Carlo simulation approach for two portfolios that are identical in every respect

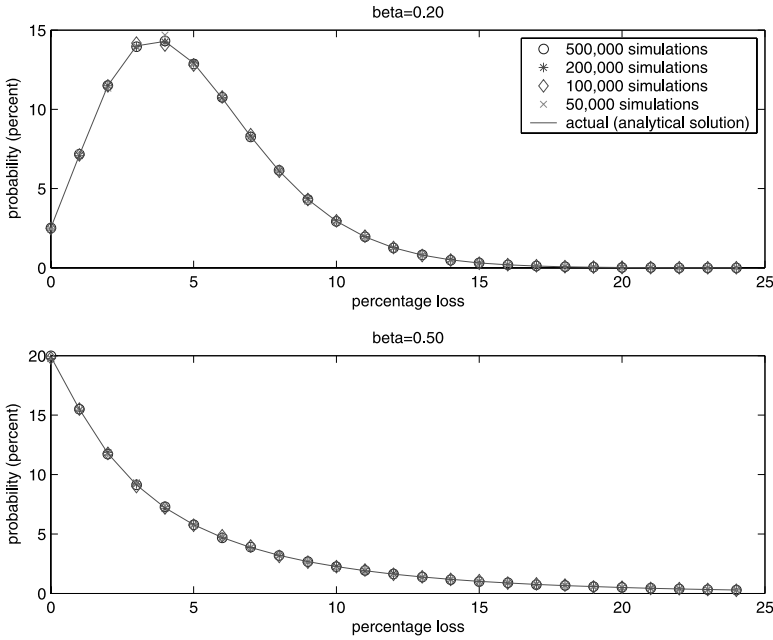


FIGURE 19.7. How Accurate is the Simulation-based Method? (Homogeneous equally weighted portfolios with 100 assets; total face value = \$100,000)

except for their degree of asset correlation. In particular, each portfolio is composed of 100 assets with individual default probabilities of 5 percent and zero recovery rates. For each portfolio, the figure shows “actual” (analytical) values of the loss distribution, as well as a set of Monte-Carlo-based values. The results suggest that, even for computations involving a number of simulations as low as 50,000—which take only a few seconds to run in a well equipped laptop—the Monte Carlo method seems to do a very good job capturing both the level and the shape of the true loss distribution function.

The results in Figure 19.7 are only illustrative, however. A more formal evaluation of the Monte Carlo approach would involve examining the results of a large number of simulation exercises—for instance, running 100,000 simulations of the model 1,000 times and computing the mean and standard deviation of all 1,000 results—so that one could look at the variability of the final results. It should also be noted, that the simple simulation method described in this chapter can be improved considerably through the use of

“variance-reduction” techniques, which are designed to reduce the amount of random noise that is inherent in simulation-based methods.²⁰

19.4.2 Evaluating the Large-Portfolio Method

We can now compare results obtained through the large-portfolio approximation method to those generated by the simulation-based approach, which, as we have just seen, can be made very accurate. Similar to the spirit of the last subsection, one can view this exercise as a very informal evaluation of the large-portfolio approximation approach’s ability to capture the main features of the loss distribution function of progressively smaller portfolios under different correlation assumptions.

Figure 19.8 summarizes the main findings. We consider several hypothetical portfolios designed along the same lines as the ones examined thus far in this chapter. The top panel shows the unconditional loss distribution

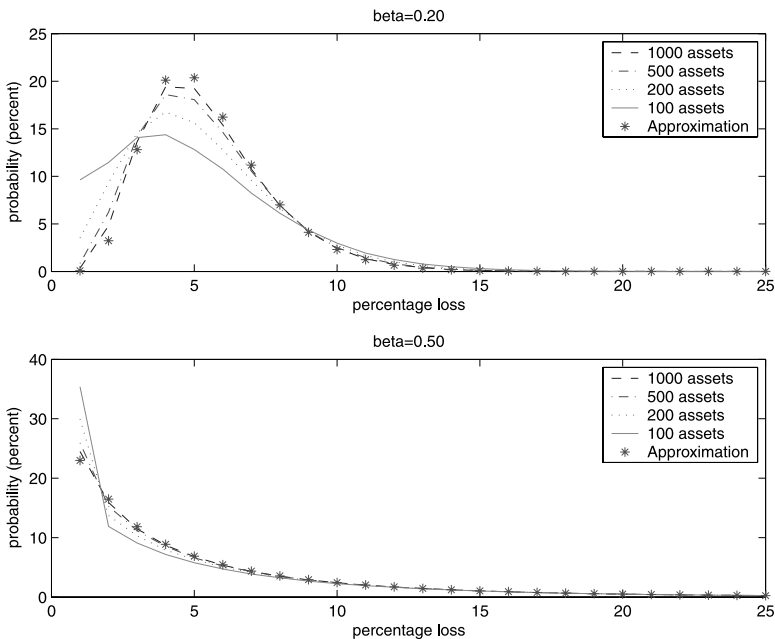


FIGURE 19.8. An Informal Evaluation of the Large-Portfolio Approximation Method

²⁰ See, for instance, Hull (2003)[41] for an overview of variance reduction techniques.

functions for four equally weighted homogeneous portfolios that are identical in every aspect, except for the number of reference entities included in the portfolio, which varies from 100 to 1,000. For all portfolios, the default probabilities of the individual entities are assumed to be 5 percent and $\beta = 0.2$. The Monte-Carlo-based results were derived based on 500,000 simulations of the model. The figure also shows the loss distribution function based on the large-portfolio approximation. Not surprisingly, the approximation works best for the largest portfolio, but it is noteworthy that even for the 200-asset portfolio, the large-portfolio approximation seems to have been able to capture the general shape and level of the loss distribution.

The lower panel of Figure 19.8 depicts results for the same cases examined in the upper panel, but we now examine scenarios with a bit more return correlation among the entities in the portfolios, $\beta = 0.5$. The results suggest that the approximation works relatively well, especially as we move away from the left end of the distribution. On the whole, the figure suggests that the large-portfolio approximation seems to work “better” for portfolios with higher correlation than for those with lower correlation. While we are being vague here about what better means in this context, our intuition would suggest that in the limit case of perfect correlation, the portfolio essentially behaves as a single asset and the loss distributions of small and large portfolios should essentially become the same.

Once again, we should emphasize that this section provides only an informal evaluation of the large-portfolio approximation approach. Our purpose here was only to provide some of the flavor of the practical and methodological issues that one is likely to face when dealing with real-world portfolios and more advanced techniques. Issues not addressed here include the fact that real-life portfolios are not strictly homogeneous, which would add another source of approximation error to the large portfolio approach.

One final point about Figure 19.8. The loss distribution functions of the various portfolios shown is remarkably similar, especially for the portfolios with more than 100 reference names, although one can see a pattern of slightly less dispersion of likely losses—more of the probability mass is in the center of the distribution—for the larger portfolios. Yet, from an investment and portfolio management perspective, it may be substantially more costly to monitor, say, a portfolio with more than 500 names, as opposed to one with 200. (Not to mention that, in most markets, one may be hard pressed to find 500+ names that fit the profile sought by the investor/manager.) One small simple lesson learned from Figure 19.8 then is that the diversification benefits, in terms of less disperse loss distributions, achieved by adding more and more names to the portfolio eventually are likely to be outweighed by the potentially higher costs of constructing and managing portfolios in which an ever larger number of entities is represented.

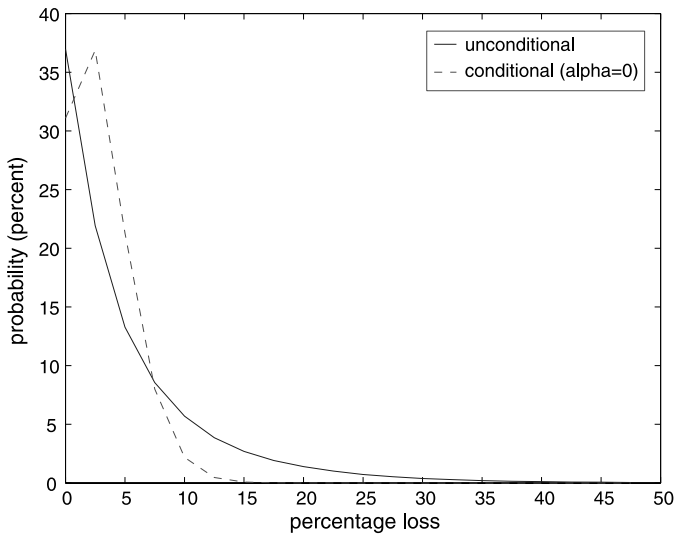


FIGURE 19.9. Conditional vs. Unconditional Loss Distribution Functions

19.5 Conditional vs. Unconditional Loss Distributions

Suppose one has a view as to where the common factor will be by a certain date and then computes the loss distribution function of a given portfolio based on that view. The resulting function is none other than the conditional loss distribution function that we discussed earlier in this chapter. Suppose, in particular, that one makes the seemingly reasonable forecast that α will be at its expected value of zero at the end of the relevant time horizon and then computes the corresponding (conditional) loss distribution of the portfolio.

How much off would one be by making risk assessments based on particular views regarding the common factor? Figure 19.9 answers this question for the portfolio described in the previous section, with β set to 0.5. It shows that conditioning the distribution of losses on the expected value of α would lead one to underestimate significantly the default risk in the portfolio, exposing the danger of what would otherwise look like a sensible assumption regarding the evolution of the common factor. More generally, ignoring the uncertainty related with future values of the common factor can lead one to understate substantially the likelihood of large losses in the portfolio.

19.6 Extensions and Alternative Approaches

We discussed a simple approach for modeling the credit risk in a portfolio and described three basic methods for deriving a portfolio's loss distribution function. The first method involved the analytical derivation of the loss distribution function. The second technique was the large-portfolio approximation motivated by the work of Vasicek (1987)[73] and others. The third was a simple simulation-based method that can also be used for small portfolios. We also highlighted the key role that default correlation plays in determining the range of likely losses that can be experienced in a portfolio.

There are several directions in which the basic model discussed in this chapter can be extended. For instance, portfolios with heterogeneous assets can be examined in a relatively straightforward manner with the help of simulation-based methods, a topic that we examine further in Chapter 20. Other extensions include allowing for defaults to take place at any time—rather than only at the maturity date of the contract—the treatment of accrued premiums, and the analysis of multi-period contracts—an issue that we address in Chapter 23.

Additional modeling approaches for examining portfolio credit risk fall along the lines of the more advanced versions of the single-default models that we studied in Part III of the book. For instance, they include both intensity- and ratings-based models. We briefly discuss examples of such models in Chapter 22.

20

Valuing Basket Default Swaps

We now start applying some of the techniques introduced in Chapter 19 to the valuation of basket default swaps, the mechanics of which we discussed in some detail in Chapter 9. We start by reviewing some of the basic ideas addressed in that chapter, which mostly centered on two-entity baskets, and then move on to the valuation of more realistic cases involving baskets with several reference entities.

20.1 Basic Features of Basket Swaps

As we saw in Chapter 9, a basket default swap is a credit derivative contract where the protection seller agrees to take on some of the credit risk in a basket (portfolio) of reference entities. For instance, in the first-to-default (FTD) basket the protection seller commits to making a payment to the protection buyer to cover the first default among the entities referenced by the basket. As with a standard credit default swap, the default-contingent payment made by the protection seller is typically equal to the face value of the defaulted asset minus its recovery value. In exchange for this kind of protection, the protection buyer makes periodic payments to the seller—the FTD premium—until the maturity date of the contract.

After payments associated with the first default in the FTD basket are settled, both parties are relieved of further obligations under the contract. Thus, assuming the protection buyer bought no additional default protection on the basket, any losses related to further defaults among the

remaining reference entities in the basket will be borne out entirely by the protection buyer. Similar to other credit derivative contracts, if no default takes place during the life of the contract, the protection seller keeps on collecting the FTD premium from the buyer until the expiration of the contract.

Second- and third-to-default baskets are similarly defined. We will take a quick look at the second-to-default basket towards the end of this chapter.

20.2 Reexamining the Two-Asset FTD Basket

Continuing with a brief review of Chapter 9, we reexamine the valuation principles surrounding the simplest basket swap, one that references only two debt issuers, but we now start to incorporate some of the results discussed in Chapter 19.

For a given time horizon, say one year, let ω_A and ω_X denote the risk-neutral default probabilities associated with XYZ Corp. and AZZ Bank, the two reference entities included in the basket. We further assume that the default correlation between the two entities is $\rho_{A,X}$, which, as we saw in Chapter 19—equation (19.5)—can be written as

$$\rho_{A,X} = \frac{\omega_{A\&X} - \omega_A\omega_X}{\sqrt{\omega_A(1 - \omega_A)}\sqrt{\omega_X(1 - \omega_X)}} \quad (20.1)$$

where $\omega_{A\&X}$ is the probability that both XYZ and AZZ will default within a specific horizon.

Let us consider a one-year FTD basket involving only one premium payment date, at the end of the contract. The probability that the protection seller will have to make a payment under the contract is simply equal to the probability $\omega_{A \text{ or } X}$ of at least one default in one year's time. From Chapter 9, we know:

$$\omega_{A \text{ or } X} = \omega_A + \omega_X - \omega_{A\&X} \quad (20.2)$$

For simplicity, we shall assume that both reference entities have a zero recovery rate and that defaults, if any, will only take place in one-year's time. We further assume that the total notional amount of the basket is \$10 million, with each reference entity accounting for exactly half of the total. Thus, the expected value of the payment made by the protection seller, denoted below as $\tilde{E}[\text{protection payment}]$, is

$$\tilde{E}[\text{protection payment}] = \omega_{A \text{ or } X} \times \$5 \text{ million} + (1 - \omega_{A \text{ or } X}) \times 0 \quad (20.3)$$

i.e., with probability $\omega_{A \text{ or } X}$ the protection seller will have to pay for one default—\$5 million given our assumptions—and with probability $1 - \omega_{A \text{ or } X}$

there will be no default in the basket, in which case the protection seller will not have to make any payment.

Given that the contract typically has zero market value at its inception (no money changes hands at the inception of the basket swap), and, again, assuming a single premium payment date, at the end of the contract, it must be the case that the premium paid by the protection buyer is equal to the expected payment made by the protection seller.¹ Thus, the value of the FTD premium payment is:

$$S_{basket} = \omega_{A \text{ or } X} \times \$5 \text{ million} \quad (20.4)$$

which is customarily quoted in terms of basis points (see Section 6.1). For instance, if $\omega_{A \text{ or } X}$ is equal to .01—a 1 percent probability that either one of the reference entities will default in one-year's time—then the premium for this one-year FTD basket would be 100 basis points, and the total premium paid by the protection buyer would be \$50,000.

A few additional points are worth remembering about the two-asset FTD basket. First, as we have just seen, the key ingredient in the determination of the FTD premium is the probability that any one of the reference entities in the basket will default during the life of the contract. Second, that probability depends importantly on the default correlation of the two entities. Indeed, as we saw in Chapters 9 and 19 (Section 19.1.1), as we approach the polar cases of (i) perfect correlation, $\rho_{A,X} = 1$, and (ii) no correlation, $\rho_{A,X} = 0$, coupled with a sufficiently low value for the probability that both reference entities will default while the contract is in force, the FTD basket premium tends to the default probability of the weaker asset and to the sum of the default probabilities of each asset, respectively.²

20.3 FTD Basket with Several Reference Entities

In practice, basket swaps are generally written on more than two reference entities. The basic results derived from the two-asset case do carry over to baskets with several entities, but the valuation exercise does become a bit more involved.

20.3.1 A Simple Numerical Example

We start with a simple example that can be used to fix some important points regarding more realistic cases involving baskets with more than two

¹ Unless otherwise indicated, throughout most of Part IV, we shall assume that future premiums are discounted at the same rate as expected future protection payments.

² The reader is invited to verify this using the diagrams discussed in Section 19.1.

TABLE 20.1
Basic Valuation of Five-Asset Basket Swaps^a
(Total notional = \$50 million; individual recovery rates = 0)

Possible outcomes (No. of defaults)	Outcome probabilities (percent)	Payment by FTD protection seller ^b (\$ millions)	Payment by STD protection seller ^b (\$ millions)
(1)	(2)	(3)	(4)
0	90.0	0	0
1	6.0	10	0
2	3.0	10	10
3	0.5	10	10
4	0.3	10	10
5	0.2	10	10

^aEach reference entity represents an equal portion of the portfolio (\$10 million).

^bValues shown in columns 3 and 4 are the total payments made by FTD and STD protection sellers under each possible default outcome.

reference entities. Table 20.1 shows all possible default outcomes for a one-year basket swap that references five entities, along with the risk-neutral probabilities, shown in column 2, associated with each outcome.³ There are six such possible outcomes, ranging from no defaults to defaults by all entities referenced by the basket. For instance, there is a 90 percent probability that none of the entities referenced by the basket will default during the life of the basket and a 6 percent probability that only one of the entities will default.

We now consider a reference portfolio with the characteristics featured in Table 20.1. The total face value of the portfolio is \$50 million, and each of the reference entities accounts for an equal piece, \$10 million, of the portfolio. We continue to assume a zero recovery rate for all the entities referenced by the basket.

The third column in Table 20.1 shows, for each possible default outcome, the payment made by a protection seller in a first-to-default basket swap written on the portfolio. Naturally, if there are no defaults, the FTD protection seller owes nothing to the protection buyer, but if at least one of the reference entities defaults, she is liable for the entire loss associated with that asset, \$10 million in this example. Note that, provided there is at least one default, the total payment made by the FTD

³Note that columns 1 and 2 essentially correspond to the loss distribution function of the portfolio.

protection seller is always \$10 million as she is only liable to cover the first default.

What would be the FTD premium for this five-asset FTD basket? Following the same logic outlined in the valuation of the two-asset basket, if, as customary, the contract has zero market value at its inception, then the FTD premium owed by the protection buyer to the protection seller should be equal to the (risk-neutral) expected value of the payment made by the protection seller. Armed with the probabilities shown in Table 20.1, this expected value is \$1 million:

$$\text{FTD premium} = .9 \times 0 + .1 \times \$10 \text{ million} = \$1 \text{ million}$$

i.e., with probability 90 percent there is no default among the entities referenced by the basket, and the protection seller makes no payment under the contract; but with probability 10 percent there is at least one default, in which case the protection seller owes the buyer the \$10 million loss associated with the first default. Thus the premium is \$1 million for \$10 million worth of protection, which amounts to 1,000 basis points.

The last column of Table 20.1 illustrates possible payments made by a protection seller in a second-to-default (STD) basket swap written on the portfolio. By definition, such a seller would owe nothing under the terms of the STD agreement if less than two defaults take place, but would be liable for \$10 in the event of a second default, and nothing more for any additional defaults. Proceeding as with the FTD basket, one would find that the STD premium amounts to 400 basis points, considerably less than that of the FTD basket, given a much smaller probability of at least two defaults than that associated with one default.

20.3.2 A More Realistic Valuation Exercise

Table 20.1 was an illustration used to provide some intuition about the valuation of basket default swaps. In real-world situations, however, one will not be presented with the risk-neutral probabilities associated with every possible default outcome regarding the entities referenced in a basket swap. Indeed, arriving at a table such as Table 20.1 is perhaps the most crucial step in the basket swap valuation exercise. We outline below a simple method for estimating the needed default probabilities.

Consider a one-year FTD basket that references five entities, each with different, but unknown, default probabilities. As in the previous section, the total notional amount of the basket is \$50 million, equally divided among the five reference entities, and all entities have a zero recovery rate. To avoid the need to model explicitly the joint probability density of default times, we follow the traditional BSM framework—see Chapter 17—and assume that defaults, if any, only occur at the maturity date of the contract,

TABLE 20.2
Reference Entities in a Hypothetical Five-Asset Basket Swap
(Total notional = \$50 million; individual recovery rates = 0)

Reference entity	CDS premium (basis points)	Notional amount (\$ million)
(1)	(2)	(3)
Entity #1	25	10
Entity #2	45	10
Entity #3	50	10
Entity #4	60	10
Entity #5	100	10

an assumption that also simplifies the treatment of accrued premiums.⁴ We also assume that one-year CDS contracts written on each reference are negotiated in a liquid market, and Table 20.2 lists the corresponding premiums.

Given the information in Table 20.2, we are then asked to value a first-to-default basket swap written on this portfolio. To do so we shall rely on the one-factor credit risk model discussed in Chapter 19, with a slight modification to allow us to leave the homogeneous portfolio case that was examined in that chapter. Namely, for each of the five entities in the basket, we assume that their returns vary in response to a common factor, α , and an entity-specific factor ϵ_i , where the subscript $i = 1, \dots, 5$ identifies each reference entity. Thus, as in equation (19.9):

$$R_{i,t} = \beta_i \alpha_t + \sqrt{1 - \beta_i^2} \epsilon_{i,t} \quad (20.5)$$

where $R_{i,t}$, α , and ϵ_i have been standardized as discussed in Chapter 19, and R_i is the return on owning reference entity i .

Recall that the model defines default as a situation where the return R_i reaches or falls below a given threshold C_i :

$$\text{Default by entity } i \iff R_i \leq C_i \quad (20.6)$$

and thus the probability ω_i of default by entity i by the maturity date of the contract is given by

$$\omega_i = \text{Prob}[R_i \leq C_i] = N(C_i) \quad (20.7)$$

⁴ Chapter 17 discussed the notion of the probability density of default time in a single-issuer context.

where $N(\cdot)$ is the cumulative density function of the standard normal distribution.

Given that our interest lies in risk-neutral probabilities, ω_i can be obtained from the observed CDS premiums for each entity—see Chapters 6 and 16—and C_i can be computed by inverting (20.7), as discussed in Chapter 19. Lastly, to parameterize β_i , one can regress stock returns for each of the entities on the common factor α , where α could correspond, for instance, to standardized stock market returns.⁵ To keep things simple, we shall assume that $\beta_i = .5$ for all companies.

We now have all ingredients necessary to run the valuation exercise. Using the simulation-based method described in Chapter 19, we obtain the results presented in Table 20.3. Such results are analogous to the information presented in Table 20.1, except that we have now derived the probabilities associated with each default outcome on the basis of market data—CDS premiums and equity returns—and of an explicit portfolio credit risk model. Carrying on with the same calculations done for Table 20.1, we arrive at the result that the FTD basket premium for this portfolio is about 265 basis points, which is well in excess of the CDS premiums for any one of the reference entities included in the basket.

TABLE 20.3
Valuation of a Five-Asset FTD Basket Swap^a
($\beta = .5$; total notional = \$50 million; individual recovery rates = 0)

Possible outcomes (No. of defaults)	Outcome probabilities (percent)	Payment by FTD protection seller ^b (\$ millions)	Loss to unhedged investor (\$ millions)
(1)	(2)	(3)	(4)
0	97.35	0	0
1	2.51	10	10
2	0.13	10	20
3	0.01	10	30
4	0.00	10	40
5	0.00	10	50

^a The referenced portfolio is the one described in Table 20.2.

^b Values shown in column 3 are the total payments made by the FTD protection seller under each possible default outcome.

⁵ This regression-based approach is only a rough approximation, however, because the resulting correlations are computed based on estimates of actual probabilities, rather than risk-neutral probabilities.

As we discussed in Chapter 9, from the standpoint of investors (protection sellers), the FTD basket swap represents an opportunity to leverage one's credit exposure: While the protection seller is exposed to the credit risk in debt instruments that total \$50 million in terms of notional amount in the example just described, the actual potential loss is limited to \$10 million, all while earning a premium that corresponds to a credit quality that is well inferior to that of each individual entity referenced in the portfolio.

To conclude this subsection, we compare the expected loss of the FTD protection seller to that of an unhedged investor who owns the entire portfolio.⁶ The total default-related losses experienced by such an investor under each possible default outcome are shown in the last column of Table 20.3. If we multiply each element of column 2 by its counterpart in column 4, and sum the resulting numbers across all possible default outcomes, we find that the unhedged investor faces an expected loss of about 0.56 percent of the entire portfolio, compared to an expected loss of 2.65 percent for the FTD protection seller—column 2 times column 3. Note, however, that the maximum loss of the unhedged investor is five times as much as that of the FTD protection seller. Moreover, for this particular portfolio, the expected (dollar) losses of the FTD protection seller—who is long only the first loss—and of the unhedged investor—who is long the entire portfolio—are \$265,000 and \$280,000, respectively.

20.4 The Second-to-Default Basket

Let us briefly revisit the second-to-default (STD) basket, which is a contract where default protection is bought and sold for the second default, and the second default only, among all the entities in a given portfolio. Column 2 in Table 20.3 has all that we need to value a STD basket written on the portfolio described in Table 20.2, once we note the fact that the potential losses of the STD protection seller are zero if the number of defaults falls below 2 and \$10 million thereafter (see Table 20.1). The reader is then invited to verify that the STD premium for our hypothetical portfolio amounts to about 14 basis points. The fact that the STD premium is so low reflects the very small probabilities associated with more than one default in the overall portfolio.

⁶ By an unhedged investor, we mean one who is holding the portfolio, but who has not hedged his or her credit risk exposure via baskets, credit default swaps, or any other hedging vehicle.

20.5 Basket Valuation and Asset Correlation

In Chapter 9, and also in Section 20.2, we used the simple two-asset basket example to suggest that default correlation can have a significant effect on the valuation of basket swaps. We can now verify this result using the more realistic basket described in Section 20.3.2. To do this we propose the following exercise: compute the one-year FTD premiums for baskets referencing 11 portfolios. The portfolios are identical to the one described in Table 20.2, except for the degree of correlation, which we vary from zero to one. We continue to assume that the premium is paid annually.

Figure 20.1 summarizes the results, which are consistent with the main conclusions drawn from the two-asset case. In particular, the figure plots one-year FTD premiums for different values of β . It shows that when $\beta = 1$, the FTD premium is equal to the default probability of the reference entity with the lowest credit quality (highest default probability), which is 1 percent or 100 basis points in this case. As for the case of no correlation, $\beta = 0$, the FTD premium is essentially equal to the sum of the CDS premiums of each individual entity referenced in the basket, or 280 basis points. As the figure shows, for intermediate values of β the FTD premium varies between these two polar cases, but the relationship between asset correlation and the FTD premium is very nonlinear.

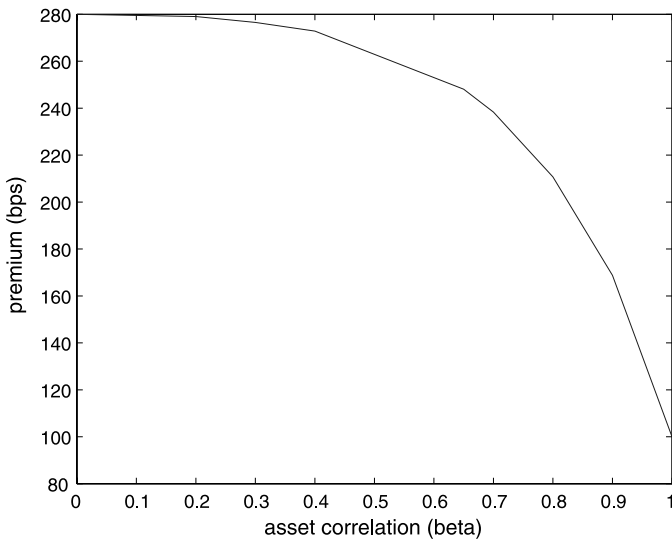


FIGURE 20.1. Asset Correlation and the FTD Premium

One practical lesson from Figure 20.1 regards the importance of trying to obtain good estimates for the degree of asset correlation in the portfolio. A large error in the estimation of β could result in a significant mispricing of the basket and, consequently, lead to a credit risk exposure that could differ substantially from the intended one. For instance, a FTD protection seller who overestimated the degree of asset correlation in the portfolio could end up charging a much lower premium than would be warranted by the risk profile of the portfolio.

20.6 Extensions and Alternative Approaches

We have mentioned already, at the end of Chapter 19, the several directions in which the simple model used here for the valuation of basket default swaps has been extended, as well as alternative modeling approaches for examining portfolio credit risk. For instance, the work of Li (2000)[55] is of particular interest in that he proposes a method that explicitly addresses issues related to the joint probability density of default times. He also illustrates an application of his method to the valuation of a basket swap.

For convenience, we have been limiting ourselves essentially to the analysis of contracts involving only one premium payment date and assets with zero recovery rates. We should note, however, that the methods described here can be extended to deal with multi-year contracts with several premium payment dates and with assets with nonzero recovery rates. Such extensions would be analogous to what we did in Part III when we went from examining zero-coupon bonds to coupon-paying bonds.

21

Valuing Portfolio Swaps and CDOs

As we discussed in Chapter 10, portfolio default swaps share many of the characteristics of basket default swaps. Some salient differences include the fact that they tend to reference a larger number of entities than do baskets and that default protection is bought and sold in terms of percentage losses in the portfolio, as opposed to with reference to the number of individual defaults. For instance, an investor might enter into a portfolio default swap where it agrees to cover the first ten percent in default-related losses in the portfolio. In exchange, the investor receives periodic payments from the protection buyer. Given these basic similarities, many methods used for the valuation of portfolio default swaps are variations of those used to value baskets.

Portfolio default swaps are important in their own right, but they are also of interest because they can be thought of as building blocks for the increasingly common synthetic CDO structure, which we discussed in Chapter 14. We conclude this chapter with a brief discussion of additional valuation issues that are relevant for the pricing of these structures.

21.1 A Simple Numerical Example

We start the discussion of valuation principles for portfolio default swaps with a simple numerical example. Similar to the analysis surrounding Table 20.1, when we examined basket swaps, the goal of this discussion is to provide some basic intuition into the valuation exercise.

TABLE 21.1
Valuation of Portfolio Default Swaps: A Simple Numerical Example
(Total notional = \$400 million; individual recovery rates = 0)

Possible outcomes (No. of defaults)	Outcome probabilities (percent)	Portfolio losses (\$ millions)	Portfolio swap losses ^a	
			first-loss (\$ millions)	second-loss (\$ millions)
(1)	(2)	(3)	(4)	(5)
0	52.73	0	0	0
1	23.28	20	20	0
2	11.21	40	40	0
3	5.80	60	40	20
4	3.03	80	40	40
5	1.80	100	40	60
6	0.92	120	40	80
7	0.56	140	40	80
8	0.29	160	40	80
9	0.19	180	40	80
10	0.10	200	40	80
11	0.05	220	40	80
12	0.03	240	40	80
13	0.01	260	40	80
14	0	280	40	80
15	0	300	40	80
16	0	320	40	80
17	0	340	40	80
18	0	360	40	80
19	0	380	40	80
20	0	400	40	80

^aValues shown in columns 4 and 5 are the total losses of FTD and STD protection sellers under each possible default outcome.

Consider a portfolio composed of bonds issued by 20 entities, all with a zero recovery rate. The first column of Table 21.1 lists all possible default outcomes associated with this portfolio over the next year, ranging from no defaults to all 20 reference entities in the portfolio defaulting together. The second column shows the corresponding risk-neutral probabilities associated with each of the 21 default outcomes.¹ As in previous chapters in this part of the book, we continue to assume that defaults, if any,

¹ Columns 1 and 2 together essentially correspond to the loss distribution function of this portfolio.

only occur at the maturity date of the contract, which is assumed to be one year. For instance, the second row of column 2 tells us that there is a 23.3 percent probability of the reference portfolio experiencing one default at the maturity date of the swap.

We assume that each of the entities represented in the portfolio accounts for an equal portion of the total value of the portfolio. Thus, for a portfolio that corresponds to a total notional amount of \$400 million, each default results in a loss of \$20 million, or 5 percent of the total. Default-related losses for each possible default outcome are shown in column 3. As a result, as we did in Chapter 20, we can multiply columns 2 and 3 on an element-by-element basis and sum the resulting numbers across all possible default outcomes to arrive at the expected (risk-adjusted) loss in the overall portfolio over the life of the contract:

$$\text{Expected loss} = \sum_{i=0}^{20} \omega_i l_i \quad (21.1)$$

which is roughly \$20 million, or 5 percent of the portfolio's face value. (In the above expression, ω_i and l_i correspond to the i th elements of the second and third columns of Table 21.1, respectively.)

Consider now a portfolio default swap written on the portfolio detailed in Table 21.1. The protection seller in this swap commits to absorb all default-related losses up to 10 percent of the notional amount of the contract, which corresponds to a maximum loss of \$40 million. The possible outcomes facing such a protection seller, in terms of his losses under different default scenarios, are listed in column 4. Multiplying columns 2 and 4 and summing across all outcomes as in equation (21.1) leads to an expected loss for this investor in the order of \$14.3 million, or 35.6 percent of the first-loss piece.

Assuming that the swap is fairly valued and that no money changes hands at its inception, and relying on arguments entirely analogous to the ones used in previous chapters, we then arrive at a premium of 35.63 percent, or 3,563 basis points, for this first-loss contract, i.e., the protection buyer promises to pay \$14.3 million ($3,563 \text{ bps} \times \40 million) in exchange for the protection provided by the contract.

Column 5 lists, for every possible default outcome, the total payment made by a second-loss protection seller who committed to cover all default-related losses falling between 10 and 30 percent of the portfolio. Again, multiplication and addition of columns 2 and 5 show that, on a risk adjusted basis, such a protection seller can expect to make a payment of \$5.2 million under the terms of this contract. Given a second-loss piece of \$80 million— $(30-10) \text{ percent} \times \400 million —the corresponding protection premium is 647 basis points, or 6.47 percent.

21.2 Model-based Valuation Exercise

As with other multi-name credit derivatives, obtaining a loss distribution function for the underlying portfolio is perhaps the most important part of the valuation exercise. In the previous section we skipped this problem altogether as our goal was simply to build some intuition on the valuation process.

Consider, as an example, an equally weighted reference portfolio with 100 companies and corresponding to a total notional amount of \$400 million. The portfolio is homogeneous with each of its constituent companies having a risk-neutral default probability of 10 percent over the life of the one-year portfolio default swaps that we will examine. For simplicity, we continue to assume a zero recovery rate. It turns out that we can value any default swap written on this portfolio by relying on essentially the same model- and simulation-based approach that we discussed in Chapter 19, much as we did in Chapter 20 in the valuation of basket default swaps. In particular, to model the credit risk in this portfolio, we assume that individual company returns and defaults behave as in equations (20.5) through (20.7). To examine the likelihood of different default scenarios associated with this portfolio, one could take the following steps:

- **Step 1:** Assign values to the key model parameters, β and C . For instance, as we noted in Chapter 20, an estimated value for β could be obtained by regressing standardized stock market returns for the companies included in the portfolio on the modeler's choice for the common factor, such as standardized changes in a marketwide stock index, and C can be obtained by inverting equation (20.7);
- **Step 2:** Based on the standard normal distribution, draw a large number of random values for the common factor α and for each of the company-specific factors ϵ_i , $i = 1, \dots, 100$;
- **Step 3:** For each reference entity i , use equation (20.5) to compute its return, R_i , for each value of α and ϵ_i and record whether or not that value of R_i constituted a default by entity i , as specified by equation (20.6);
- **Step 4:** Once individual defaults have been counted and recorded for all reference entities under all values of α and ϵ_i , the probabilities associated with different default outcomes can be computed by taking the ratio of the number of occurrences of that outcome over the total number of random draws in Step 1.²

² Each outcome corresponds to a given number of defaults in the portfolio, as in Table 21.1, and each random draw corresponds to a value for the vector $[\alpha, \epsilon_1, \epsilon_2, \dots, \epsilon_{100}]$.

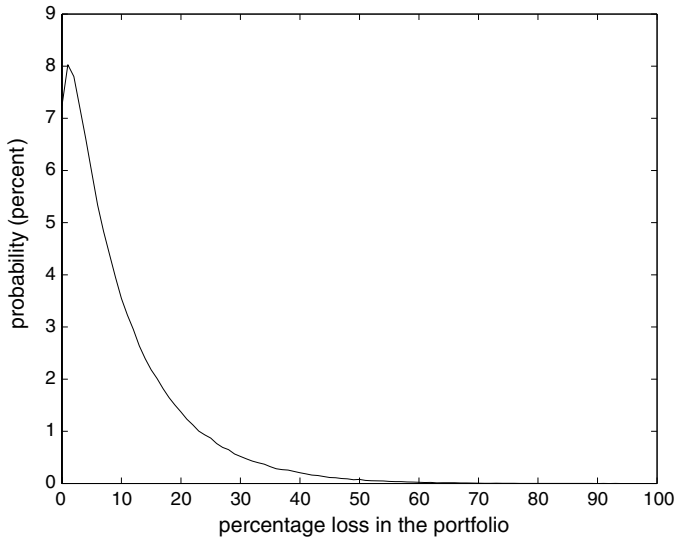


FIGURE 21.1. Loss Distribution Function for a Portfolio with 100 Assets (one-year horizon)

Once the above steps are followed, one has essentially obtained the loss distribution function associated with the portfolio, which is shown in Figure 21.1 for $\beta = .5$. Given the loss distribution function, which effectively corresponds to columns 1 and 2 in Table 21.1, the rest of the valuation exercise proceeds in exactly the same fashion as in the numerical example illustrated in that table.

Take, for instance, the valuation of two one-year portfolio default swaps written on the portfolio underlying the loss distribution function shown in Figure 21.1. The first swap is a first-loss contract covering losses of up to 20 percent of the portfolio, and the second is a second-loss contract covering losses between 20 and 50 percent. We will discuss the valuation results by examining the maximum and expected default-related losses of three investors, one who is long the entire portfolio without having bought any default protection, one who has sold protection via the first-loss contract, and another who has sold protection via the second-loss contract. We shall call each of these the unhedged investor, the first-loss investor, and the second-loss investor, respectively.

Table 21.2 displays key statistics for each investor. Column 2—rows 1 through 3—shows the corresponding maximum losses, which are \$400 million for the unhedged investor, \$80 million—20 percent of \$400 million—for the first-loss investor, and \$120 million— $(50 - 20 = 30)$ percent of \$400 million—for the second-loss investor. Thus, similar to basket swaps, portfolio swaps give investors an opportunity to lever up their

TABLE 21.2
Valuation of Portfolio Default Swaps^a
($\beta = .5$; total notional = \$400 million; individual recovery rates = 0)

Investor type	Maximum loss (\$ millions)	Expected loss (\$ millions) (percent ^b)	
(1)	(2)	(3)	(4)
1. unhedged investor	400	40	10
2. first-loss investor	80	34	43
3. second-loss investor	120	5	4.4

Memo:
 Prob[portfolio loss $\geq 20\%$] = 14.7 percent
 Prob[portfolio loss $\geq 50\%$] = 0.6 percent

^aResults based on 500,000 simulations of the credit risk model.

^bPercentage losses for first- and second-loss investors are reported relative to their maximum losses—see column 2.

credit exposure. For instance, while the first-loss investor is exposed to all reference entities in this \$400 million portfolio, he or she is liable only for the first \$80 million of default-related losses, all while enjoying, as we shall see below, a substantial premium. Of course this “substantial premium” is there for a reason. For instance, as the memo lines in Table 21.2 show, there is a 14.7 percent chance that the first-loss investor will lose the entire first-loss piece, whereas the probabilities of complete losses for the second-loss and unhedged investors are, respectively, 0.6 percent and virtually zero.

To compute the fair value of the premiums owed to the first- and second-loss investors in Table 21.2, we use the loss distribution function associated with the reference portfolio—Figure 21.1—and carry out the same calculations described for Table 21.1. The results are shown in columns 3 and 4. For instance, the expected losses of the first-loss and unhedged investors are relatively close, at \$34.4 million and \$40 million, respectively. However, relative to their maximum losses, expected percentage losses are much higher for the first-loss investor (about 43 percent) than for the unhedged investor (10 percent).

Thus assuming that no money changed hands at the inception of the first-loss contract and that market forces acted to rule out any arbitrage opportunities, the first-loss investor receives a premium from the protection buyer in that contract that amounts to 43 percent of the first-loss piece, or 4,300 basis points. Likewise, the fair value of the premium for the second-loss portfolio default swap is about 440 basis points as the

risk-adjusted expected loss of the second-loss investor is approximately 4.4 percent.³

21.3 The Effects of Asset Correlation

As with the default basket, the degree of correlation among the entities included in the reference portfolio is an important determinant of the portfolio swap premium. We illustrate this point in Figure 21.2, which shows the premiums for three one-year portfolio default swaps written on the portfolio described in Section 21.2 for values of β varying from 0 to 1. The first two contracts—labeled first- and second-loss—are the ones examined in Section 21.2. The third one—labeled third-loss—is essentially a contract that would absorb any residual losses after the protection provided by both the first- and second-loss contracts is exhausted. One can think of a protection-selling position in the third contract as equivalent to the residual risk exposure of an investor who owns the entire portfolio, but who has bought protection through the first two contracts.

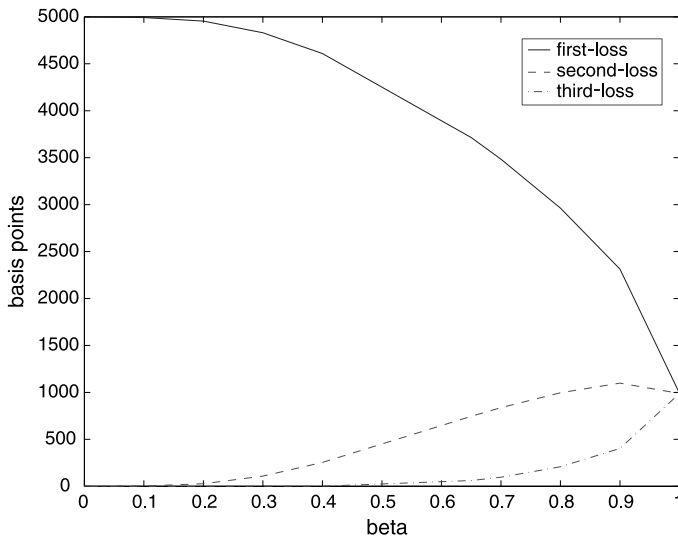


FIGURE 21.2. Portfolio Swap Premiums and Asset Correlation

³ Recall, as noted in Chapter 19, that we are assuming that all parties to the contract discount expected future payments at the same rate and that the premium is paid at the maturity date of the contract.

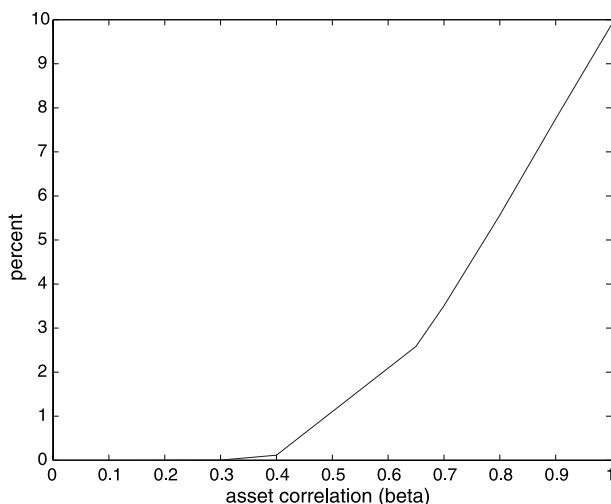


FIGURE 21.3. Probability of Portfolio Losses Equal to or Greater than 50 Percent

Figure 21.2 shows that as the degree of correlation in the portfolio increases, the premium owed to first-loss investors decreases. Although this may seem counterintuitive at first, a closer look at the nature of the contract proves otherwise. As the degree of correlation in the portfolio increases—see Figure 21.3—so does the probability of large losses. This happens because higher correlation increases the chances that several reference entities default together. Thus greater correlation increases the risk that second- and third-loss investors will be called upon to cover default-related losses, i.e., that they will have to pay up for losses beyond those covered by the first-loss investors. Thus, the second- and third-loss investors need to be compensated with larger premiums as more of the total risk in the portfolio is being borne by them as β increases. Indeed, for very high correlation, even the premium owed to second-loss investors starts declining in this example as a substantial portion of the total risk in the portfolio is now also shared with the third-loss investors. (The premium that would be owed to a third-loss investor is shown as the dash-dotted line in Figure 21.2.)

One final note: When $\beta = 1$, the portfolio essentially behaves like a single asset because either all reference entities survive or all default together. In this extreme case, Figure 21.2 shows that all investors earn the same premium, 1,000 basis points, which corresponds to the 10 percent default probability of the individual entities included in the portfolio.⁴

⁴ As noted in Chapter 10, with nonzero recovery rates, the equality of premiums across different classes of investors when $\beta = 1$ generally does not hold. The positive

21.4 The Large-Portfolio Approximation

In Chapter 19 we discussed the large-portfolio approximation method for computing loss distribution functions for a given portfolio. In particular, see equation (19.33), we found that the following expression approximates the probability that default-related losses in a large homogeneous portfolio will not exceed θ percent over a given time period:

$$\text{Prob}[L \leq \theta] = N\left(-\frac{C - N^{-1}(\theta)\sqrt{1-\beta^2}}{\beta}\right) \quad (21.2)$$

where L is the percentage default-related loss in the portfolio; $N(\cdot)$ is the standard normal cumulative distribution function; $N^{-1}(\cdot)$ is its inverse; and C and β have their usual definitions.

We can then use (21.2) to compute an approximate value for the probability that the first-loss investor described in Section 21.2 will lose the entire first-loss piece as a result of defaults in the underlying portfolio. That probability is approximately

$$1 - N\left(-\frac{C - N^{-1}(.2)\sqrt{1-\beta^2}}{\beta}\right)$$

or 13.5 percent, which is close to the 14.7 percent probability reported in Table 21.2. The analogous figures for the second-loss contract are 0.5 percent and 0.6 percent, respectively.

One can also compute approximate expressions for the premiums for the first- and second-loss portfolio swaps described in Section 21.2, but these calculations involve some mathematical manipulations that go beyond the scope of this book. Interested readers will find the following formula in O'Kane and Schlogl (2001)[64] for the premium owed under an arbitrary portfolio swap written on a large homogeneous portfolio:

$$\frac{N_2\left(-N^{-1}(L_{lb}), C, -\sqrt{1-\beta^2}\right) - N_2\left(-N^{-1}(L_{ub}), C, -\sqrt{1-\beta^2}\right)}{L_{ub} - L_{lb}} \quad (21.3)$$

where $N_2(\cdot)$ is the bivariate normal cumulative distribution function, and L_{lb} and L_{ub} are, respectively, the lower and upper bounds of the range of losses covered by the contract. For instance, for the first-loss contract featured in Section 21.2, $L_{lb} = 0$ and $L_{ub} = .2$, whereas, for the second-loss contract, $L_{lb} = .2$ and $L_{ub} = .5$.

recovery value of the assets in the reference portfolio provides an additional cushion to higher-order investors, potentially reducing their expected losses.

Using the above formula, we obtain (approximate) premiums of 4,356 and 417 basis points for the first- and second-loss portfolio swaps, respectively, which are indeed close to the results reported in Table 21.2.⁵

21.5 Valuing CDOs: Some Basic Insights

The basic framework laid out in this chapter for the valuation of portfolio default swaps easily translates into the foundation for a simple method for valuing the tranches of CDOs (synthetic or cash flow). Indeed, as we shall see in Chapter 22, some of the methods used in practice in the valuation of CDO structures can be cast as extensions of this simple framework.

For the sake of illustration, let us take a closer look at the synthetic CDO structure. From our discussion in Chapter 14 we know that we can think of a synthetic CDO approximately as a structure made up of one or more portfolio default swaps combined, in the case of a funded synthetic CDO, with an outright position in highly rated assets (the SPV collateral).

It is relatively straightforward to see that equity investors in a CDO structure have a position that is akin to that of first-loss investors in a portfolio default swap (or of second-loss investors if the institution that originated the CDO retained the first-loss piece). Likewise, mezzanine and senior tranche investors are long positions that are analogous to that of second- and third-loss investors in portfolio swaps.

Indeed, Figure 21.2 would be very informative for someone considering a CDO where equity tranche investors absorb the first 20 percent in default-related losses, mezzanine tranche investors absorb losses between 20 and 50 percent, and senior investors absorb any remaining losses. Essentially, one can reinterpret Figure 21.2 as showing the expected default-related losses for each class of investors in this CDO under varying degrees of asset correlation, assuming that the individual one-year default probability of the underlying entities is 10 percent.⁶

21.5.1 *Special Considerations for CDO Valuation*

Crucial as they are, default correlation and the credit quality of the underlying reference entities are not the only factors determining the valuation of CDO structures. For instance, an important difference between CDOs and

⁵ Not surprisingly, the large-portfolio approximation works better for larger portfolios. For instance for a 500-asset portfolio, the second-loss premium based on the simulation method is 420 basis points.

⁶ A reading of, say, 2,500 basis points in Figure 21.2 can be thought of as a (risk-neutral) expected loss of 25 percent of the notional amount represented by each CDO tranche.

portfolio default swaps is that the former typically incorporate so-called “coverage tests” provisions, but the latter do not. These tests are part of the legal structure of CDOs and are intended to protect investors in more senior tranches against a deterioration in the credit quality of the pool of collateral assets. In particular, a CDO structure may specify that its senior tranches will be provided a certain cushion such that the ratio of the structure’s total par value to that of that tranche’s will not fall below a certain “overcollateralization” level.

Should defaults occur among the assets in the collateral pool and bring overcollateralization ratios for senior tranches below the prescribed minimum (“trigger”) levels, the CDO is said to have failed its overcollateralization tests. As a result, the CDO structure may require the diversion of principal and interest cash flows from lower tranches to pay down enough of the principal of more senior tranches to bring the structure back into compliance with its overcollateralization requirements. In the context of this book, it suffices to say that coverage tests bring an additional level of complexity to the valuation of CDOs, one that was not captured by the portfolio default swap valuation exercise discussed in this chapter.

Some CDO structures call for the diversion of cash flows away from lower tranches even in the absence of default, if, for instance, the credit quality of the underlying assets is deemed to have deteriorated significantly. In contrast, if no defaults have occurred, a deterioration in the credit quality of the reference portfolio has no cash-flow implications for protection sellers in portfolio default swaps.

We mentioned other salient differences between CDOs and portfolio default swaps in Chapter 14. These include the credit quality of the SPV collateral in a synthetic CDO, reinvestment and “manager” risk associated with structures that include ramp-ups, removals, and replenishments, as well as other aspects of the CDO’s legal structure. Along with coverage tests, these factors should not be ignored in real-world attempts to value CDO structures.

21.6 Concluding Remarks

The techniques described in this chapter are intended to serve as introductory illustrations of some of the key factors that influence the valuation of multi-name credit derivatives. Before using these or any other methods to value portfolio products in the real world, the reader should consider several key questions:

- Is the model a good description of return and default dynamics of the underlying reference entities?

- Should I allow for more than one common factor to determine the extent of correlation among the reference entities?
- To which extent can I rely solely on market prices, such as CDS premiums, as a proxy for the risk-neutral default probabilities that are fed to the model?

Other issues such as non-normal shocks, uncertain recovery rates, and time-varying correlations (e.g., greater default correlation during economic downturns), which were not addressed by the simple modeling framework described in this chapter, should also be taken into account and serve as a reality check to would-be portfolio credit risk modelers. Still, imperfect as the basic modeling framework described in this chapter may be, it constitutes the basis for understanding more complex models that are used in commercial applications. For instance, as we shall see in Chapter 22, the CreditMetrics model, developed by the RiskMetrics Group, is essentially a more elaborate version of the modeling framework discussed here and in Chapters 19 and 20.⁷

Lastly, as we noted elsewhere in this part of the book, our basic model represents only one of several approaches to assessing the extent of default-related losses in a portfolio of credit-related instruments. For instance, a well-known alternative method for valuing CDOs is the intensity-based model of Duffie and Garleanu (2001)[22].

⁷ CreditMetrics is a trademark of J.P. Morgan.

22

A Quick Tour of Commercial Models

As the credit markets have grown in both size and sophistication so have the technical skills required to assess the risk-reward characteristics of an ever-expanding array of new products and structures, such as multi-name credit derivatives. Rather than developing in-house the analytical tools and databases required to fully understand and examine these new products, many investors have turned to outside experts for technical assistance. Indeed, several firms have come to be known as leading providers of analytical services regarding portfolio credit risk. In this chapter we briefly discuss some of the better-known models developed and marketed by these firms and compare them to the basic credit risk model discussed in Chapters 19 through 21.

We shall focus on four commercially available modeling approaches to the analysis of portfolio credit risk: Moody's Investors Service's Binomial Expansion Technique (BET), J.P. Morgan/RiskMetrics Group's CreditMetrics model, Moody's KMV's KMV model, and Credit Suisse Financial Products' CreditRisk⁺ model.¹ Given the number of approaches just mentioned, however, as well as the length and scope of this introductory book, our discussion of each modeling framework will be brief and, for the most part, non-technical. Basic sources for each approach are cited throughout the chapter. In addition, Crouhy, Galai, and Mark (2000)[17] provide a

¹ CreditMetrics is a trademark of J.P. Morgan; KMV is a trademark of Moody's KMV Corp.; CreditRisk⁺ is a trademark of Credit Suisse Financial Products.

comprehensive comparison involving most of the models summarized in this chapter.

22.1 CreditMetrics

Of the modeling approaches discussed in this chapter, this is the one that is most closely related to the basic model described in Chapter 19 and used in Chapters 20 and 21 for the valuation of basket and portfolio swaps. Indeed, one can think of our basic model, described by equations (19.7) through (19.9), as a simplified version of the CreditMetrics model. The CreditMetrics approach is described in detail by Gupton et al. (1997)[37].

Similar to the basic portfolio credit risk model, the CreditMetrics model is a Merton-style model that specifies defaults as situations where a variable R_i , which is assumed to measure the creditworthiness of a given entity i , falls below some threshold C_i . Another similarity regards the evolution of R_i , which CreditMetrics also assumes to be a function of both common (marketwide) and idiosyncratic random factors, where the former are the main determinants of the extent of default correlation in the portfolio.

An important feature of CreditMetrics that was not captured by the basic model discussed in Chapters 19–21 is that it is also designed to examine the likelihood of “ratings transitions,” or the probability, for instance, that an A-rated corporate borrower will be downgraded to, say, a BBB rating over a given time horizon. Thus, whereas the basic model only allowed for a reference entity to be in one of two states—solvency and default—CreditMetrics allows for as many states as the number of credit ratings under consideration. The way ratings transitions are modeled in CreditMetrics is similar to the manner in which we described the passage from solvency into default in the basic model. In particular, continuing with the same example just mentioned, the probability of firm i being downgraded from A to BBB is modeled as the probability that R_i falls below the threshold $C_{i,BBB}$. More generally, one can write

$$\text{Prob}[\text{downgrade to J-rating}] = \text{Prob}[R_i \leq C_{i,J}] \quad (22.1)$$

where, as always, the probability is defined with respect to a given time horizon.

22.2 The KMV Framework

We described the basic features of Moody’s KMV’s single-default model in Part III in our introduction to structural models of credit risk. Moody’s

KMV also offers a related tool for analyzing portfolio credit risk, called Portfolio Manager. The main output of Portfolio Manager is the loss distribution of the portfolio under consideration, from which, as we saw in Chapters 20 and 21, one can value a wide array of multi-name credit derivatives.

Portfolio Manager is similar to CreditMetrics in that it incorporates a model where default correlations are captured through the dependence of individual entities' returns on common factors. In addition, as in the CreditMetrics model, individual returns depend on a firm-specific factor, and defaults and credit migrations are characterized as situations where individual returns fall below certain prescribed thresholds. The basic Moody's KMV framework is described by Crosbie (2002)[16].

Differences between the CreditMetrics and Moody's KMV approaches include the fact that the latter uses an empirical distribution for returns based on proprietary data, as opposed to the normal distribution used in the CreditMetrics framework. Another difference regards the fact that the RiskMetrics Group makes the details of its model specification publicly available, where Moody's KMV does not.

22.3 CreditRisk⁺

Unlike the full versions of the CreditMetrics and Moody's KMV models, and similar to the basic model used in this book, the CreditRisk⁺ model allows for firms to be in only one of two states, solvency or default. Thus CreditRisk⁺ is essentially a model of default risk that is not designed, for instance, to mark portfolios to market when one or more of their components are downgraded.

The incorporation of default correlation into the CreditRisk⁺ framework is, in some aspects, analogous to the approach adopted by CreditMetrics and Moody's KMV. In particular, individual default probabilities are assumed to be functions of several "risk factors" that are common across different assets in the portfolio.

A key difference between CreditRisk⁺ and the models discussed in Sections 22.1 and 22.2 regards technical assumptions regarding the random nature of defaults. Rather than relying either on the normal or empirical distributions for modeling the evolution of risk factors, CreditRisk⁺ assumes the factors to be independently distributed according to the gamma distribution, which, subject to some approximations, allows the model to produce analytical results for unconditional probabilities of various losses in the portfolio. In contrast, the reader may recall that the derivation of the loss distribution function of a given portfolio generally involved either Monte Carlo simulations or, for large portfolios, approximation methods.

Methodologically, CreditRisk⁺ also differs from CreditMetrics and Moody's KMV in that, while these two latter models are structural models that follow the spirit of the BSM framework discussed in Chapter 17, CreditRisk⁺ is closer to the reduced-form approach, also discussed in that chapter. In particular, CreditRisk⁺ is based on actuarial methods that have long been used in the insurance industry to analyze event risk. Similar to intensity-based models, CreditRisk⁺ does not explicitly link the likelihood of default to the fundamentals of the firm. Instead, defaults occur exogenously according to the probabilities implied by the model.

Credit Suisse Financial Products (1997)[67] describes the CreditRisk⁺ model in further detail. The model is also summarized by Gordy (2000)[35], who compares and contrasts it to the CreditMetrics approach.

22.4 Moody's Binomial Expansion Technique

In Chapter 19 we showed that, conditioned on the common factor α , individual defaults were uncorrelated in the basic credit risk model, and the (conditional) loss distribution function of the homogeneous portfolio was given by the binomial probability distribution—see equations (19.20) and (19.21). As we saw then, one attractive feature of the conditional loss distribution function implied by the basic model was that its computation involved no Monte Carlo simulations, which can be computer-intensive, and no approximation methods, such as the large-portfolio method. Moody's binomial expansion technique is designed to take advantage of this convenient feature of portfolios with zero default correlation.

For a given credit portfolio with J potentially correlated assets, Moody's BET essentially aims at arriving at an otherwise equivalent homogeneous equally weighted portfolio with D uncorrelated assets, where D is dictated by the “diversity score” of the original portfolio. We say that this idealized portfolio is otherwise equivalent to the actual portfolio we are interested in because it shares many of the fundamental characteristics of the original portfolio. For instance, the individual default probability of each constituent of the idealized portfolio is set to the average default probability of the assets included in the original portfolio.

The diversity score of the original portfolio is a measure conceived by Moody's to capture the degree of industry diversification represented in the portfolio. For instance, a portfolio with a high degree of concentration in a given industry would be one with a low diversity score, resulting in an idealized portfolio with a number of assets, D , that could fall well short of the actual number of assets in the original portfolio. Intuitively, the smaller number of assets in the idealized portfolio controls for the fact that the large number of assets in the original portfolio tends to overstate the portfolio's true degree of diversification when two or more of its constituents have a tendency to default together.

Once the idealized portfolio is constructed, and the main characteristics of the actual portfolio are mapped into it, one can derive its loss distribution function in a way that is entirely analogous to the derivation of the conditional loss distribution function in Chapter 19. In particular, the probability of, say, k defaults in the idealized portfolio can be derived from the binomial distribution as in

$$\text{Prob}[k \text{ defaults}] = \frac{D!}{k!(D-k)!} \omega^k (1-\omega)^{D-k} \quad (22.2)$$

where ω is the probability of default of each reference entity represented in the idealized portfolio.

Armed with equation (22.2), as well as Moody's estimates of the diversity score and of individual default probabilities in the portfolio, it is then relatively straightforward to value, for instance the tranches of a CDO. Indeed, the BET method is an important component of Moody's rating methodology for rating both traditional and synthetic CDOs—see, for instance, Cifuentes and O'Connor (1996)[12] and Yoshizawa (2003)[75]. Other components include qualitative adjustments made to take other features of individual CDOs into account, such as the legal aspects of the structure.

Moody's has found the BET approach to work well for relatively homogeneous portfolios, but less so for portfolios where individual assets' default probabilities are markedly different. For such situations, Moody's has developed a modified approach, dubbed the double binomial expansion method, whereby the portfolio is divided with different portions that are mutually uncorrelated, but where each portion has a different individual default probability—see Cifuentes and Wilcox (1998)[13]. A further variation on the method, called the multiple binomial method, is employed for portfolios with further heterogeneity in individual default probabilities (Yoshizawa, 2003)[75].

22.5 Concluding Remarks

There is obviously much more to the credit risk models discussed in this chapter than has been covered in this brief overview. In addition, there are a number of other well-regarded commercially available models that we did not address here. Our choice of models, as well as the coverage provided for each model discussed in this chapter, was driven by a main guiding principle, which was to provide some insight into how the key concepts and methods described in Chapters 15 through 21 have actually been incorporated into credit risk analytical services that are bought and sold in the marketplace.

This Page Intentionally Left Blank

23

Modeling Counterparty Credit Risk

In the context of the credit derivatives market, counterparty credit risk refers mainly to the chance that a protection seller will fail to make good on its promise to make previously agreed-upon payments in the event of qualified defaults by reference entities.¹ We have thus far mostly sidestepped the issue of counterparty credit risk when discussing the valuation of credit derivative contracts. We have done so in part because existing arrangements among market participants—such as collateralization agreements and netting—help mitigate such a risk, but also for analytical convenience—factoring counterparty credit risk into the valuation exercise often adds a layer of complexity to the analysis that, for the most part, goes beyond the scope of this book. We say “for the most part” because we can in fact use a variant of the simple model discussed in Chapters 19 through 21 to capture some of the key insights regarding the role of counterparty credit risk in the valuation of credit derivatives.

¹ Strictly speaking, the protection seller is also subject to the risk that the buyer will fail to make the agreed-upon protection premium payments. The seller’s potential exposure, however, is essentially limited to the marked-to-market value of the contract, which, as we saw in Chapter 16, is a function of the difference between the premium written into the contract and the one prevailing in the market place at the time of default by the protection buyer. Thus the contract could well have negative market value to the seller, which would be the case if the credit quality of the reference entity had deteriorated since the inception of the contract. Under such circumstances, the seller would in principle experience a windfall upon default by the buyer, although, before defaulting, the buyer would likely have a strong incentive to seek to monetize the positive market value of the contract.

Our focus will be on the ubiquitous single-name credit default swap, which we discussed in some detail in Chapter 6. We shall assume that the CDS contracts examined in this chapter are uncollateralized agreements that are not subject to netting and that do not include any other credit enhancement mechanism. Towards the end of the chapter we outline ways to extend the model for the analysis of more complex contracts, such as baskets and portfolio default swaps.

23.1 The Single-Name CDS as a “Two-Asset Portfolio”

One might wonder why include a discussion of counterparty credit risk—especially one that focuses on single-name credit default swaps—in this part of the book, which, after all, deals with portfolio credit risk. The answer lies in the following simple insight: In the presence of counterparty credit risk, and from the perspective of the protection buyer, one can think of a single-name CDS as being akin to a portfolio involving risk exposures to two entities: the one referenced in the CDS and the protection seller. Indeed, the protection buyer has a default-contingent exposure to the protection seller, in that it will have to rely on the seller to cover any losses resulting from a default by the reference entity. Thus, the protection buyer effectively also has some residual exposure to the reference entity because, if the protection seller does not make good on its commitment, the buyer will have to bear any losses associated with a default by the reference entity.

We have to be careful not to take the portfolio analogy too far, however. In contrast to a traditional portfolio setting, the protection buyer only really bears a loss upon default by the reference entity if that entity happens to default at around the same time as the protection seller. (Or if the reference entity defaults after a default by the seller, and the original contract was not replaced.) Still, the basic insight that an uncollateralized single-name CDS shares some of the basic characteristics of a two-asset portfolio has some insightful implications for the analysis of counterparty credit risk. Indeed, as we shall see below, we can examine the effects of counterparty credit risk on the valuation of CDS contracts by relying on a modified version of the portfolio credit risk model that we discussed in Chapters 19 through 21.

23.2 The Basic Model

To admit explicitly the possibility that the protection seller can default on its obligations under the CDS contract, we assume that the seller is a

risky entity whose standardized returns $R_{p,t}$ follow the same basic model introduced in Chapter 19:

$$R_{p,t} = \beta_p \alpha_t + \sqrt{1 - \beta_p^2} \epsilon_{p,t} \quad (23.1)$$

where, as indicated in that chapter, α_t is a common factor (systematic risk) driving the returns on the protection seller; $\epsilon_{p,t}$ represents a risk factor that is specific to the protection seller (idiosyncratic risk); and β_p denotes the degree of correlation between $R_{p,t}$ and the common factor α_t .

We assume that any protection payment owed by the protection seller will be made only on one of the premium payment dates of the CDS. In addition, we define a default by the protection seller as the first instance when its return R_p is equal to or below its default barrier C_p on any of those dates.²

Given equation (23.1), and assuming that α and ϵ_p have zero mean and unit variance and are mutually independent and normally distributed, we saw in Chapter 19 that, conditional on all information available at time t , and given survival through that time, the risk-neutral probability that $R_{p,T}$ will be at or below $C_{p,T}$ at some future date T is

$$H_p(t, T) \equiv \text{Prob}_t[R_{p,T} \leq C_{p,T}] = N(C_{p,T}) \quad (23.2)$$

where $N(\cdot)$ is the standard normal cumulative distribution function. More generally, $H_p(t, T_j)$ is the time- t probability that R_{p,T_j} will be at or below C_{p,T_j} at time T_j .

The evolution of returns associated with the entity referenced in the CDS is modeled in an entirely analogous way. If we let $R_{r,t}$ denote the standardized return on the reference entity, and make the same assumptions made for the protection seller, we have:

$$R_{r,t} = \beta_r \alpha_t + \sqrt{1 - \beta_r^2} \epsilon_{r,t} \quad (23.3)$$

$$H_r(t, T) \equiv \text{Prob}[R_{r,T} \leq C_{r,T}] = N(C_{r,T}) \quad (23.4)$$

Equations (23.1) through (23.4) constitute our basic framework for analyzing the effects of counterparty credit risk in credit default swaps. As we will see later in this chapter, these equations capture two important determinants of counterparty credit risk effects: the credit quality of the protection seller (H_p) and the degree of default correlation between the seller

² In contrast to Chapters 19 through 22, where we limited ourselves to one-year contracts, we are now dealing with multi-period contracts. That is why we are characterizing the time of default as the first premium payment date in which R_p touches the default barrier from above.

and the reference entity, which is mainly determined by their respective values of β .

We should note that, although it may not be clear from the simplified notation used thus far, the basic model is flexible enough to allow for time-varying, even stochastic, probabilities of default. Such features could be incorporated, for instance, by modeling the evolution of the default thresholds accordingly.

23.3 A CDS with No Counterparty Credit Risk

If collateralization and other credit enhancement mechanisms embedded in the contract are such that the protection seller poses no risk to the protection buyer, we can proceed as if $H_p(t, T) = 0$ for all T , and we can thus ignore equations (23.1) and (23.2) when valuing the CDS. In essence, this is what we did in Part III of this book. We shall take the case of no counterparty credit risk as a benchmark against which to compare valuations derived from our counterparty credit risk model, but first we recast some of the main results derived in Part III in terms of the modeling framework described in the previous section.

Let us consider a J -year CDS written on the firm described by equations (23.3) and (23.4). We assume that the CDS has a notional amount of \$1, is entered into at time t , and involves the annual payment of premiums, at dates $T_1, T_2, \dots, T_J$, i.e.,

$$\delta_j \equiv T_j - T_{j-1} = 1 \text{ year}$$

for all j . For notational convenience we set $T_0 = t$. As we saw in Chapter 16, valuing such a swap involves, first, finding the expected present values (PV) of its protection and premium legs and, second, determining the value of the premium $S_{r,J}$ such that the CDS has zero market value at its inception. We follow the spirit of Merton's (1974)[59] model and assume that a default by the reference entity, if any, only occurs at specific times. In the context of this chapter, those times are the premium payment dates of the CDS.

As we saw in Chapter 16, the expected (risk-adjusted) present value of the premium leg for this CDS can be written as

$$\text{PV}[\text{premiums}]_t = \sum_{j=1}^J Z(t, T_j) Q_r(t, T_j) S_{r,J} \quad (23.5)$$

where $Q_r(t, T_j)$ is the risk-neutral probability, conditional on all information available at time t and given no default by that time, that the reference

entity will survive through time T_j , and $Z(t, T_j)$ is the time- t price of a riskless zero-coupon bond that matures at time T_j — $Z(t, T_j)$ is the time- t value of a dollar that will be received/paid at time T_j .³

The relationship between Q_r and H_r at time T_1 is straightforward:

$$Q_r(t, T_1) = 1 - H_r(t, T_1) = 1 - N(C_{r, T_1}) \quad (23.6)$$

and, generalizing for $j > 1$,

$$Q_r(t, T_j) = \prod_{i=1}^j [1 - N(C_{r, T_i})] \quad (23.7)$$

where the last equation follows from the fact that R_{r, T_j} is serially uncorrelated, i.e.,^{4,5}

$$\begin{aligned} Q_r(t, T_j) &\equiv \text{Prob}_t[R_{r, T_1} > C_{r, T_1} \text{ and } R_{r, T_2} > C_{r, T_2} \text{ and } \dots R_{r, T_j} > C_{r, T_j}] \\ &= \text{Prob}_t[R_{r, T_1} > C_{r, T_1}] \text{Prob}_t[R_{r, T_2} > C_{r, T_2}] \dots \text{Prob}_t[R_{r, T_j} > C_{r, T_j}] \\ &= \prod_{i=1}^j \text{Prob}_t[R_{r, T_i} > C_{r, T_i}] \end{aligned}$$

In Chapter 16 we also showed that the expected risk-adjusted present value of the protection leg can be written as

$$\text{PV}[\text{protection}]_t = \sum_{j=1}^J Z(t, T_j) [Q_r(t, T_{j-1}) - Q_r(t, T_j)] (1 - X_r) \quad (23.8)$$

where X_r is the recovery rate associated with the reference entity ($0 \leq X_r < 1$).

³ Equation (23.5) implicitly assumes that no accrued premium is due to the protection seller upon a default by the reference entity. This simplifying assumption, and a simple approach to relax it, was discussed in Chapter 16.

⁴ As we saw in Chapter 19, serial uncorrelation of R_r , which is normally distributed with zero mean and unit variance, means that $R_{r, T}$ and $R_{r, T-s}$ are uncorrelated random variables for all nonzero values of s . Thus, for instance,

$$\text{Prob}_t[R_{r, T} > C \text{ and } R_{r, T-s} > C] = \text{Prob}_t[R_{r, T} > C] \text{Prob}_t[R_{r, T-s} > C]$$

⁵ The notation $\prod_{i=1}^j G(i)$, for any function $G(i)$ of i , denotes the product operator:

$$\prod_{i=1}^j G(i) \equiv G(1)G(2) \dots G(j)$$

Given equations (23.5) and (23.8), the fair value of the CDS premium when there is no counterparty credit risk is

$$S_{r,J|H_p=0} = \frac{\sum_{j=1}^J Z(t, T_j)[Q_r(t, T_{j-1}) - Q_r(t, T_j)](1 - X_r)}{\sum_{j=1}^J Z(t, T_j)Q_r(t, T_j)} \quad (23.9)$$

and, thus, the model-implied CDS premium in the absence of counterparty credit is

$$S_{r,J|H_p=0} = \frac{\sum_{j=1}^J Z(t, T_j) \prod_{i=1}^{j-1} [1 - N(C_r, T_i)] N(C_r, T_j) (1 - X_r)}{\sum_{j=1}^J Z(t, T_j) \prod_{i=1}^j [1 - N(C_r, T_i)]} \quad (23.10)$$

23.4 A CDS with Counterparty Credit Risk

A first step in understanding how the model presented in Section 23.2 can be used to value a CDS that involves counterparty credit risk is to lay out the possible default outcomes regarding the contract and then to compute the risk-neutral probabilities associated with each outcome, just as we did with baskets and portfolio default swaps in Chapters 20 and 21. As noted earlier, we shall approach this problem from the perspective of the protection buyer in a CDS.

The protection buyer will pay the premium due on the dates T_j specified in the contract, $j = 1, \dots, J$, for as long as both the protection seller and the reference entity remain solvent on those dates. Let $Q_{rp}(t, T_j)$ denote the probability of such an event at time T_j , conditional on all available information at time t and given survival by both entities through that time. If we also assume that the protection seller is entitled to no accrued premium upon its own default, we can write the expected present value of the premium leg of a CDS that is subject to counterparty credit risk as:

$$\text{PV}[\text{premiums}]_t = \sum_{j=1}^J Z(t, T_j) Q_{rp}(t, T_j) S_{r,J} \quad (23.11)$$

which is entirely analogous to equation (23.5), except that we have replaced the survival probability of the reference entity with the probability that both the reference entity and the protection seller will survive through different dates in the future. The intuition is quite clear. If either the protection seller or the reference entity defaults at T_j , the protection buyer has no reason to continue to make premium payments. In the former case (seller's default), the default protection provided by the contract becomes worthless; in the latter case (reference entity's default) the contract is triggered and it is the seller that owes a payment to the buyer.

As for the protection leg of the swap, the buyer will receive the protection payment at a given time T_j only if two events occur at that time: the reference entity defaults *and* the protection seller is solvent. Let $H(t, T_j)$ denote the probability that these events take place at T_j , conditional on all information available at time t and on survival by both entities through t . Then, similar to equation (23.8), the expected present value of the protection leg is:

$$\text{PV}[\text{protection}]_t = \sum_{j=1}^J Z(t, T_j) H(t, T_j) (1 - X_r) \quad (23.12)$$

where, for simplicity, we assume a zero recovery rate associated with the protection seller.⁶

Given equations (23.11) and (23.12), the fair value of the credit default swap premium in the presence of counterparty credit risk is:

$$S_{r,J|H_p>0} = \frac{\sum_{j=1}^J Z(t, T_j) H(t, T_j) (1 - X_r)}{\sum_{j=1}^J Z(t, T_j) Q_{rp}(t, T_j)} \quad (23.13)$$

Equation (23.13) is a model-independent expression for the premium for a CDS that is subject to counterparty credit risk. To obtain expressions for the probabilities $H(t, T_j)$ and $Q_{rp}(t, T_j)$, we need to go back to the credit risk model. We discuss below two methods for doing so. The first is based on deriving explicit solutions for these probabilities in terms of parameters of the model— C_{p,T_j} , C_{r,T_j} , β_p , and β_r . The second is based on Monte Carlo simulation methods and is similar in spirit to the approach emphasized in Chapters 20 and 21.

23.4.1 Analytical Derivation of Joint Probabilities of Default

Unlike the case of no counterparty credit risk, where the probabilities included in the formula for the CDS premium depended only on the

⁶ Extending the framework to allow for a nonzero recovery rate for the protection seller would be relatively straightforward. We would add terms involving $X_p(1 - X_r)$ —where $0 \leq X_p < 1$ is the recovery rate of the protection seller—and the risk-neutral conditional probabilities, $G(t, T_j)$, associated with a default by both entities at time T_j , as in

$$\text{PV}[\text{protection leg}]_t = \sum_{j=1}^J Z(t, T_j) H(t, T_j) (1 - X_r) + \sum_{j=1}^J Z(t, T_j) G(t, T_j) X_p (1 - X_r)$$

To keep things simpler, however, and because this extension is relatively trivial, we choose to set X_p to zero and leave the nonzero X_p case as an exercise for the reader.

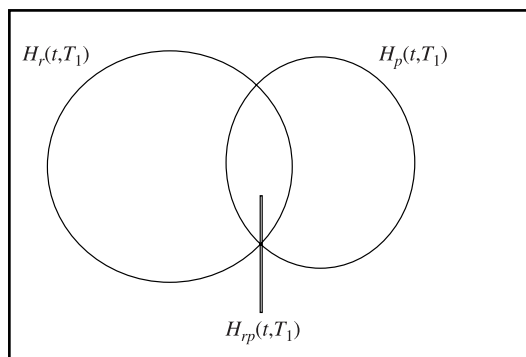


FIGURE 23.1. Diagrammatic Representation of Probabilities of Defaulting at Time T_1

distribution of future returns for the reference entity, we now need to use equations (23.1) through (23.4) to compute joint probabilities of various default and survival scenarios involving the reference entity and the protection seller. As we saw in Chapter 19, the derivation of these joint probabilities can quickly become very messy for a portfolio with several assets, even for the simplest case of a homogeneous portfolio. But computing joint default and survival probabilities for the “two-asset portfolio” implicit in a CDS with counterparty credit risk can be a significantly simpler exercise.

Consider, for instance, the derivation of $Q_{rp}(t, T_1)$ and $H(t, T_1)$. Figure 23.1 shows a diagrammatic representation of the probabilities attached to possible default and survival outcomes involving the reference entity and the protection seller at T_1 .⁷ In particular, the area labeled $H_r(t, T_1)$ represents the conditional risk-neutral probability that the reference entity will default at time T_1 , and the one labeled $H_p(t, T_1)$ is analogously defined for the protection seller. The region of overlap between the two areas, labeled $H_{rp}(t, T_1)$ in the figure, is the probability that both the reference entity and the protection seller will default at time T_1 .⁸

Relying on the same arguments laid out in the beginning of Chapter 19, the probability that both the protection seller and the reference entity will survive through time T_1 is

$$Q_{rp}(t, T_1) = 1 - [H_r(t, T_1) + H_p(t, T_1) - H_{rp}(t, T_1)] \quad (23.14)$$

⁷ We first discussed the use of such diagrams in Chapter 19.

⁸ We continue to assume that these probabilities are conditional on all information available at time t , given survival by both entities through t .

which is simply one minus the probability that at least one of the two entities—the protection seller and the reference entity—will default by time T_1 .

Given the definition of default events—equations (23.2) and (23.4)— $H_{rp}(t, T_1)$ can also be characterized as the probability that both R_{p,T_1} and R_{r,T_1} will fall below their respective default thresholds. Because both R_{p,T_1} and R_{r,T_1} are normally distributed and, as we saw in Chapter 19, have a coefficient of correlation equal to $\beta_p\beta_r$, this probability is given by

$$H_{rp}(t, T_1) = N_2(C_{r,T_1}, C_{p,T_1}, \beta_p\beta_r) \quad (23.15)$$

where $N_2()$ is the cumulative distribution function of the bivariate normal distribution.⁹

Thus, in terms of the model parameters, the probability that both the reference entity and the protection seller will not default at time T_1 is:¹⁰

$$Q_{rp}(t, T_1) = 1 - [N(C_{r,T_1}) + N(C_{p,T_1}) - N_2(C_{r,T_1}, C_{p,T_1}, \beta_p\beta_r)] \quad (23.16)$$

Going back to Figure 23.1, it is relatively straightforward to see that the probability that the reference entity will default at time T_1 and that the protection seller will remain solvent is

$$H(t, T_1) = H_r(t, T_1) - H_{rp}(t, T_1) = N(C_{r,T_1}) - N_2(C_{r,T_1}, C_{p,T_1}, \beta_p\beta_r) \quad (23.17)$$

The derivation of expressions for $Q_{rp}(t, T_j)$ and $H(t, T_j)$ for $j \geq 2$ requires only a bit more work than the computations just described for $Q_{rp}(t, T_1)$ and $H(t, T_1)$. Take, for instance, the computation of $Q_{rp}(t, T_2)$. If we let PS and RE denote the protection seller and reference entity, respectively, and define $R_{T_j} \equiv [R_{r,T_j}, R_{p,T_j}]$ and $C_{T_j} \equiv [C_{r,T_j}, C_{p,T_j}]$ we can write

$$\begin{aligned} Q_{rp}(t, T_2) &\equiv \text{Prob}_t[\text{PS and RE survive through } T_2] \\ &= \text{Prob}_t[R_{T_1} > C_{T_1} \text{ and } R_{T_2} > C_{T_2}] \\ &= \text{Prob}_t[R_{T_1} > C_{T_1}] \text{ Prob}_t[R_{T_2} > C_{T_2}] \end{aligned}$$

⁹ Appendix B provides a brief overview of the bivariate normal distribution.

¹⁰ Those familiar with the bivariate normal distribution have probably noticed that the right-hand side of (23.16) is equivalent to $N_2(-C_{r,T_1}, -C_{p,T_1}, \beta_p\beta_r)$.

where, as in the case of no counterparty credit risk, the last equality follows from the fact that returns are serially uncorrelated.¹¹

Thus, given

$$\text{Prob}_t[R_{T_i} > C_{T_i}] \equiv \text{Prob}_t[R_{r,T_i} > C_{r,T_i} \text{ and } R_{p,T_i} > C_{p,T_i}] \quad (23.18)$$

it is straightforward to see that we can write:

$$Q_{rp}(t, T_2) = \prod_{i=1}^2 \{1 - [H_r(t, T_i) + H_p(t, T_i) - H_{rp}(t, T_i)]\} \quad (23.19)$$

As for the derivation of a model-implied expression for $H(t, T_2)$, we once again recall its definition:

$$\begin{aligned} H(t, T_2) &\equiv \text{Prob}_t[\text{RE defaults at } T_2 \text{ and PS survives through } T_2] \\ &= \text{Prob}_t[(\text{RE def. and PS surv. at } T_2) \text{ and} \\ &\quad (\text{PS and RE surv. through } T_1)] \end{aligned}$$

Serial uncorrelation of returns implies that the two events in parenthesis above are independent. Thus we can write:

$$\begin{aligned} H(t, T_2) &= \text{Prob}_t[\text{RE def. and PS surv. at } T_2] \\ &\quad \times \text{Prob}_t[\text{PS and RE surv. through } T_1] \end{aligned}$$

The expressions for the two probabilities on the right-hand side of the above equation are entirely analogous to the ones derived for T_1 . Thus,

$$H(t, T_2) = [H_r(t, T_2) - H_{rp}(t, T_2)]Q_{rp}(t, T_1) \quad (23.20)$$

Generalizing for T_j , $j \geq 2$, and expressing all probabilities in terms of the parameters of the model, we obtain:

$$Q_{rp}(t, T_j) = \prod_{i=1}^j \{1 - [N(C_{r,T_i}) + N(C_{p,T_i}) - N_2(C_{r,T_i}, C_{p,T_i}, \beta_r \beta_p)]\} \quad (23.21)$$

$$H(t, T_j) = [N(C_{r,T_j}) - N_2(C_{r,T_j}, C_{p,T_j}, \beta_r \beta_p)]Q_{rp}(t, T_{j-1}) \quad (23.22)$$

¹¹ R_p and R_r are only contemporaneously correlated. Serial uncorrelation of Z , which is the only source of contemporaneous correlation between R_p and R_r implies that returns on the protection seller and the reference entity are intertemporally uncorrelated.

which we can substitute into equation (23.13) to write an explicit formula for the CDS premium in the presence of counterparty credit risk.

23.4.2 Simulation-based Approach

Rather than using the model to derive explicit formulae for the probabilities in equation (23.13), one may choose to rely on a Monte Carlo simulation approach similar to the one described in Chapter 19 and used in Chapters 20 and 21. In the context of the J -year CDS studied thus far, each simulation of the model involves relying on equations (23.1) through (23.4) to generate J values for the vector $R_{T_j} \equiv [R_{p,T_j}, R_{r,T_j}]$, for $j = 1$ to J . For each generated pair of returns, we record whether any of the two basic default outcomes of interest—survival by both entities and default by the reference entity while the protection seller survives—took place.

Thus, for $j = 1$ to J , we define

- $q(j) \equiv$ total number of simulations where both the reference entity and the protection seller survive through time T_j
- $h(j) \equiv$ total number of simulations where the reference entity defaults at time T_j and the protection seller survives through T_j .¹²

After running a sufficiently large number of simulations, we can compute approximate values of $Q_{rp}(t, T_j)$ as

$$Q_{rp}(t, T_j) \approx \frac{q(j)}{M} \quad (23.23)$$

Likewise, we can approximate $H(t, T_j)$ as

$$H(t, T_j) \approx \frac{h(j)}{M} \quad (23.24)$$

where M is the total number of simulations of the model. Thus, the method involves generating M values of R_r and R_p for each of the J premium payment dates of the CDS.

To improve the accuracy of the results, it is common practice to perform these computations a large number of times and then report the average of the results obtained. For instance, we may run $M = 500,000$ simulations of the model 200 times and compute the average of the 200 results obtained.

¹² Recall that the time of default is defined as the first time that the asset return reaches or falls below the default barrier on a premium payment date.

23.4.3 An Example

We examine the effects of counterparty credit risk on the valuation of a five-year CDS written on a reference entity with a 5 percent risk-neutral probability of default over the next year and a flat credit curve. We assume that the swap involves no credit enhancement mechanisms such as collateralization and netting, that $\beta_r = \beta_p = \beta$, and that the premium is paid annually.

Figure 23.2 shows premiums for the CDS just described under various assumptions regarding the credit quality of the protection seller and for different values of β . Accordingly, β varies from zero to .99, and $H_p(t, T_j)$ ranges from zero to 4 percent. These results are based on the analytical results derived in Section 23.4.1 and were confirmed by Monte Carlo simulations.¹³ Before we proceed, we should note that this exercise involves several simplifying assumptions, such as a zero recovery rate for the protection seller and no accrued premiums or interest in the event of default by the reference entity. In more realistic settings, these factors should not be ignored.¹⁴

Each curve in the figure corresponds to a given value of β . For instance, the solid line shows values of the CDS premium when $\beta = 0$, which corresponds to the case of no default correlation between the protection seller and the reference entity. The values along the horizontal axis correspond to risk-neutral default probabilities of potential protection sellers in this contract.

Consistent with one's intuition, Figure 23.2 shows that, for a given degree of default correlation between the reference entity and the protection seller, the fair value of the CDS premium declines as the credit quality of the protection seller deteriorates. Nonetheless, such an effect is barely noticeable for very low levels of default correlation. This occurs as the protection and premium legs of the CDS are about equally affected by counterparty credit risk when the coefficient of default correlation between the protection seller and the reference entity is low. Take the case of a protection seller that has a risk-neutral default probability of 4 percent, which is only marginally below the risk-neutral default probability of the reference entity. With $\beta = .2$ —a coefficient of default correlation of about 1 percent—the fair value of the CDS premium that such a seller would be able to charge is only about 3 basis points lower than that charged in a contract that involves no counterparty credit risk.

¹³ We assume a flat credit curve for the protection seller.

¹⁴ For instance, a nonzero recovery rate associated with the protection seller would lead to a smaller effect of counterparty credit risk on CDS premiums than the one suggested by this example.

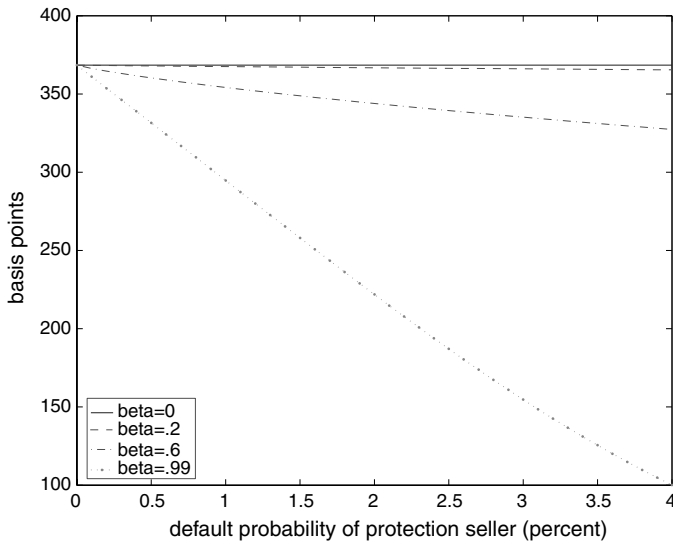


FIGURE 23.2. Effects of Counterparty Credit Risk on Five-year CDS Premiums. (The reference entity is assumed to have a recovery rate of 30 percent. The riskless yield curve is assumed to be flat at 5 percent.)

Figure 23.2 also illustrates the effect of default correlation on the CDS premium. As one would expect, higher correlation (higher β) increases the likelihood that the seller and the reference entity will default together and, thus, makes the CDS less valuable to the protection buyer, resulting in a lower premium. For instance, with $\beta = .6$, which amounts to a default correlation of approximately 12 percent, the drop-off in premiums as the credit quality of the protection seller declines is much steeper than in comparable cases with lower default correlation. Under this scenario, the same protection seller described in the previous paragraph—one with a risk-neutral default probability of 4 percent—would have to offer roughly a 40 basis point concession on the five-year premium relative to the premium charged in a comparable contract with no counterparty credit risk.

Taken together, the results in Figure 23.2 suggest that, although counterparty credit risk can have a significant effect on the extent of default protection effectively provided by a CDS, protection buyers can substantially mitigate their exposure to sellers by being mindful of the potential degree of default correlation between the seller and the reference entity. That mitigation would be over and above that conferred by relatively common contractual arrangements such as collateralization and netting, and

by the fact that protection sellers tend to have a high credit quality to begin with.

23.5 Other Models and Approaches

In keeping with the introductory nature of this book, we have focused on a very simple approach to analyzing the effects of counterparty credit risk in credit derivatives contracts. Nonetheless, our basic framework captures many of the key elements and insights of other approaches developed in the credit risk literature. For instance, a model that is closely related to the one described in this chapter is the one developed by Hull and White (2001)[42].

Similar to our basic model, the Hull-White specification follows the structural approach to modeling credit risk and assumes that there is a variable $X_j(t)$ that describes the creditworthiness of entity j at time t and that the entity defaults at t if $X_j(t)$ falls below a certain level $K_j(t)$. Hull and White also used their model to examine the valuation of credit default swaps that are subject to counterparty credit risk. Their model allows for defaults to occur at any time, not just at the premium payment dates specified in the CDS, and also allows for accrued premiums and interest rate payments to be factored into the valuation exercise. These features are not difficult to incorporate into the model discussed in Sections 23.1 through 23.4. For instance, accrued premiums can be introduced as noted in Chapter 16.

Many other methods and approaches for modeling the effects of counterparty credit risk exist and have varying degrees of complexity and effectiveness. For instance, a well-known framework rooted on the intensity-based approach was proposed by Jarrow and Yu (2001)[47]. Additional methods are discussed by Arvanitis and Gregory (2001)[2].

One issue that was not explicitly addressed by our basic model is the replacement value of a credit derivative contract in the event of default by the protection seller while the reference entity is still solvent. By replacement cost we mean the cost, to the protection buyer, of replacing a contract where the protection seller has defaulted with another one written on the same reference entity and with the same remaining maturity.

If the credit quality of the reference entity has not changed since the inception of the original contract, then the replacement cost is zero. Thus, for instances of low asset default correlation between the protection seller and the reference entity, the replacement cost should generally be very small. For high (positive) correlation, however, the replacement cost of the contract may be non-trivial, in that the credit quality of the reference entity is more likely to have deteriorated than improved when that of the protection seller has worsened to the point of leading it to default on its

obligations. For negative default correlation between the protection seller and the reference entity, the replacement cost could actually turn out to be negative, in which case the protection buyer could experience a windfall upon a default by the seller.

23.6 Counterparty Credit Risk in Multi-name Structures

We have thus far focused on approaches for assessing the effects of counterparty credit risk on the pricing of single-name credit default swaps. It turns out that the basic framework described in Sections 23.1 through 23.4 can be extended for the analysis of more complex structures. For instance, in the case of the first-to-default basket, we can expand the set of possible default outcomes—shown in Table 20.1 for a five-asset example that involves no counterparty credit risk—to allow for scenarios where the protection seller defaults during the life of the basket. By carefully keeping track of all possible outcomes and their respective risk-neutral probabilities implied by the model, one can rely on Monte Carlo simulation methods similar to the one described in Section 23.4.2. A similar logic applies to the valuation of portfolio default swaps, which we discussed in Chapter 21.

23.7 Concluding Thoughts

As we mentioned in the beginning of Part IV, the portfolio credit risk literature has been growing rapidly and is technically demanding. Indeed, we have barely scratched its surface in this introductory book. We do hope, however, that we have managed to provide a broad overview of some of the main issues that are germane to the valuation of multi-name credit derivatives and of counterparty credit risk. More important, we hope we have been able to provide a base from which one can expand one's knowledge of the subject and grow into this important segment of the credit derivatives market.

This Page Intentionally Left Blank

Part V

A Brief Overview of Documentation and Regulatory Issues

This Page Intentionally Left Blank

24

Anatomy of a CDS Transaction

This chapter provides an overview of legal and documentation issues involving credit default swaps (CDS), the most prevalent of all credit derivatives.¹ Similar to other over-the-counter derivatives instruments, credit default swaps are typically initiated with a phone call in which the basic terms of the transaction are agreed upon by the two prospective counterparties.² That initial oral agreement is then followed up by a confirmation letter, which, together with any supporting documentation referenced in the letter, spells out the rights and obligations of each counterparty, as well as the procedures for fulfilling them.

From a legal standpoint, the confirmation letter and related documents are the core of any CDS transaction and jointly constitute what we shall refer here as a CDS contract. In this chapter, we take a closer look at the main features and provisions of CDS contracts and the role that they have played in the rapid growth of the credit derivatives market. We also look at how developments in the marketplace have helped shape the contracts.

In keeping with the scope of this book, we limit ourselves to providing a broad, and thus necessarily incomplete, overview of documentation

¹ Many other credit derivatives instruments are negotiated on the basis of a similar documentation framework.

² In recent years, it has become increasingly common for transactions to be initiated “on-line” via various electronic platforms.

issues regarding CDS transactions. Needless to say, this chapter does not constitute legal advice regarding CDS, or any other credit derivatives contracts.

24.1 Standardization of CDS Documentation

As we mentioned in Chapter 2, CDS contracts are largely standardized, with the marketplace mostly relying on documentation sponsored by the International Swaps and Derivatives Association (ISDA), a trade group whose members include major dealers and end-users of over-the-counter derivatives products ranging from interest rate swaps to credit derivatives. We say “largely” because there are some contractual variations across national borders. Indeed, while the ISDA documentation for credit derivatives is preeminent worldwide, it coexists, in a few countries, with alternative, locally drawn, documentation frameworks.

To appreciate the role that the so-called ISDA contracts for CDS have played in the development of the credit derivatives market, it is useful to imagine the counterfactual. Suppose that, to this date, market participants still had to contend with different forms of CDS contracts, both within and across jurisdictions, depending on the counterparty, each with its own stipulations and definitions of key terms of the agreement. To reduce the inherent “legal risk” that would prevail in a world with a multitude of contract types, participants, and their lawyers, would have to devote valuable time and resources to scrutinizing each agreement, often negotiating its terms on a case-by-case basis. Such a situation would hardly be conducive to the impressive growth in liquidity and size that the credit derivatives market has experienced in recent years.

The fact is that, starting in the early days of the credit derivatives market, participants came to realize the need for a common set of contractual provisions and practices, following the example of other successful over-the-counter derivatives markets, most notably that for interest rate swaps. In essence, that is how the ISDA framework has become the most prevalent standard for documenting CDS transactions both at the national and international levels. Thus, our discussion of legal and documentation issues involving CDS transactions is centered entirely on the ISDA legal framework.

Broadly speaking, the ISDA framework for credit default swap contracts revolves around the following main components:

- The **Master Agreement** is a contract between the two prospective counterparties that often preexists the CDS transaction in which they are about to enter. The master agreement governs those aspects of the legal relationship between the two counterparties that are not specific

to the CDS transaction at hand. For instance, the agreement may specify that the laws of the state of New York should be the applicable law to any contracts entered into by the two counterparties. The master agreement may also specify that netting provisions should be applicable to any over-the-counter derivatives contract entered by the two counterparties and covered by the master agreement.³ Procedures relating to default by one of the counterparties in any of the types of contracts covered by the master agreement are generally also dealt with in this document. Highlighting the general nature of the master agreement, it is not uncommon for the same agreement to cover several types of over-the-counter contracts, not just credit derivatives.

- The **Confirmation Letter** is the document that follows up on the initial (generally oral) agreement between two prospective counterparties to enter into a specific CDS. As its name indicates, this document confirms the “economic” terms of the swap such as the identity of the reference entity, the notional amount of the contract, and the protection premium. In the early days of the credit derivatives market, the confirmation letter was a relatively lengthy document that described the terms of the contract in great detail. That so-called long-form of the CDS confirmation letter has effectively been replaced by its short-form, which is basically a template that allows the counterparties to “fill in the blanks” with the appropriate information and “check” the boxes that apply for the transaction at hand. For instance, the parties might have to agree on which types of default events are covered by the contract. The use of the short-form of the confirmation letter became widespread as the degree of standardization of acceptable provisions and definitions became significant enough that those terms and provisions could be listed in a separate document, called the ISDA Credit Derivatives Definitions.
- The **ISDA Credit Derivatives Definitions**, as just noted, is a list that defines key terms referred to in the confirmation letter. Examples include definitions of the credit events allowable in the confirmation letter, as well as additional detail on deliverable obligations and settlement procedures.
- **Supplements** are separate documents issued by ISDA that amend, update, or clarify terms in the Definitions.
- **Credit Support Documentation** involves agreements that call for the collateralization of net exposures between the counterparties in

³ See Chapter 2 for a brief discussion of netting provisions.

order to mitigate counterparty credit risk considerations. Not all CDS contracts include credit support documentation, but contracts that do are becoming increasingly common, especially when a lower-rated counterparty is involved.

The legal framework sponsored by ISDA is primarily intended to promote standardization of legal provisions and market practices, but it does recognize that some of the legal stipulations of the transaction may need to be tailored to the needs of the counterparties. For instance, the master agreement has two parts, the so-called “printed form” and the “schedule.” While the former contains key standard provisions of the master agreement, the latter allows any two counterparties to agree to make certain choices specified in the printed form and/or to amend any provisions.

We could go on at length in a discussion of master agreements and other legal issues involving credit derivatives, but that would go beyond the intent of this book. Those interested in learning more about the ISDA documentation framework may want to visit ISDA’s website, www.isda.org.

24.1.1 Essential Terms of a CDS Transaction

In addition to the names of the two parties in the contract—the buyer and seller of default protection—essential terms that need to be specified front and center in the confirmation letter include, obviously, the identity of the reference entity, the types of obligations of that entity that are covered by the contract, the specific events that will trigger a payment by the protection seller, and the procedures for settling the contract in the event of default by the reference entity. As basic and self-evident as some of these terms may seem, they need to be carefully defined in the text of the agreement. Any ambiguity regarding these key terms can result in significant legal and financial headaches down the road.

24.1.1.1 The Reference Entity

With regard to the reference entity, if all that the contract did were to name it, one might reasonably ask whether a default by a fully owned subsidiary of that entity is enough to trigger payments by protection sellers, or whether following the merger or demerger of the reference entity, the successor entity or entities become the new reference entity in the CDS. These questions, and other related issues, have been addressed in the Definitions and in various Supplements over the years. For instance, a default by a subsidiary generally does not trigger a contract written on the parent company, provided that company itself has not defaulted. Only contracts referencing the subsidiary are triggered. On the second question, the documentation determines that if an entity assumes 75 percent or more of the

bonds and loans of the original reference entity, then the assuming entity becomes the new reference entity.⁴

24.1.1.2 Reference and Deliverable Obligations

The reference obligation is a debt instrument issued by the reference entity that is designated in the CDS contract. The characteristics of the reference obligation are important for several reasons. First, it dictates the level of the reference entity's capital structure at which default protection is being bought and sold. Typically, the reference obligation is a senior unsecured debt instrument of the reference entity, although some contracts are written with reference to subordinated debt. Second, in the case of physically settled CDSs, the reference obligation is always a deliverable obligation (see Section 24.2.2), but a deliverable obligation need not be a reference obligation. For instance, most contracts in the United States accept obligations that have the same rank in the reference entity's capital structure as the reference obligation, generally subject to certain restrictions such as the currency in which the instrument is denominated and, in the case of bank loans, the ability to transfer the obligations to someone else.

The confirmation letter generally also specifies the range of obligations in which a default must occur in order for a credit event to take place. Most contracts simply specify the catch-all category "borrowed money," although, as we shall see below, there are some safeguards to prevent the contract from being triggered prematurely, for instance, because of certain events involving small dollar amounts or short delays in repayment.

24.1.1.3 Settlement Method

As we saw in Chapter 6, upon default by the reference entity, a CDS can be either physically or cash settled. The choice of settlement method is specified in the confirmation letter, which, as noted above, also determines the types of debt securities that can be delivered in the case of physically settled contracts.

Cash-settled CDS contracts are more common in Europe than in the United States, where physical settlement is the method of choice. We discuss settlement procedures in greater detail in Section 24.2.

24.1.1.4 Credit Events

One of the most important definitions in a credit default swap contract is that of "default." Which event, or events, must take place for the protection

⁴ If no single entity assumes 75 percent of the bonds and loans of the original reference entity, then the notional amount of the CDS contract is split pro rata among all entities assuming at least 25 percent of the bonds and loans of the original entity.

payment to be triggered? There are several “credit events” that constitute default in the ISDA documentation framework for credit derivatives and each are detailed in the Credit Derivatives Definitions. The following is a brief description of each event:

- **Bankruptcy** constitutes a situation where the reference entity becomes insolvent or unable to repay its debts. This credit event does not apply to CDS written on sovereign reference entities.
- **Obligation Acceleration** occurs when an obligation has become due and payable earlier than it would have otherwise been.
- **Failure to Pay** means essentially what it says. It occurs when the reference entities fails to make due payments.
- **Repudiation/Moratorium** is deemed to have occurred when the reference entity rejects or challenges the validity of its obligations.
- **Restructuring** is a change in the terms of a debt obligation that is adverse to creditors, such as a lengthening of the maturity of debt. We discuss restructuring in greater detail later in this chapter.

Not all of the above events apply to all contracts. The parties to the contract can agree to exclude certain events. For instance, the market consensus has moved towards excluding obligation acceleration as a credit event in newly entered CDS contracts in the US. In addition, as we shall see in Section 24.3, there is a certain degree of bifurcation in the marketplace, where some contracts allow for restructuring to be included in the list of credit events, whereas others do not.

24.1.2 Other Important Details of a CDS Transaction

Other important terms and provisions included in the confirmation letter and accompanying documentation include:

- The maturity of the contract, also referred to as the “scheduled termination date” of the contract.
- The notional amount of the contract, called the “fixed rate payer calculation amount” in the language of the confirmation letter.
- the CDS premium, which the contract calls the “price” or “fixed rate,” normally expressed in terms of basis points per annum.⁵

⁵ The protection payment and the protection seller are often referred to as the floating-rate payment and floating-rate payer, respectively, borrowing on language commonly used in the interest rate swap market. One interpretation of the use of this terminology

- The day-count conventions that should be applied, for instance, to the calculation of accrued premiums in the event of default by the reference entity while the contract is in force.
- The frequency with which CDS premium payments are made—payments are typically made on a quarterly basis on dates specified in the confirmation letter.

24.1.3 A Few Words of Caution

The standardization of CDS documentation has worked to reduce, but not eliminate, the legal risks associated with CDS transactions. Nonetheless, standardized documents are no guarantee of universal agreement as to the interpretation of those documents. Indeed, the effort to improve on the existing documentation framework is an ongoing one, especially as unanticipated market developments have come to expose deficiencies and limitations of earlier contracts, an issue that we discuss in Section 24.3.

Moreover, given that the CDS market is, as discussed in Chapter 2, still a relatively young marketplace, there is potentially much to be learned regarding how CDS documentation will be interpreted by the parties involved and by the courts. For instance some market observers have expressed concerns about the eventual enforceability of the language in ISDA contracts in national courts, especially in cases involving non-OECD parties.

24.2 When a Credit Event Takes Place...

The contract specifies in detail all the procedures that must be followed in the event of default by the reference entity while the CDS is in place. But before the settlement phase of the contract goes fully into gear, the ISDA documentation provides for some safeguards to ensure that a bona fide credit event has indeed taken place.

24.2.1 Credit Event Notification and Verification

A credit event is not a credit event until it meets certain minimum payment and default requirements under the terms of the CDS contract. For instance, a qualifying failure-to-pay event must take into account any applicable grace periods and must involve a minimum dollar threshold—e.g., a

is that, unlike the protection buyer who knows exactly when and how much to pay for as long as the contract is in force and the reference entity is solvent, uncertainty about the recovery rate means the seller does not know in advance the timing or size of the protection payment, if any, that it will have to make to the buyer.

loss of at least \$1 million. Repudiation, moratorium, and restructuring events also are subject to minimum dollar thresholds. In addition to meeting the minimum payment and default requirements, a credit event has to be verifiable through at least two sources of public information, such as Bloomberg, Reuters, or similar services. The event notification and verification process is generally formalized via the delivery of the Credit Event Notice and the Notice of Public Information by the party triggering the contract.⁶

These pre-settlement requirements are designed to protect the interests of both sellers and buyers of protection. From the seller's perspective, a premature triggering of the contract involving an otherwise solvent reference entity may mean that the seller will end up making a payment in a situation where no bona fide credit event ever takes place during the life of the CDS. From the buyer's perspective, the seller may end up receiving a small protection payment now while foregoing valuable protection down the road when a bona fide event takes place.

24.2.2 Settling the Contract

Once the occurrence of a credit event is verified, usually via acceptance of the Credit Event Notice and Notice of Publicly Available Information, the contract goes into its settlement phase.

For cash-settled contracts, upon default by the reference entity the protection buyer is entitled to receive from the seller the difference between the par and market values of the reference obligation. The latter tends to be a bond issued by the reference entity as bond prices are typically easier to determine in the marketplace than those of loans. The market value of the reference obligation is commonly determined by a dealer poll typically conducted a few days after the credit event. The contract generally allows the settlement price to be determined on the basis of a similar obligation of the reference entity if the original reference obligation is no longer available, which could be the case if, for instance, it was prepaid by the reference entity.

For physically settled contracts, the protection buyer has the right to decide which of the eligible obligations of the reference entity will be delivered to the seller. Once that decision is made, the buyer delivers a Notice of Intended Physical Settlement to the seller and then commits to delivering the designated obligations. The physical settlement period is capped at 30 business days. If delivery is not successful during that period because,

⁶ Commonly, the Credit Event Notice and the Notice of Publicly Available Information can be delivered up to 14 days after the maturity date of the contract, provided the event itself took place while the contract was still in place.

say, the buyer was not able to buy the obligations in the marketplace, the contract usually allows for “fallback” settlement procedures, which are described in ISDA’s 2003 Credit Derivatives Definitions.

24.3 The Restructuring Debate

One of the most prominent documentation issues in the history of the credit derivatives market has evolved around the definition of restructuring as a credit event. The scope of the issues debated by market participants over the years has ranged from the very definition of restructuring to the question of whether restructuring should be included in the list of allowable credit events in the first place.

In the early days of the marketplace, restructuring was defined only in very broad terms in ISDA’s standard CDS documentation. Indeed, the then-prevailing definition, which had been adopted by ISDA back in 1991, characterized restructuring essentially as any change in the terms of the obligations of a reference entity that was “materially less favorable” to its creditors. The Asian crisis of 1997 and the Russian default of 1998 brought to light some of the problems with the subjectivity of the 1991 definition, and, in 1999, market participants adopted a new definition of structuring.⁷ That definition basically provided a narrower characterization of which circumstances would trigger the restructuring clause in CDS contracts. In particular, the 1999 ISDA definition characterized restructuring as:

- a reduction in the rate or amount of interest payable;
- a reduction in the amount of principal;
- a postponement of interest or principal payment;
- a change in the seniority of the debt instrument;
- a change in the currency composition of any payment.

In addition, market participants, through ISDA, adopted the condition that, for any of the above to constitute a restructuring event for the purposes of a CDS contract, they must result “directly or indirectly from a deterioration in the creditworthiness or financial condition of the reference entity.”

⁷ Rule (2001)[68] discusses the events that led up to the adoption of the 1999 definition of restructuring, as well as other important cases in the history of the marketplace, such as the Consecro restructuring episode in the United States, summarized below, and the National Power demerger case in the United Kingdom, which helped shape the successor provisions outlined in Section 24.1.1.1 of this book.

24.3.1 *A Case in Point: Conseco*

Conseco Inc. is an insurance and financial services company based in the United States. The company was facing a deteriorating financial outlook in the late 1990s and eventually lost access to the commercial paper market, a situation that led it to rely on its back-up lines of bank credit to repay maturing debt. In 2000, even though its business condition was showing some signs of improvement, Conseco's lines of credit were fully utilized and about to become due. Conseco's bankers agreed that it would not be able to repay the maturing bank loans and decided to restructure the loans, extending their maturity while charging a higher interest rate and obtaining some collateralization.

The loan restructuring helped Conseco remain solvent in 2000.⁸ Moreover, given the higher coupon and degree of collateralization on the restructured debt, the affected lenders were thought to have been at least partially compensated for the maturity extension. Yet, according to the 1999 definition of restructuring a qualifying restructuring event had taken place, and CDS contracts written on Conseco were consequently triggered.

The Conseco case helped expose one shortcoming associated with the 1999 definition of restructuring. Unlike an outright default or bankruptcy, where the market prices of the bonds and loans of the reference entity broadly converge regardless of their maturity, restructurings often affect the prices of bonds and loans of the reference entity differently. For instance, in Conseco's case, the restructured loans were trading at a substantially smaller discount from their face value shortly after the triggering of the corresponding CDS contracts than were longer-dated senior unsecured bonds previously issued by Conseco. Yet, both the long-dated bonds and the loans were deliverable obligations under the terms of the contract!

As a result, some protection buyers in the CDS market that had made loans to Conseco were more than compensated for their restructuring-related losses regarding Conseco. They suffered a relatively small marked-to-market loss in the restructured loans, but that loss was more than offset by gains derived from their CDS positions, where they could deliver the cheaper longer-dated bonds, as opposed to the loans, and receive their full face value from their CDS counterparties.

To sum up, the 1999 definition of restructuring had the effect of giving protection buyers a potentially valuable cheapest-to-deliver option in their CDS positions. While protection sellers honored the letter of the CDS contracts written on Conseco and made good on their commitments under the contracts, many wondered whether the Conseco episode was consistent

⁸The company's troubles were not fully resolved by the 2000 restructuring of its bank debt, however. In late 2002, Conseco filed for Chapter 11 bankruptcy protection. It emerged from its Chapter 11 reorganization process in September 2003.

with the spirit of a CDS agreement. Indeed, most market participants point to the Consecro case, as we do in this book, as a major catalyst of the debate that culminated with the adoption of a new set of provisions regarding restructuring in CDS contracts.

24.3.2 Modified Restructuring

The main thrust of the so-called modified restructuring provision of CDS contracts—which was introduced in the standard CDS documentation in an ISDA Restructuring Supplement adopted in 2001 and later incorporated into the 2003 Credit Derivatives Definitions—was to mitigate the cheapest-to-deliver problem that was at the heart of the Consecro case. Thus, without substantially changing the 1999 definition of restructuring as a credit event, the new provisions had mostly the effect of limiting the range of obligations that would become deliverable when a contract is triggered by a restructuring event. The new provisions also disallowed the applicability of the restructuring clause in cases where restructuring is limited to bilateral loans.

The motivation for the first set of main changes associated with modified restructuring—limiting the basket of deliverables in contracts triggered by restructuring—is very straightforward in light of the Consecro case. Indeed, modified restructuring works primarily by capping the maturity of obligations that are eligible to be delivered in the event of restructuring. As for the second main innovation brought about by modified restructuring—the bilateral-loan exclusion—its primary intent was to address potential “moral hazard” problems, whereby a lender may force the restructuring of a loan on which it bought protection and then attempt to gain from “exercising” any remaining cheapest-to-deliver option embedded in a CDS contract.

Remaining provisions of the modified restructuring clause sought to fine tune other issues related to restructuring as a credit event, including the transferability of any deliverable obligations and the treatment of changes in the seniority of an obligation in the reference entity’s capital structure.

24.3.3 A Bifurcated Market

Modified restructuring went a long way towards meeting the interests of both buyers and sellers in the CDS market. Many sellers, especially banks who count on the credit derivatives market to obtain regulatory capital relief, wanted the broadest possible default protection and thus preferred to keep restructuring in the list of allowable credit events. Protection sellers, on the other hand, wanted to minimize the cheapest-to-delivery problem, which, as we discuss below, introduces some tough valuation issues. The modified restructuring clause thus emerged as a compromise between these two positions.

Still, the restructuring debate is most likely not over. Indeed, some degree of bifurcation persists in the marketplace, with some contracts being negotiated with restructuring and some without. In addition, there are some variations in the standard contracts used in North American, Europe, and Asia when it comes to restructuring. For instance, in Europe, contracts typically allow for less stringent maturity limitations than in the United States.

24.4 Valuing the Restructuring Clause

A contract that provides protection, for instance, against restructuring, bankruptcy, and failure-to-pay events ought to cost more than one that provides protection only against the latter two. How much extra should a protection buyer have to pay to have restructuring added to the list of credit events? Drawing an analogy with the vast default risk literature, theory would suggest that the value of the restructuring clause should depend on the probability of a restructuring taking place. Unfortunately, however, there are no well-known models of restructuring probabilities so the analogy between modeling default and restructuring risks does not take us very far in terms of valuing the restructuring clause.

One simple approach to gauging the value of the restructuring clause is to compare premiums actually charged in CDS contracts that include restructuring to those that do not. Based on the limited amount of data generated since the modified structuring clause became part of the standard CDS documentation, some market participants have estimated that premiums that correspond to contracts with the modified restructuring clause are about 5 to 10 percent higher than premiums for contracts without restructuring.

24.4.1 *Implications for Implied Survival Probabilities*

In Chapter 16 we discussed a simple method for inferring risk-neutral probabilities of default from CDS premiums quoted in the marketplace. Suppose now that all that we have are premiums that correspond to contracts that include the modified restructuring clause, but that we are interested in the risk-neutral probabilities of an outright default over various horizons—where by outright default we mean events like bankruptcy and failure to pay. The discussion in the previous section suggests a simple two-step approach to computing those probabilities. First, we reduce the observed premiums by 5 to 10 percent in order to obtain a rough estimate of premiums that would prevail in the absence of the modified restructuring clause. Second, we follow the same method described in Section 16.1.

Table 24.1 shows the results of the two-step approach when applied to AZZ Bank, one of the hypothetical reference entities examined in

TABLE 24.1
CDS-implied Survival Probabilities and Modified Restructuring

Horizon/ Maturity	CDS premium (basis points)	Survival Probabilities		Bias (3) – (4)
		Raw quotes (percent)	Adjusted quotes (percent)	
(1)	(2)	(3)	(4)	(5)
One year	29	99.42	99.48	– 0.06
Two years	39	98.45	98.60	– 0.15
Three years	46	97.26	97.53	– 0.27
Four years	52	95.88	96.29	– 0.40
Five years	57	94.37	94.92	– 0.55

Note. Assumed recovery rate: 50 percent. All probabilities shown are risk-neutral. All other assumptions are as in Table 16.1.

Chapter 16. Column 2 shows the CDS premiums quoted in the market place, which we are now assuming to be for contracts that include the modified restructuring clause. Column 3 shows CDS-implied survival probabilities based on the unadjusted (raw) quotes shown in column 2—these are the same probabilities shown in Table 16.2. Column 4 shows survival probabilities obtained from the two-step approach, where we reduced the premiums shown in column 2 by 10 percent. The results suggest that, at least for highly rated reference entities such as the one examined in this exercise, the survival probabilities based on the adjusted premiums differ very little from those based on the raw market quotes.

The differences between results derived from raw and restructuring-adjusted premiums—shown in the column labeled “bias”—would be bigger for lower-rated entities such as XYZ Corp., the other entity examined in Chapter 16. Still, the maximum absolute value of the bias for XYZ Corp. would amount to only minus 2.75 percentage points. We should also note that it is often the case that CDS contracts written on lower-rated entities tend not to include restructuring in the list of allowable credit events so the “restructuring bias” in derived survival probabilities is often not an issue for these reference entities.

This Page Intentionally Left Blank

25

A Primer on Bank Regulatory Issues

We mentioned in Section 3.2 that bank regulatory considerations spelled out in the 1988 Basel Bank Capital Accord—also known as the Basel I Accord—played an important role in the early days of the credit derivatives markets. In particular, banks’ use of credit derivatives was, at least initially, significantly motivated by the desire to better align the notion of regulatory capital—the share of risk-adjusted assets that bank regulators require to be set aside to support risk-taking activities—with that of economic capital—a prudent bank’s own estimate of the needed capital reserve. For instance, in that chapter we examined the case of a hypothetical bank that made a loan to a corporation and subsequently bought protection from an OECD bank in the CDS market in a contract that referenced that corporation. The protection-buying bank would essentially see its capital charge drop from 8 percent of the total exposure associated with the corporate loan to only 1.6 percent of the exposure, provided the bank could demonstrate to its regulators that the terms of the contract—maturity of the CDS vs. that of the loan, definition of credit events, etc.—effectively granted it appropriate protection against default-related losses on the loans.

The interplay between bank regulatory issues and banks’ usage of credit default swaps and related instruments has changed considerably since the formative days of the credit derivatives market, however. Not only have banks increased their use of credit derivatives for reasons other than just regulatory capital management, but the regulatory environment itself has also evolved. We discussed, in Chapter 3, banks’ other (non-regulatory-driven) uses of credit derivatives—such as applications to risk management and portfolio diversification. We will now take a quick look at the new

regulatory environment facing banks and examine its possible implications for the future development of the credit derivatives market.

Similar to the treatment of documentation considerations in Chapter 24, the coverage of regulatory issues provided in this chapter is of a very broad nature and thus does not constitute advice about what is permissible under current international accords or even under the banking laws of any single country.

25.1 The Basel II Capital Accord

In 1999, member countries of the Basel Committee on Banking Supervision formally started working on a new set of guidelines and standards for their national bank regulators and supervisors.¹ That effort culminated with the New Basel Capital Accord—also known as the Basel II Accord. The Accord was finalized at the Basel Committee level in 2004, and the process now is largely in the hands of signatory countries, which have agreed to start implementing the terms of the new accord by the end of 2006. The process towards implementation will likely include further deliberation at the national level, and, possibly, additional fine-tuning in the language of the accord at the international level. As it was the case with Basel I, the Basel II Accord only provides a framework for the regulation of bank capital. Each national authority then has the discretion to adapt the accord’s stipulations to its own needs and reality.

A main drive behind the genesis of the Basel II Accord has been the need to strengthen the regulatory framework for large banking organizations, especially those that are active in the international markets, and, in the process, devise capital requirements that are more reflective of these organizations’ somewhat unique risk profiles. We say “unique” because, more so than their smaller cousins, large banking organizations have tended to rely on newer instruments and approaches, including credit derivatives, in measuring and managing risk. Many of these instruments either did not exist or were not widely used when Basel I was conceived and thus were not directly mentioned by the framers of that accord. In addition, the new Basel Accord seeks to address some of the limitations of its predecessor,

¹ The Basel Committee on Banking Supervision was established in 1974. Its members include representatives of central banks or other supervisory authorities from Belgium, Canada, France, Germany, Italy, Japan, Luxembourg, the Netherlands, Spain, Sweden, Switzerland, the United Kingdom, and the United States. The committee has no supra-national supervisory authority and its conclusions have no legal force. Its goals are to formulate and propose standards and guidelines to encourage and facilitate the adoption of best practices regarding bank regulation among member countries (see Federal Reserve Board (2003))[27].

such as, as noted in Chapter 3, the lack of granularity in the way the original accord assigned risk weights to different categories of borrowers.

As we saw in Chapter 2, despite the phenomenal growth of the credit derivatives market, as well as the fact that banks are major users and dealers of credit derivatives, these instruments represent only a small portion of the total notional amount of derivatives at banks. Thus, it is both logical and natural that the scope and motivation for revamping the Basel I Accord goes well beyond issues directly related to the emergence of the credit derivatives market. Given the subject matter of this book, however, we shall highlight those aspects of the new accord that are most relevant for credit derivatives, especially credit default swaps. Towards the end of the chapter we provide a list of basic sources for readers interested in delving deeper into bank regulatory issues involving credit derivatives.

Similar to the 1988 Accord, the provisions of the Basel II Accord call for regulatory capital to be determined according to the credit risk associated with the range of debt instruments held by a bank. But the new accord is much more discriminating than its predecessor with regard to the credit quality of a banks' debtors when it comes to determining the appropriate risk weights that go into the calculation of regulatory capital.² The computation of regulatory capital is analogous to that devised under Basel I, in that it involves multiplying the product of the regulatory risk weight and the basic capital requirement of 8 percent by the notional exposure, as discussed in Chapter 3.

The new accord allows the risk weights to be based on either ratings provided by outside credit-rating agencies, such as Standard and Poor's and Moody's—the so-called standardized approach—or on two methods derived from banks' own internal ratings systems—the so-called internal ratings-based approaches—provided, in the case of the latter, those systems have been explicitly approved by the bank regulator. National bank regulators are free to specify which risk weighting approach(es) are applicable to their jurisdiction and under which conditions.

Based on the standardized approach, Table 25.1 shows Basel II risk weights that correspond to sovereign, bank, and corporate (non-bank) borrowers. These weights stand in sharp contrast to the weighting scheme specified in the Basel I Accord—see Table 3.1 in Chapter 3 for a comparison. In particular, whereas the original accord made no distinction among corporate borrowers, and differentiated banks and sovereign borrowers only according to their OECD status, the new accord allows for a closer correspondence between regulatory risk weights and the credit quality of

² Similar to the discussion of Basel I, in Chapter 3, we limit ourselves here to the regulatory treatment of obligations on banks' banking books. The new accord also addresses requirements associated with obligations held on the trading book (see Basel Committee on Bank Regulation (2003)[1]).

TABLE 25.1
Risk Weights Specified in the Basle II Accord for Selected Obligors^a
(percent)

Obligor	Credit Rating					
	AAA to AA–	A+ to A–	BBB+ to BBB–	BB+ to B–	Below B–	Unrated
Sovereign	0	20	50	100	150	100
Bank ^b	20	50	50	100	150	50
Corporate	20	50	100	100	150	100

^aAs specified in the April 2003 version of the Consultative Document on the New Basel Capital Accord (Basel Committee on Banking Supervision, 2003[1]). Weights shown are for the so-called standardized approach to weighting banking book exposures.

^bThe Accord also provides for an alternative risk weighting of claims on banks. The alternative weights are based on the credit assessment of the country in which the bank is incorporated (see Basel Committee on Banking Supervision, 2003[1]).

obligors, a provision that helped narrow the gap that existed in the original accord between regulatory and economic capital. For instance, under Basel I, loans to corporations rated either AAA or B would be subject to the same regulatory risk weight of 100 percent, despite the fact that these corporations are nearly at opposite ends of the credit quality spectrum. In contrast, the Basel II Accord assigns risk weights of 20 and 100 percent to these obligors, respectively.

25.2 Basel II Risk Weights and Credit Derivatives

With regard to credit default swaps, the new accord generally allows protection-buying banks to continue to substitute the risk weight of the protection seller for that of the reference entity in instances where the bank is long an obligation of that entity and where the bank can demonstrate that effective protection has been bought. Thus, banks continue to have an incentive to seek regulatory capital relief in the credit derivatives market, but that incentive is arguably not as strong as under the Basel I provisions, given the closer alignment between Basel II risk weights and true economic risk.

One might be tempted to conclude that the closer alignment between regulatory and economic capital might lead banks to become less important users of credit derivatives. Yet, as we saw in Chapter 3, even before the

adoption of Basel II, banks' use of such derivatives had been increasingly driven by nonregulatory considerations, such as the need to manage credit risk exposures in response to greater scrutiny by investors. Indeed, given its closer correspondence between regulatory and economic capital, Basel II may well end up reinforcing banks' incentives to use credit derivatives as a risk management tool.

Another implication of the greater granularity of the risk weighting scheme of the Basel II Accord regards the attractiveness of non-banks as potential sellers of protection to banks. Under Basel I, banks would be granted no capital relief, say, for buying protection from an AA-rated corporation in a credit default swap intended to hedge exposure to a BBB-rated corporate reference entity. They would simply be replacing exposure to one 100-percent risk-weighted obligor—the BBB-rated entity—with another—the AA-rated protection seller. Under Basel II, however, this same contract would result, assuming other regulatory conditions are met, in a decrease in the corresponding regulatory risk weight from 100 percent to 20 percent, a development that bodes well for the greater participation of highly-rated nonbank entities as protection sellers in credit derivatives. (A similar argument applies to highly rated non-OECD banks.)

On the whole, Basel II has multiple, and sometimes offsetting, implications for future developments in the credit derivatives market. On one hand, banks' incentives to use of credit derivatives purely as a regulatory capital management tool may wane further. On the other hand, the new accord is clearly intended to reinforce banks' incentives to monitor closely and manage their credit risk exposures, which could be accomplished with greater use of instruments such as credit derivatives. Outside the banking sector, the risk weighting provisions of the new accord may well promote greater participation of other market players, such as highly-rated corporations, as sellers of protection, which may help facilitate the transfer of credit risk from banks to other institutions.

25.3 Suggestions for Further Reading

A complete discussion of the full regulatory treatment of credit derivatives under the Basel II Accord would go well beyond the scope of this book. Indeed, we have omitted the discussion of several topics, such as the treatment of basket swaps and synthetic securitization. In addition, the new accord covers situations where the default protection bought in a credit derivative contract only partially offsets a bank's exposure to the reference entity, such as is the case when the maturity of a loan to the reference entity is longer than that of the corresponding credit default swap. Readers interested in these and other related topics are encouraged to consult

the text of the accord itself, which can be downloaded from the Bank for International Settlements' website (www.bis.org).

While the focus of this chapter has been on international bank regulatory standards as they relate to credit derivatives, we should once again remind the reader that the Basel II Accord, like its predecessor, only provides a general framework for national bank regulators in signatory countries. The implementation of the accord's provisions varies from country to country according to local concerns and market characteristics. Moreover, credit derivatives have regulatory implications that go beyond the setting of capital requirements, and these too vary across national borders. The following is a partial list of national bank regulator websites, some of which provide further on-line information on issues related to credit derivatives in their respective jurisdictions:

- Australia: Australian Prudential Regulation Authority (www.apra.gov.au);
- Canada: Office of the Superintendent of Financial Institutions (www.osfi-bsif.gc.ca);
- France: Commission Bancaire (www.commission-bancaire.org);
- Germany: Bundesanstalt für Finanzdienstleistungsaufsicht (www.bafin.de);
- Ireland: Central Bank of Ireland and Financial Services Authority (www.centralbank.ie);
- Italy: Banca D'Italia (www.bancaditalia.it);
- Japan: Bank of Japan (www.boj.or.jp);
- Sweden: Swedish Financial Supervisory Authority (www.fi.se);
- United Kingdom: Financial Services Authority (www.fsa.gov.uk);
- United States: Board of Governors of the Federal Reserve System (www.federalreserve.gov), Office of the Comptroller of the Currency (www.occ.treas.gov), Federal Deposit Insurance Corporation (www.fdic.gov), and the Office of Thrift Supervision (www.ots.treas.gov).

Appendix A

Basic Concepts from Bond Math

We briefly discuss in this appendix some of the basic bond and term-structure concepts featured in the book. Hull (2003)[41] provides further details on these concepts at the introductory level.

A.1 Zero-coupon Bonds

A zero-coupon bond, as its name suggests, is a bond that makes no coupon payments. Its yield comes from the fact that it is sold at a discount from its face value. For instance, a one-year zero-coupon bond initially sold for 95 cents on the dollar can be said to have an annual yield of 5.26 percent, or $(1 - .95)/.95$. Indeed, it is straightforward to see that, for this one-year bond:

$$1 = .95(1 + 0.0526)$$

i.e., the annual yield is the annual rate at which the price of the bond will have to grow so that it will converge to its face value at the maturity date of the bond. This is also called the yield-to-maturity of this bond.

The yield-to-maturity, Y_n , of a n -year zero-coupon bond is likewise defined as the constant annual rate at which the bond's price will have to grow so that the bond will be valued at par at maturity:

$$\text{face value} = (1 + Y_n)^n \text{ price} \tag{A.1}$$

A.2 Compounding

The zero-coupon yields discussed above assume annual compounding. If we were to compound twice per annum, the yield-to-maturity of the one-year zero-coupon bond considered above would be:

$$Y_1^{(2)} = 2[(1/.95)^{1/2} - 1] = 0.0520$$

or 5.20 percent, where the above equation comes from the fact that $Y_1^{(2)}$ is such that

$$1 = .95 \left(1 + \frac{Y_1^{(2)}}{2} \right)^2$$

For compounding j times per annum, the one-year yield is:

$$Y_1^{(j)} = j[(1/.95)^{1/j} - 1]$$

and, as j tends to infinity, we obtain the following expression for the yield to maturity:

$$Y_1^{(\infty)} = -\log(.95) = 0.05129$$

which is the continuously compounded yield-to-maturity for the one-year bond.

Generalizing for n -year bonds, the yield-to-maturity with compounding j times per year is

$$Y_n^{(j)} = j \left[\left(\frac{\text{face value}}{\text{price}} \right)^{1/(jn)} - 1 \right] \quad (\text{A.2})$$

and the continuously compounded yield is:

$$Y_n^{(\infty)} = -\log \left(\frac{\text{price}}{\text{face value}} \right) / n \quad (\text{A.3})$$

The last two equations trivially allow us to write the price of a zero-coupon bond in terms of its yield-to-maturity. With compounding j times per annum:

$$\text{price of } n\text{-year bond} = \frac{\text{face value}}{\left(1 + \frac{Y_n^{(j)}}{j} \right)^{jn}} \quad (\text{A.4})$$

and, with continuous compounding:

$$\text{price of } n\text{-year bond} = \left[e^{-nY_n^{(\infty)}} \right] [\text{face value}] \quad (\text{A.5})$$

A.3 Zero-coupon Bond Prices as Discount Factors

Zero-coupon bond prices can be thought of as discount factors that can be applied to future payments in order to express them in today's dollars. For instance, the present value of \$1 to be received in one-year's time is, by definition, simply today's price of a zero-coupon bond with a face value of \$1.

If there is no chance that a given future payment will be missed (no default risk) then the receiver of that payment should discount it on the basis of the price of a zero-coupon bond that involves no credit risk. By the same token, future payments that are subject to default risk should be discounted with prices of zero-coupon bonds that are subject to comparable default risk. This is essentially what we do in equation (4.2) in the text, where we discounted the future cash flows of a corporate security using prices of zero-coupon bonds derived from the security issuer's yield curve.

A.4 Coupon-paying Bonds

Consider a n -year bond that pays a fixed annual coupon C at dates $T_1, T_2, \dots, T_n$. In addition, the bond pays its face value F at its maturity date T_n . The bond has no default risk. Given the discount-factor interpretation of zero-coupon bond prices, we can write:

$$V(t, T_n) = \sum_{h=1}^n Z(t, T_h)C + Z(t, T_n)F \quad (\text{A.6})$$

where $V(t, T_n)$ is the price of the coupon-paying bond, and $Z(t, T_h)$ is the price of a riskless zero-coupon bond that pays \$1 at date T_h .

If the coupon-paying bond has some default risk, the discount factors should correspond to the prices of zero-coupon bonds issued by the same entity issuing the coupon-paying bond. Assuming that the bond has no recovery value upon default, we can write:

$$V_0^d + -(t, T_n) = \sum_{h=1}^n Z_0^d(t, T_h)C + Z_0^d(t, T_n)F \quad (\text{A.7})$$

TABLE A.1
Par Yield Curve for a Hypothetical Issuer^a

Maturity (years)	Yield to Maturity (percent)
1	2.0
2	2.2
3	2.5
4	2.7

^aYields shown are compounded annually and coupons are paid once a year.

where $V^d(t, T_n)$ is the price of the defaultable coupon-paying bond, and $Z_0^d(t, T_h)$ is the price of a defaultable zero-coupon bond that promises to pay \$1 at date T_h . Thus, given the zero-coupon curve for a given issuer, we can price other debt instruments of that same issuer, as we do, for instance, in Chapter 11 when valuing principal-protected notes.

A.5 Inferring Zero-coupon Yields from the Coupon Curve

Zero-coupon bonds with long maturities are rarely issued in practice. Instead, actual longer-dated bonds traded in the marketplace are of the coupon-paying variety. Using (A.7), however, we can back out the zero-coupon yields and prices embedded in an issuer’s coupon curve. To see how this can be done, we consider a simple numerical example for a hypothetical issuer. Table A.1 summarizes the main inputs. The table shows the “par” yield curve for the issuer, where by par we mean that the prices of the bonds shown are all equal to their face values, or, equivalently, that the yield-to-maturity of each bond is equal to its respective coupon rate. We show four bonds, with maturities ranging from one to four years, each with a face value of \$1 and a zero recovery rate.¹

The time- t (today’s) price of the one-year zero-coupon bond associated with this issuer, which would be used to discount payments to be made in

¹ In a more realistic setting, the issuer may not have outstanding bonds along the entire maturity spectrum, and one may need to resort to interpolation and smoothing methods. See, for instance, James and Webber (2000)[44] for a discussion of these methods.

one-year's time, is simply the price of the first bond shown. Thus, given

$$1 = Z_0^d(t, t+1)(1.02)$$

we have

$$Z_0^d(t, t+1) = 1/1.02 = 0.9804$$

To derive the price of the two-year zero-coupon bond, recall that the price of the two-year coupon-paying bond can be written as

$$1 = Z_0^d(t, t+1)0.022 + Z_0^d(t, t+2)(1.022)$$

Thus

$$Z_0^d(t, t+2) = \frac{1 - Z_0^d(t, t+1)0.022}{1.022}$$

where the value of $Z_0^d(t, t+1)$ was derived in the prior step.

For the three-year bond:

$$1 = Z_0^d(t, t+1)0.025 + Z_0^d(t, t+2)0.025 + Z_0^d(t, t+3)(1.025)$$

and thus

$$Z_0^d(t, t+3) = \frac{1 - \sum_{h=1}^2 Z_0^d(t, t+h)0.025}{1.025}$$

Generalizing, the price of a n -year zero-coupon bond can be written as a function of the prices of shorter-dated zeros and the coupon of a n -year coupon-paying bond sold by the same issuer:

$$Z_0^d(t, t+n) = \frac{1 - \sum_{h=1}^{n-1} Z_0^d(t, t+h)C_n}{1 + C_n} \quad (\text{A.8})$$

where C_n is the coupon payment of the n -year bond.

A.6 Forward Rates

A forward rate $F(t, T_1, T_2)$ is the annual interest rate agreed upon in the marketplace at today's date (time t) for lending and borrowing during a future period $[T_1, T_2]$, but involving no net cash outlay at time t . One can obtain the fair value of $F(t, T_1, T_2)$ entirely in terms of time- t prices of zero-coupon bonds. Consider the following simultaneous transactions at time t :

- sell one zero-coupon bond that matures at time T_1 ;
- buy $\frac{Z(t, T_1)}{Z(t, T_2)}$ zero-coupon bonds that mature at time T_2 .

Assume that the face value of the bonds is \$1. Note that this transaction results in no net cash flow at time t . Subsequent cash flows are as follows:

- Time T_1 : Pay \$1 to the buyer of the T_1 -bond sold at t ;
- Time T_2 : Receive $\$ \left(\frac{Z(t, T_1)}{Z(t, T_2)} \right)$ from the issuer of the T_2 -bonds bought at time t .

Thus we can see that the time- t transaction involves no initial cost and is akin to contracting to lend \$1 at time T_1 in exchange for receiving $\$ \left(\frac{Z(t, T_1)}{Z(t, T_2)} \right)$ at time T_2 . The implicit interest payment in this forward loan is $\frac{Z(t, T_1)}{Z(t, T_2)} - 1$. As already noted, the annualized interest rate implicit in this forward loan, which we denote below as $F(t, T_1, T_2)$, is defined as the time- t forward rate for the future time period $[T_1, T_2]$. For instance, in Chapter 4, we examine an example where the length of the future period is 6 months, $T_2 - T_1 = .5$ year, which corresponds to the following expression for $F(t, T_1, T_2)$:

$$F(t, T_1, T_2) = \frac{1}{T_2 - T_1} \left(\frac{Z(t, T_1)}{Z(t, T_2)} - 1 \right) \quad (\text{A.9})$$

Using equation (A.5), the time- t continuously compounded forward rate for the future period $[T_1, T_2]$ is the rate $f(t, T_1, T_2)$ that solves the following equation:

$$e^{f(t, T_1, T_2)(T_2 - T_1)} = \frac{Z(t, T_1)}{Z(t, T_2)}$$

Thus:

$$f(t, T_1, T_2) = \frac{\log(Z(t, T_1)) - \log(Z(t, T_2))}{T_2 - T_1} \quad (\text{A.10})$$

A.7 Forward Interest Rates and Bond Prices

The rate $f(t, T)$ such that

$$f(t, T) \equiv \lim_{\Delta T \rightarrow 0} \frac{\log(Z(t, T)) - \log(Z(t, T + \Delta T))}{\Delta T} = - \frac{\partial \log(Z(t, T))}{\partial T} \quad (\text{A.11})$$

is called the instantaneous continuously compounded forward rate. It can be thought of as the time- t interest rate that applies to the future period $[T, T + dt]$, where dt is an infinitesimal time increment.

Integrating both sides of equation (A.11), we obtain:

$$Z(t, T) = e^{-\int_t^T f(t, v) dv} \quad (\text{A.12})$$

Given the definition of the continuously compounded yield to maturity as the rate $R(t, T)$ such that $Z(t, T) = e^{-R(t, T)(T-t)}$:

$$R(t, T) = \frac{\int_t^T f(t, v) dv}{T - t} \quad (\text{A.13})$$

which says that the time- t yield-to-maturity on a zero-coupon bond can be thought of as the average of all time- t instantaneous forward rates that span the remaining life of the bond.

This Page Intentionally Left Blank

Appendix B

Basic Concepts from Statistics

This appendix provides a brief review of some key statistical concepts used in the text. We have skipped over some technical details and mathematical proofs involving these concepts. Bain and Engelhardt (1987)[5], Hogg and Tanis (1983)[39], and Grimmett and Stirzaker (1998)[36] provide a more complete coverage of the topics discussed herein.

B.1 Cumulative Distribution Function

The cumulative distribution function (c.d.f.) of a random variable X is defined as

$$F(x) = \text{Prob}[X \leq x] \tag{B.1}$$

$F(x)$ is also commonly called the distribution function of X . We highlight two of its basic properties below:

- Because $F(x)$ is a probability: $0 \leq F(x) \leq 1$;
- $F(x)$ is a non-decreasing function of x , i.e., if $x_1 < x_2$ then if $X \leq x_1$ we also have $X \leq x_2$, and thus:

$$F(x_1) = \text{Prob}[X \leq x_1] \leq \text{Prob}[X \leq x_2] = F(x_2)$$

B.2 Probability Function

A random variable X is said to be discrete if it can only take on one of the discrete values $x_1, x_2, x_3, \dots$. If X is a discrete random variable:

$$F(x) = \sum_{x_i \leq x} p(x_i) \quad (\text{B.2})$$

where the sum is computed over all x_i such that $x_i \leq x$, and where $p(x)$ is called the probability function of X .

The function $p(x)$ assigns a probability to each one of the possible values of X :

$$p(x) = \text{Prob}[X = x] \quad x = x_1, x_2, \dots \quad (\text{B.3})$$

Note that, for $p(x)$ to be a probability function, it must satisfy the following two properties:

$$\begin{aligned} p(x_i) &\geq 0 && \text{for all } x_i \\ \sum_{\text{all } x_i} p(x_i) &= 1 \end{aligned}$$

B.3 Probability Density Function

Let X be a continuous random variable, in that it can take *any* value, for instance, in the interval $[x_1, x_n]$. The c.d.f. of X can be written as:

$$F(x) = \int_{-\infty}^x f(s) ds \quad (\text{B.4})$$

where $f(x)$ is called the probability density function (p.d.f.) of X .

Similar to the probability function in the discrete case, a p.d.f. must satisfy the following conditions

$$\begin{aligned} f(x) &\geq 0 && \text{for all real } x \\ \int_{-\infty}^{\infty} f(x) dx &= 1 \end{aligned}$$

In the light of (B.4), it can be shown that:

$$\text{Prob}[a \leq X \leq b] = F(b) - F(a) = \int_a^b f(x) dx \quad (\text{B.5})$$

Because $F(x)$ is a continuous function:

$$\text{Prob}[X = a] = 0 \quad (\text{B.6})$$

which is consistent with the basic result from classic calculus that $\int_a^a f(x)dx = 0$. Thus:

$$\text{Prob}[a < X < b] = \text{Prob}[a < X \leq b] = \text{Prob}[a \leq X < b] = \text{Prob}[a \leq X \leq b]$$

We also highlight the following useful relationship:

$$\text{Prob}[x < X \leq x + dx] = F(x + dx) - F(x) \approx f(x)dx \quad (\text{B.7})$$

where the approximation error is negligible for sufficiently small values of dx . Indeed, given (B.4), from classic calculus we know that, for values of x for which $\frac{\partial F(x)}{\partial x}$ exists:

$$\lim_{dx \rightarrow 0} \frac{F(x + ds) - F(x)}{dx} = \frac{\partial F(x)}{\partial x} = f(x) \quad (\text{B.8})$$

B.4 Expected Value and Variance

The expected value of a discrete random variable is simply the probability-weighted average of all of its possible values. If the possible values of X are $x_1, x_2, \dots, x_n$, its expected value, $E[X]$, is given by

$$E[X] = \sum_{i=1}^n p(x_i)x_i \quad (\text{B.9})$$

where $p(\cdot)$ is the probability function of X .

The variance of X is likewise defined as

$$V[X] = \sum_{i=1}^n p(x_i)(x_i - E[X])^2 \quad (\text{B.10})$$

If X is a continuous random variable we have

$$E[X] = \int_{-\infty}^{\infty} sf(s)ds \quad (\text{B.11})$$

and

$$V[X] = \int_{-\infty}^{\infty} (s - E[X])^2 f(s)ds \quad (\text{B.12})$$

Note. The square root of the variance of X is called the standard deviation of X .

B.5 Bernoulli Trials and the Bernoulli Distribution

Throughout this book, we often deal with situations where a company can be in only one of two possible states at some given future date, default or survival, and where there are probabilities associated with each state. Statistically, one way to characterize such situations is through the concept of Bernoulli trials.

A Bernoulli trial is a random experiment that can result in only one of two possible outcomes—e.g., a given company will either default or survive. A sequence of independent Bernoulli trials is one where the probabilities associated with the two outcomes are the same from trial to trial.

Let X be a random variable associated with a Bernoulli trial. For instance:

$X = 1$ if the company defaults and $X = 0$ if it survives

If the probability of default is denoted as ω , the probability function of X can be written as

$$p(x) = \omega^x(1 - \omega)^{1-x} \quad x = 0, 1 \quad (\text{B.13})$$

and one can say that X has a Bernoulli distribution.

The expected value and variance of X are:

$$\begin{aligned} E[X] &= \sum_{x=0}^1 x\omega^x(1 - \omega)^{1-x} = (1)\omega + (0)(1 - \omega) = \omega \\ V[X] &= \sum_{x=0}^1 (x - \omega)^2\omega^x(1 - \omega)^{1-x} = \omega(1 - \omega) \end{aligned}$$

B.6 The Binomial Distribution

Consider a sequence of n independent Bernoulli trials. For instance, given n corporate borrowers, each trial may involve either the default or survival of an individual borrower, where defaults among the n borrowers are mutually independent. Let the default probability for each borrower be denoted as ω .

Let Y be the random variable that represents the number of defaults among the n borrowers over a given period of time. The probability function of Y is

$$b(y; n, \omega) = \frac{n!}{y!(n-y)!} \omega^y (1-\omega)^{n-y} \quad (\text{B.14})$$

where y denotes the possible values of Y — $y = 0, 1, 2, \dots, n$ —and Y is said to be binomially distributed.

The c.d.f. of the binomial distribution is

$$B(y; n, \omega) \equiv \text{Prob}[Y \leq y] = \sum_{s=0}^y b(s; n, \omega) \quad y = 0, 1, \dots, n \quad (\text{B.15})$$

which, continuing with our example, is the probability that at most y companies will default over a given time horizon.

In Part IV we use the results just derived to examine expected default-related losses in an equally weighted homogeneous portfolio where defaults among the issuers represented in the portfolio are mutually independent. With default independence the question of how many issuers are likely to default or survive reduces to a sequence of independent Bernoulli trials, in which case the binomial distribution applies. Without default independence—for instance, a default by one company changes the default probabilities of the others—we cannot directly appeal to results based on the binomial distribution.

B.7 The Poisson and Exponential Distributions

Most reduced-form models make use of the Poisson distribution to characterize the “arrival” process of defaults over time for a given borrower.¹ For this borrower, let X be the discrete random variable that corresponds to the number of defaults arriving (occurring) over a given continuous time interval. We assume that defaults occur randomly at the mean rate of λ per year, with $\lambda > 0$.

From a credit risk modeling perspective, in most cases all that we care about is the first occurrence of default. Nonetheless, in some applications—such as when modeling defaults and corporate reorganizations—one may also be interested in the notion of second default, third default, etc.

¹ We will measure time in terms of non-negative real numbers, with 0 denoting “the beginning of time” and time t representing the present time. The unit of measurement will be years so that, for instance, $t = 1.25$ means that we are one and a quarter years away from the beginning of time.

If X is Poisson distributed, the following conditions must be satisfied—see, for instance, Hogg and Tanis (1983)[39]:

1. the numbers of defaults occurring in nonoverlapping time intervals are independent;
2. the probability of exactly one default occurring during a short time interval of length s is approximately λs ;
3. the probability of more than one default during a sufficiently short time period is essentially zero.

Assuming the above conditions are met, the probability function of X can be shown to be:

$$\begin{aligned} p(x; 1) &\equiv \text{Prob}[x \text{ defaults in a one-year interval}] \\ &= \frac{\lambda^x e^{-\lambda}}{x!} \quad \text{for } x = 0, 1, 2, \dots \end{aligned} \quad (\text{B.16})$$

which, indeed, has the property, stated above, that the expected number of defaults during a time interval of one year is λ :

$$E[X] = \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} = \lambda \quad (\text{B.17})$$

The variance of X can also be shown to be λ .

More generally, let us change the reference time interval to be s years—where $s > 0$ —so that, for instance, $s = \frac{1}{12}$ indicates a time interval of about a month. If the mean arrival rate of defaults for this borrower in a one-year interval is λ , then the mean arrival of defaults in a s -year interval is λs . As a result, we can write:

$$\begin{aligned} p(x; s) &\equiv \text{Prob}[x \text{ defaults in a } s\text{-year interval}] \\ &= \frac{(\lambda s)^x e^{-\lambda s}}{x!} \quad \text{for } x = 0, 1, 2, \dots \end{aligned} \quad (\text{B.18})$$

provided, of course, conditions 1 through 3 are satisfied.

Thus, using time 0 as our vantage point, the unconditional probability that no default will take place during a time interval of length s years is:

$$\begin{aligned} p(0; s) &\equiv \text{Prob}[0 \text{ defaults in a } s\text{-year interval}] \\ &= e^{-\lambda s} \end{aligned} \quad (\text{B.19})$$

Similarly, the unconditional probability of exactly one default occurring during a time interval of length s years is

$$p(1; s) = \lambda s e^{-\lambda s}$$

which, consistent with condition 2, is approximately λs for small s .

As discussed in Chapter 17, one concept that is of great interest in the pricing of credit derivatives is the expected time of first default by a given reference entity. Let τ denote the time of first default, which can be thought of as a continuous random variable with c.d.f. $\bar{G}(s)$. (Alternatively, τ can also be thought of as the “waiting time” until the first default, as seen from time 0.) Clearly, for $s < 0$, $\bar{G}(s) = 0$, given that a waiting time cannot be negative. For $s \geq 0$ we can write

$$\bar{G}(s) \equiv \text{Prob}[\tau \leq s] = 1 - \text{Prob}[\tau > s]$$

but note that $\text{Prob}[\tau > s]$ is simply the probability, as seen at time 0, that there will be no default by time s . This probability is given by equation (B.19). Thus, we can write:

$$\bar{G}(s) = 1 - e^{-\lambda s} \quad (\text{B.20})$$

which can be thought of as the unconditional probability of a first default by time s .

The unconditional p.d.f. of τ , defined as $\frac{\partial \bar{G}(s)}{\partial s}$, can be written as:

$$\bar{g}(s) = \lambda e^{-\lambda s} \quad (\text{B.21})$$

Readers with some familiarity with statistics will recognize $\bar{g}(s)$ as the p.d.f. of an exponentially distributed random variable with mean $\frac{1}{\lambda}$ and variance $\frac{1}{\lambda^2}$. Thus, the time of first default is exponentially distributed when defaults occur according to a Poisson process.

The conditional probability of first default by time s , given no default through time t , for $s \geq t$, can be written as

$$\begin{aligned} G_t(s) &\equiv \text{Prob}[\tau \leq s | \tau > t] \\ &= 1 - \text{Prob}[\tau > s | \tau > t] \\ &= 1 - \frac{\text{Prob}[\tau > s]}{\text{Prob}[\tau > t]} \\ &= 1 - e^{-\lambda(s-t)} \end{aligned} \quad (\text{B.22})$$

where, to arrive at (B.22), we used the Bayes rule and equation (B.19).

The conditional p.d.f. of τ , given no default through time t , is the function $g_t(s)$ such that

$$g_t(s) = e^{-\lambda(s-t)}\lambda \quad (\text{B.23})$$

which is simply $\frac{\partial G_t(s)}{\partial s}$.

Armed with (B.23), we can, for instance, compute the expected time of the first default, conditional on no default through time t :

$$E[\tau|\tau > t] = \int_t^\infty s g_t(s) ds \quad (\text{B.24})$$

which can be shown to be equal to $t + \frac{1}{\lambda}$.

B.8 The Normal Distribution

If X is a continuous random variable that is normally distributed with mean μ and variance σ^2 , its p.d.f. is

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-[(x-\mu)/\sigma]^2/2} \quad (\text{B.25})$$

and the normal c.d.f. is accordingly given by

$$F(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} e^{-[(s-\mu)/\sigma]^2/2} ds \quad (\text{B.26})$$

If we define $Y \equiv \frac{X-\mu}{\sigma}$, then it can be shown that Y is normally distributed with a mean of zero and a variance of one. Y is commonly called the standardized value of X , and its density is referred to as the standard normal p.d.f.

$$n(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \quad (\text{B.27})$$

The standard normal c.d.f. is

$$N(y) = \int_{-\infty}^y \frac{1}{\sqrt{2\pi}} e^{-s^2/2} ds \quad (\text{B.28})$$

which is used extensively in this book, especially in discussion of structural credit risk models in Part III and in the treatment of portfolio credit risk issues in Part IV.

A key result regarding the normal distribution is that

$$F(x; \mu, \sigma) = N\left(\frac{x - \mu}{\sigma}\right) = N(y) \quad (\text{B.29})$$

so one can rely exclusively on $N(\cdot)$ when handling normally distributed variables. This is especially convenient because there is no analytical formula for the normal c.d.f., and one often has to rely on tabulated values. In addition, most statistical and mathematical software have embedded functions to generate values of $N(\cdot)$ for any given y . Alternatively, Hull (2003)[41] provides a simple approximation that is very accurate.

A useful property of the standard normal distribution, and one that we rely upon in Part IV, is that of symmetry. By symmetry we mean that the standard normal p.d.f. is such that for any real number y :

$$n(y) = n(-y) \quad (\text{B.30})$$

which can be shown mathematically, or verified visually if one recalls that $n(\cdot)$ has a bell-shape and is perfectly symmetric around its mean of zero. It can also be shown that, again for any real number y :

$$N(-y) = 1 - N(y) \quad (\text{B.31})$$

a result that we use in the derivation of the loss distribution of a large homogeneous portfolio in Chapter 19.

B.9 The Lognormal Distribution

If X is normally distributed, with mean μ and variance σ^2 , then $Y \equiv e^X$ is lognormally distributed. The p.d.f. of Y is

$$f(y) = \frac{1}{\sqrt{2\pi\sigma y}} e^{-(\log(y) - \mu)^2 / (2\sigma^2)} \quad \text{for } y > 0 \quad (\text{B.32})$$

The distribution of Y is called lognormal because, given $Y = e^X$ and the fact that X is normally distributed, $\log(Y)$ is itself normally distributed with mean μ and variance σ^2 .

The lognormal c.d.f. can be expressed in terms of the normal c.d.f. In particular, given that $\log(Y)$ is an increasing function of Y , it can be shown that

$$F(y) \equiv \text{Prob}[Y \leq y] = \text{Prob}[\log(Y) \leq \log(y)]$$

but the term on the right-hand side of the above equation can be rewritten as $\text{Prob}[X \leq \log(y)]$, and thus:

$$F(y) = N\left(\frac{\log(y) - \mu}{\sigma}\right) \quad (\text{B.33})$$

The mean and variance of Y are:

$$E[Y] = e^{\mu + \sigma^2/2} \quad (\text{B.34})$$

$$V[Y] = e^{2(\mu + \sigma^2)} - e^{2\mu + \sigma^2} \quad (\text{B.35})$$

We use the lognormal distribution in Chapters 17 and 18 when discussing structural models of credit risk and in the valuation of credit options, respectively. Lognormality of the value of individual firms is also an important assumption in the basic portfolio credit risk model discussed in Part IV.

B.10 Joint Probability Distributions

The joint probability function of the discrete random variables $X_1, X_2, \dots, X_n$ is defined as the function $f(\cdot)$ such that

$$f(x_1, x_2, \dots, x_n) = \text{Prob}[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n] \quad (\text{B.36})$$

for all possible values $x_1, x_2, \dots, x_n$ of $X_1, X_2, \dots, X_n$. In words, $f(x_1, x_2, \dots, x_n)$ is the probability that $X_1 = x_1$ and $X_2 = x_2, \dots$, and $X_n = x_n$.

The joint c.d.f. of these X_i s is the function $F(\cdot)$ such that

$$F(x_1, x_2, \dots, x_n) = \text{Prob}[X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n] \quad (\text{B.37})$$

i.e., $F(x_1, \dots, x_n)$ is the probability that $X_1 \leq x_1$, and $X_2 \leq x_2, \dots$, and $X_n \leq x_n$.

The joint p.d.f. of the continuous random variables $Y_1, Y_2, \dots, Y_n$ is the function $g(\cdot)$ such that the joint c.d.f. $G(\cdot)$ can be written as

$$G(y_1, y_2, \dots, y_n) = \int_{-\infty}^{y_n} \dots \int_{-\infty}^{y_2} \int_{-\infty}^{y_1} g(s_1, s_2, \dots, s_n) ds_1 ds_2 \dots ds_n \quad (\text{B.38})$$

for all possible values of $y_1, y_2, \dots, y_n$.

B.11 Independence

The random variables $Y_1, Y_2, \dots, Y_n$ are independent if and only if one of the following holds:

$$G(y_1, y_2, \dots, y_n) = G_1(y_1)G_2(y_2), \dots, G_n(y_n) \quad (\text{B.39})$$

$$g(y_1, y_2, \dots, y_n) = g_1(y_1)g_2(y_2), \dots, g_n(y_n) \quad (\text{B.40})$$

where $G_i(y_i)$ and $g_i(y_i)$ are the c.d.f. and p.d.f., respectively, of Y_i . (This definition applies to both discrete and continuous random variables.)

A closely related concept is that of serial independence. For instance, if R is a random variable that changes its value over time, R is serially independent if R_t , the value of R at time t , is independent of R_s , its value at time s , for s different from t .

Note: For normally distributed variables, events that are uncorrelated are also independent and vice versa. For such variables, the terms uncorrelation and independence can generally be used interchangeably.

B.12 The Bivariate Normal Distribution

If X_1 and X_2 are two normally distributed random variables with means μ_1 and μ_2 and variances σ_1 and σ_2 , respectively, and ρ is the correlation coefficient of X_1 and X_2 , then the joint distribution of X_1 and X_2 is bivariate normal.

The bivariate normal p.d.f. is

$$n_2(x_1, x_2, \rho) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \times e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]} \quad (\text{B.41})$$

and the bivariate normal c.d.f. has the usual definition

$$N_2(x_1, x_2, \rho) = \int_{-\infty}^{x_2} \int_{-\infty}^{x_1} n_2(s_1, s_2, \rho) ds_1 ds_2 \quad (\text{B.42})$$

Similar to the univariate normal, there is no analytical formula for the bivariate c.d.f. Hull (2003)[41] describes a numerical approximation that is reasonably accurate.

We use the bivariate normal distribution in Chapter 19, in the treatment of default correlation, in Chapter 21, when discussing premiums on portfolio default swaps, and in Chapter 23, in the valuation of credit default swaps that are subject to counterparty credit risk.

This Page Intentionally Left Blank

Bibliography

- [1] BASEL COMMITTEE ON BANKING SUPERVISION. The New Basel Capital Accord, Consultative Document. April 2003.
- [2] A. ARVANITIS AND J. GREGORY. *Credit: The Complete Guide to Pricing, Hedging and Risk Management*. Risk Books, 2001.
- [3] A. ARVANITIS, J. GREGORY, AND J. LAURENT. Building Models for Credit Spreads. *Journal of Derivatives*, 1:27–43, 1999.
- [4] BRITISH BANKERS ASSOCIATION. *Credit Derivatives Report 2000/2002*. British Bankers Association, 2002.
- [5] L. BAIN AND M. ENGELHARDT. *Introduction to Probability and Mathematical Statistics*. PSW, 1987.
- [6] M. BAXTER AND A. RENNIE. *Financial Calculus: An Introduction to Derivative Pricing*. Cambridge University Press, 2001.
- [7] T. BJORK. *Arbitrage Theory in Continuous Time*. Oxford University Press, 1998.
- [8] F. BLACK. The Pricing of Commodity Futures Contracts. *Journal of Financial Economics*, 3:167–79, 1976.
- [9] F. BLACK AND M. SCHOLES. The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81:637–54, 1973.

- [10] R. BLACK AND J. COX. Valuing Corporate Securities: Some Effects of Bond Indenture Provisions. *Journal of Finance*, 31:351–67, 1976.
- [11] A. BOMFIM. Credit Derivatives and their Potential to Synthesize Riskless Assets. *Journal of Fixed Income*, December: 6–16, 2002.
- [12] A. CIFUENTES AND G. O’CONNOR. The binomial expansion method applied to cbo/clo analysis. Technical report, Moody’s Investors Service, December 1996.
- [13] A. CIFUENTES AND C. WILCOX. The double binomial method and its applications to a special case of cbo structures. Technical report, Moody’s Investors Service, March 1998.
- [14] P. COLLIN-DUFRESNE, R. GOLDSTEIN, AND J. MARTIN. The Determinants of Credit Spread Changes. *Journal of Finance*, 56:2177–207, 2001.
- [15] J. COX, J. INGERSOLL, AND S. ROSS. A Theory of the Term Structure of Interest Rates. *Econometrica*, 53:385–407, 1985.
- [16] P. CROSBIE. Modeling Default Risk. KMV Corporation, 2002.
- [17] M. CROUHY, D. GALAI, AND R. MARK. A Comparative Analysis of Current Credit Risk Models. *Journal of Banking and Finance*, 24:59–117, 2000.
- [18] S. DAS. *Credit Derivatives and Credit-Linked Notes*. John Wiley & Sons, second edition, 2000.
- [19] S. DAS AND P. TUFANO. Pricing Credit Sensitive Debt when Interest Rates, Credit Ratings and Credit Spreads are Stochastic. *Journal of Financial Engineering*, 5:161–98, 1996.
- [20] G. DELIANEDIS AND R. GESKE. Credit Risk and Risk Neutral Default Probabilities: Information about Rating Migrations and Defaults. Working Paper 19-98, Anderson Graduate School of Business, University of California, Los Angeles, 1998.
- [21] G. DUFFEE. The Relationship between Treasury Yields and Corporate Bond Yield Spreads. *Journal of Finance*, 53:2225–41, 1998.
- [22] D. DUFFIE AND N. GARLEANU. Risk and Valuation of Collateralized Debt Obligations. *Financial Analysts Journal*, 57:41–62, 2001.
- [23] D. DUFFIE AND D. LANDO. Term Structures of Credit Spreads with Incomplete Accounting Information. *Econometrica*, 69:633–64, 2001.

- [24] D. DUFFIE AND K. SINGLETON. Modeling Term Structures of Defaultable Bonds. *Review of Financial Studies*, 12:687–720, 1999.
- [25] D. DUFFIE AND K. SINGLETON. *Credit Risk*. Princeton University Press, 2003.
- [26] P. EMBRECHTS, A. MCNEIL, AND D. STRAUTMAN. Correlation and Dependency in Risk Management: Properties and Pitfalls. Working Paper, ETH Zurich, 1999.
- [27] FEDERAL RESERVE BOARD. Capital Standards for Banks: The Evolving Basel Accord. *Federal Reserve Bulletin*, September 2003; 395–405.
- [28] FITCHRATINGS. Credit derivatives: Risk management or risk? Technical report, FitchRatings, March 2003.
- [29] J. FONS. Using Default Rates to Model the Term Structure of Credit Risk. *Financial Analysts Journal*, Sept/Oct: 25–33, 1994.
- [30] R. GESKE. The Valuation of Corporate Liabilities as Compound Options. *Journal of Financial and Quantitative Analysis*, 12:541–52, 1977.
- [31] R. GESKE AND R. JOHNSON. The Valuation of Corporate Liabilities as Compound Options: A Correction. *Journal of Financial and Quantitative Analysis*, 19:231–2, 1977.
- [32] K. GIESECKE. Default and Information. Working paper, Cornell University, 2001.
- [33] K. GIESECKE AND L. GOLDBERG. Forecasting Default in the Face of Uncertainty. Working paper, Cornell University, 2004.
- [34] L. GOODMAN AND F. FABOZZI. *Collateralized Debt Obligations: Structures and Analysis*. Wiley, 2002.
- [35] M. GORDY. A Comparative Anatomy of Credit Risk Models. *Journal of Banking and Finance*, 24:119–49, 2000.
- [36] G. GRIMMETT AND D. STIRZAKER. *Probability and Random Processes*. Oxford University Press, 1998.
- [37] G. GUPTON, C. FINGER, AND M. BHATIA. CreditMetrics—technical document. Technical report, Morgan Guaranty Trust Co., 1997.
- [38] D. HAMILTON AND L. CARTY. Debt recoveries for corporate bankruptcies. Technical report, Moody’s Investors Service, June 1999.

- [39] R. HOGG AND E. TANIS. *Probability and Statistical Inference*. Macmillan, 1983.
- [40] C. HUANG AND R. LITZENBERGER. *Foundations for Financial Economics*. Prentice Hall, 1988.
- [41] J. HULL. *Options, Futures, and Other Derivatives*. Prentice Hall, fifth edition, 2003.
- [42] J. HULL AND A. WHITE. Valuing Credit Default Swaps II: Modeling Default Correlations. *Journal of Derivatives*, 8:12–22, 2001.
- [43] P. J. HUNT AND J. E. KENNEDY. *Financial Derivatives in Theory and Practice*. John Wiley & Sons, 2000.
- [44] J. JAMES AND N. WEBBER. *Interest Rate Modelling*. Wiley, 2000.
- [45] R. JARROW, D. LANDO, AND S. TURNBULL. A Markov Model for the Term Structure of Credit Risk Spreads. *Review of Financial Studies*, 10:481–523, 1997.
- [46] R. JARROW AND S. TURNBULL. Pricing Options on Financial Securities Subject to Default Risk. *Journal of Finance*, 50:53–86, 1995.
- [47] R. JARROW AND F. YU. Counterparty Risk and the Pricing of Defaultable Securities. *Journal of Finance*, 56:1765–99, 2001.
- [48] E. JONES, S. MASON, AND E. ROSENFELD. Contingent Claims Analysis of Corporate Capital Structures: An Empirical Investigation. *Journal of Finance*, 39:611–25, 1984.
- [49] S. KEENAN, D. HAMILTON, AND A. BERTHAULT. Historical default rates of corporate bond issuers, 1920–1999. Technical report, Moody’s Investors Service, January 2000.
- [50] M. KIJIMA AND K. KOMORIBAYASHI. A Markov Chain Model for Valuing Credit Risk Derivatives. *Journal of Derivatives*, 6:97–108, 1998.
- [51] D. LANDO. Cox Processes and Credit-Risky Securities. *Review of Derivatives Research*, 2:99–120, 1998.
- [52] D. LANDO AND T. SKODEBERG. Analyzing Rating Transitions and Rating Drift with Continuous Observations. *Journal of Banking and Finance*, 26:423–44, 2002.
- [53] H. LELAND. Corporate Debt Value, Bond Covenants, and Optimal Capital Structure. *Journal of Finance*, 49:371–87, 1994.

- [54] S. LEROY AND J. WERNER. *Principles of Financial Economics*. Cambridge University Press, 2001.
- [55] D. LI. On Default Correlation: A Copula Function Approach. *Journal of Fixed Income*, March: 43–54, 2000.
- [56] F. LONGSTAFF AND E. SCHWARTZ. A Simple Approach to Valuing Risky Fixed and Floating Rate Debt. *Journal of Finance*, 50:789–819, 1995.
- [57] D. LUCAS. Default Correlation and Credit Analysis. *Journal of Fixed Income*, March: 76–87, 1995.
- [58] D. MADAN AND H. UNAL. A Two-factor Hazard Rate Model for Pricing Risky Debt and the Term Structure of Credit Spreads. *Journal of Financial and Quantitative Analysis*, 35:43–65, 2000.
- [59] R. MERTON. On the Pricing of Corporate Debt: The Risk Structure of Interest Rates. *Journal of Finance*, 29:449–70, 1974.
- [60] T. MIKOSCH. *Elementary Stochastic Calculus with Finance in View*. World Scientific, 1999.
- [61] M. MUSIELA AND M. RUTKOWSKI. *Martingale Methods in Financial Modeling*. Springer Verlag, 1998.
- [62] S. NEFTCI. *An Introduction to the Mathematics of Financial Derivatives*. Academic Press, 2002.
- [63] D. O’KANE. Credit derivatives explained: Market, products, and regulations. Technical report, Lehman Brothers, March 2001.
- [64] D. O’KANE AND L. SCHLOGL. Modeling credit: Theory and practice. Technical report, Lehman Brothers, February 2001.
- [65] G. PAN. Equity to Credit Pricing. *Risk*, November: 107–10, 2001.
- [66] N. PATEL. Credit Derivatives Survey: Flow Business Booms. *Risk Magazine*, February: 20–3, 2003.
- [67] CREDIT SUISSE FINANCIAL PRODUCTS. Creditrisk+, a credit risk management framework. Technical report, Credit Suisse Financial Products, 1997.
- [68] D. RULE. The Credit Derivatives Market: Its Development and Possible Implications for Financial Stability. *Financial Stability Review (Bank of England)*, June: 117–40, 2001.

- [69] O. SARIG AND A. WARGA. Some Empirical Estimates of the Risk Structure of Interest Rates. *Journal of Finance*, XLIV:1351–60, 1989.
- [70] P. SCHONBUCHER. Term Structure Modeling of Defaultable Bonds. *Review of Derivatives Research*, 2:161–92, 1998.
- [71] P. SCHONBUCHER. A Tree Implementation of a Credit Spread Model for Credit Derivatives. Working paper, University of Bonn, 1999.
- [72] D. SHIMKO, N. TEJIMA, AND D. VAN DEVENTER. The Pricing of Risky Debt when Interest Rates are Stochastic. *Journal of Fixed Income*, 3:58–65, 1993.
- [73] O. VASICEK. Probability of Loss on Loan Portfolio. KMV Corporation, 1987.
- [74] P. WILMOTT, S. HOWISON, AND J. DEWYNNE. *The Mathematics of Financial Derivatives: A Student Introduction*. Cambridge University Press, 1999.
- [75] Y. YOSHIKAWA. Moody’s approach to rating synthetic CDOs. Technical report, Moody’s Investors Service, July 2003.
- [76] C. ZHOU. A Jump-Diffusion Approach to Modeling Credit Risk and Valuing Defaultable Securities. Finance and Economics Discussion Series 1997-15, Board of Governors of the Federal Reserve System, 1997.

Index

- asset correlation
 - basket default swaps 247–8
 - FTD 247–8
 - PDS 255–6
- asset swaps 53–65
 - bond piece 59
 - credit risk 57–8
 - diagram 55
 - embedded interest rate
 - swap 59–62
 - cf. FRNs 62–5
 - hedge funds 57
 - market size 20
 - mechanisms 54–6
 - cf. par floaters 62–5
 - uses 56–8
 - valuation considerations
 - 58–65
- balance-sheet CDOs 136, 137
- bank regulatory issues 299–304
 - see also* regulatory issues
- bankruptcy, CDS 290
- banks
 - Basle Accord (1988) 31–5
 - credit risk 29–31
 - market participants 23–5
 - OECD 31–5
 - regulatory capital
 - management 31–5
- Basel II Capital Accord,
 - regulatory issues 300–2
- Basel II Risk Weights and Credit Derivatives,
 - regulatory issues 302–3
- basket default swaps 99–106
 - alternative approaches 248
 - asset correlation 247–8
 - basic features 239–40
 - extensions 248
 - FTD 99–106, 240–8
 - mechanisms 99–101
 - STD 246
 - two-asset FTD basket 240–1
 - uses 101
 - valuation considerations
 - 101–5
 - valuing 239–48
 - variations 105–6
- basket swap, market size 20
- Basle Accord (1988) 31–5

- BBA survey *see* British Bankers Association survey
- benefits, credit derivatives 5
- Bernoulli distribution, statistics 316
- Bernoulli trials, statistics 316
- bifurcated market, CDS 295–6
- binomial distribution, statistics 316–17
- bivariate normal distribution, statistics 323
- Black-Scholes-Merton (BSM) model
 - credit modeling 172–8
 - empirical validation 178–81
 - extensions 178–81
 - practical implementation 178
 - solving 176–8
- bond math 305–11
 - compounding 306–7
 - coupon-paying bonds 307–8
 - forward bond prices 310–11
 - forward interest rates 310–11
 - forward rates 309–10
 - zero-coupon bonds 305, 307–9
- bond options, valuing 211
- bond piece, asset swaps 59
- British Bankers Association (BBA) survey
 - credit derivatives market 17–22
 - protection sellers 24–5
- BSM *see* Black-Scholes-Merton model
- buyer of protection 4, 24–5
- cash flows, PPNs 118–19
- cash spreads, vs. CDS 76–81
- CDOs *see* collateralized debt obligations
- CDS *see* credit default swaps
- cheapest to deliver (CTD), CDS 79
- CLN *see* credit-linked notes
- collateralization, market, credit derivatives 27
- collateralized debt obligations (CDOs) 7
 - balance-sheet 136, 137
 - mechanisms 134–6
 - synthetic 133–41
 - synthetic securitization 137–41
 - traditional 133–7
 - uses 136–7
 - valuation considerations 137
 - valuing 258–60
- commercial modeling 261–5
- common market practices 25–7
- compounding, bond math 306–7
- conditional loss distribution, portfolio credit risk 225–6, 237
- confirmation letter, documentation issues 287
- Conseco Inc., CDS 294–5
- corporate debt, synthesising long positions 36
- corporates, market participants 23–5
- costs, transaction 5
- counterparty credit risk 15
 - CDS 270–80
 - example 278–80
 - Hull-White approach 280
 - modeling 267–81
 - multi-name structures 281
 - simulation-based approach 277
- coupon-paying bonds
 - bond math 307–8
 - valuing defaultable bonds 150–2
- credit curves 157–69

- applications, other 169
- CDS-implied 158–64
- flat CDS curve assumption 162–3
- implied survival
 - probabilities 159–64
- marking to market a CDS
 - position 164–6
- PPNs 166–9
- PPNs vs. vanilla notes 168–9
- rule of thumb 163–4
- credit default swaps (CDS) 4, 67–82
 - bankruptcy 290
 - basic model 268–70
 - bifurcated market 295–6
 - vs. cash spreads 76–81
 - caution 291
 - Conseco Inc. 294–5
 - contract settling 292–3
 - counterparty credit risk 272–80
 - credit curves 158–64
 - credit events 289–90
 - CTD 79
 - default events 69
 - deliverable obligations 289
 - documentation issues 285–97
 - examples 68–9, 72–3
 - failure to pay 290
 - forward-starting contracts 205–8
 - implied survival
 - probabilities 296–7
 - ISDA 69
 - market size 20
 - mechanisms 68–70
 - modified restructuring 295–7
 - no counterparty credit risk 270–2
 - notification 291–2
 - obligation acceleration 290
 - portfolio diversification 73
 - protection buyers 70–1
 - protection sellers 71–2
 - reference entities 288–9
 - repudiation/moratorium 290
 - restructuring 290
 - restructuring debate 293–6
 - risks 72–3
 - safeguards 291–3
 - settlement method 289
 - single-name instruments 6–7, 138
 - static replication 76–8
 - terms 288–90
 - transaction anatomy 285–97
 - two-asset portfolio 268
 - unfunded instruments 35–6
 - uses 70–3
 - valuation considerations 73–81, 181–3, 197–8
 - valuing restructuring clause 296–7
 - variations 82
 - verification 291–2
- credit default swaptions, valuing 208–10
- credit derivatives market *see* market, credit derivatives
- credit events, CDS 289–90
- credit-linked notes (CLN) 8, 123–6
 - market size 20
 - mechanisms 123–5
 - uses 125
 - valuation considerations 125
 - variations 125
- credit modeling 171–204
 - Black-Scholes-Merton (BSM) model 172–8
 - CDS valuation 181–3
 - JLT model 200–4

- credit modeling (*continued*)
 - ratings-based models 200–4
 - reduced-form approach 183–98
 - structural approach 172–83
- credit options, valuing 205–11
- credit protection
 - buyers 4, 24–5
 - sellers 4, 24–5
- credit quality, reference entities 21–3
- credit ratings, reference entities 22
- credit risk, asset swaps 57–8
- credit risk management, banks 29–31
- credit support documentation,
 - documentation issues 287–8
- CreditMetrics, modeling 262
- CreditRisk, modeling 263–4
- CTD *see* cheapest to deliver
- cumulative distribution function,
 - statistics 313
- default correlation
 - loss distribution function 113–15, 230–1
 - modeling 219–22
 - pairwise 223–4
 - portfolio credit risk 215–24, 230–1
- default events, CDS 69
- default intensity, credit
 - modeling 188–90
- defaultable bonds *see* valuing
 - defaultable bonds
- deliverable obligations, CDS 289
- demand/supply/market frictions 14–15
- documentation issues
 - CDS 285–97
 - confirmation letter 287
 - credit support
 - documentation 287–8
 - ISDA 286–9
 - ISDA credit derivatives
 - definitions 287
 - market, credit derivatives 26
 - master agreement 286–7
 - supplements 287
- embedded interest rate swap,
 - asset swaps 59–62
- end-users, credit derivatives 23–5
- expected value and variance,
 - statistics 315–16
- exponential distribution,
 - statistics 317–20
- failure to pay, CDS 290
- first-to-default basket swap 7
- first-to-default (FTD) basket
 - contract 99–106
 - asset correlation 247–8
 - reference entities 241–6
 - two-asset 240–1
- FitchRatings survey (2003),
 - credit derivatives
 - market 17–18, 21–2
- floating-rate notes (FRNs) 43–51
 - cf.* asset swaps 62–5
 - interest rates 46–51
 - LIBOR 43–5
 - reset risk 46–51
 - uses 45
 - valuation considerations 45–51
- forward bond prices, bond math 310–11
- forward default probabilities,
 - credit modeling 185–6
- forward default rates, credit
 - modeling 186–8
- forward interest rates, bond
 - math 310–11
- forward rates, bond math 309–10

- forward-starting contracts,
 - valuing credit options 205–8
- FRNs *see* floating-rate notes
- FTD *see* first-to-default basket
 - contract
- further reading 303–4
- gains from trade 5
- hedge funds
 - asset swaps 57
 - market participants 23–5
- hedging
 - convertible bond investors 38–9
 - vendor-financed deals 38
- Hull-White approach,
 - counterparty credit risk 280
- implied survival probabilities
 - CDS 296–7
 - credit curves 159–64
- independence, statistics 323
- insurers, market participants 23–5
- interest rates, FRNs 46–51
- International Swaps and
 - Derivatives Association (ISDA)
 - CDS 69
 - documentation issues 286–9
- Jarrow-Lando-Turnbull (JLT)
 - model, credit modeling 200–4
- joint probability distributions,
 - statistics 322
- KMV Framework, modeling 262–3
- large-portfolio approximation
 - loss distribution function 228–30, 235–7
 - PDS 257–8
- legal risk 11–12
- leveraging credit exposure 35–6
- LIBOR *see* London Interbank
 - Offered Rate
- liquidity, credit market 5
- loan origination alternative,
 - protection sellers 39
- lognormal distribution, statistics 321–2
- London Interbank Offered Rate (LIBOR)
 - FRNs 43–5
 - LIBOR curve 93–4
- long positions, corporate debt 36
- loss distribution function
 - conditional 225–6, 237
 - default correlation 113–15, 230–1
 - large-portfolio
 - approximation 228–30, 235–7
 - PDS 111–15
 - portfolio credit risk 224–31
 - unconditional 226–8, 237
- market, credit derivatives 17–27
 - activity 19–23
 - BBA survey 17–22, 24–5
 - collateralization 27
 - common market practices 25–7
 - documentation issues 26
 - evolution 18–19
 - FitchRatings survey (2003) 17–18, 21–2
 - global 18
 - instrument type, activity by 19–23
 - netting 27
 - participants 23–5

- market, credit derivatives
 - (*continued*)
 - Risk Magazine* surveys 17–22
 - size 18–19
 - size, activity by 19–23
 - US 19
- market frictions/supply/demand 14–15
- market indicators, credit
 - derivatives as 39–40
- master agreement,
 - documentation issues 286–7
- maturities, commonly negotiated contracts 23
- model-based valuation, PDS 252–5
- model risk 12
- modeling
 - commercial 261–5
 - counterparty credit risk 267–81
 - CreditMetrics 262
 - CreditRisk 263–4
 - KMV Framework 262–3
 - Moody's binomial expansion technique 264–5
 - vs. static replication 12–14
- modeling, credit *see* credit modeling
- Monte Carlo Simulation,
 - portfolio credit risk 231–7
- Moody's binomial expansion technique, modeling 264–5
- multi-name instruments 7
 - market activity 20–1
- netting, market, credit derivatives 27
- nonzero recovery, valuing default-able bonds 152–3, 192–3
- normal distribution, statistics 320–1
- obligation acceleration, CDS 290
- overview, credit derivatives 3–15
- pairwise default correlation 223–4
- par floaters, cf. asset swaps 62–5
- PDS *see* portfolio default swaps
- Poisson distribution, statistics 317–20
- portfolio credit risk 215–38
 - alternative approaches 238
 - conditional loss distribution 225–6, 237
 - default correlation 215–24, 230–1
 - extensions 238
 - loss distribution function 224–31
 - Monte Carlo Simulation 231–7
 - unconditional loss distribution 226–8, 237
- portfolio default swaps (PDS) 7, 107–16
 - asset correlation 255–6
 - example, valuation 249–51
 - large-portfolio approximation 257–8
 - loss distribution function 111–15
 - market size 20
 - mechanisms 107–10
 - model-based valuation 252–5
 - uses 110
 - valuation considerations 110–15
 - valuing 249–60
 - variations 116

- portfolio diversification 35–6
 - CDS 73
- potential benefits, credit
 - derivatives 5
- PPNs *see* principal-protected notes
- principal-protected notes
 - (PPNs) 117–22
 - cash flows 118–19
 - credit curves 166–9
 - mechanisms 117–19
 - uses 119
 - valuation considerations 119–22
 - valuing 166–9
 - vs. vanilla notes 168–9
 - variations 122
 - zero-coupon bonds 120–2
- principal-protected structures 117–22
- probability density function, statistics 314–15
- probability function, statistics 314
- protection buyers 4, 24–5
 - CDS 70–1
- protection sellers 4, 24–5
 - CDS 71–2
 - loan origination alternative 39
 - risks 10–11
- ratings-based models, credit
 - modeling 200–4
- rationale, credit derivatives 3–4
- recovery rates, valuing
 - defaultable bonds 154–6
- reduced-form approach
 - credit modeling 183–98
 - cf. structural approach 198–200
- reference entities 4, 8–9
 - CDS 288–9
 - credit quality 21–3
 - credit ratings 22
 - FTD 241–6
 - market size 21–3
 - risks 10–11
 - sovereign reference entities 8–9
- regulatory issues 299–304
 - Basel II Capital Accord 300–2
 - Basel II Risk Weights and Credit Derivatives 302–3
 - regulatory barriers 5
 - regulatory capital
 - management, banks 31–5
- reinsurers, market participants 23–5
- repackaging vehicles *see* special-purpose vehicles
- repudiation/moratorium, CDS 290
- reset risk, FRNs 46–51
- restructuring, CDS 290
- restructuring debate, CDS 293–6
- Risk Magazine*, surveys, market 17–22
- risk-neutral
 - valuation/probability, valuing defaultable bonds 147–50
- risks 3–4
 - CDS 72–3
 - counterparty credit risk 15
 - credit risk 29–31, 57–8
 - credit risk management, banks 29–31
 - legal 11–12
 - model 12
 - protection sellers 10–11
 - reference entities 10–11
 - reset risk 46–51
 - types 10–12

- risky bond spreads, valuing
 - defaultable bonds 153–4
- second-to-default (STD) basket 246
- settlement method, CDS 289
- shorting corporate bonds 37–8
- single-name CDS, two-asset portfolio 268
- single-name instruments 6–7
 - market activity 20–1
- sovereign reference entities 8–9
 - market size 21
- special-purpose vehicles (SPVs) 127–30
 - market participants 23–5
 - mechanisms 127–30
 - reasons 129–30
 - synthetic securitization 137–41
 - valuation considerations 130
 - variations 130–1
- spread and bond options 91–7
 - mechanisms 91–3
 - uses 93–5
 - valuation considerations 95–6
 - variations 96–7
- spread option, market size 20
- SPVs *see* special-purpose vehicles
- static replication
 - CDS 76–8
 - vs. modeling 12–14
- statistics 313–23
 - Bernoulli distribution 316
 - Bernoulli trials 316
 - binomial distribution 316–17
 - bivariate normal distribution 323
 - cumulative distribution function 313
 - expected value and variance 315–16
 - exponential distribution 317–20
 - independence 323
 - joint probability distributions 322
 - lognormal distribution 321–2
 - normal distribution 320–1
 - Poisson distribution 317–20
 - probability density function 314–15
 - probability function 314
- STD *see* second-to-default basket
- stochastic interest rates, credit modeling 184–5
- structural approach
 - credit modeling 172–83
 - cf. reduced-form approach 198–200
- supplements, documentation issues 287
- supply/demand/market frictions 14–15
- surveys, market
 - BBA survey 17–22, 24–5
 - FitchRatings survey (2003) 17–18, 21–2
 - Risk Magazine* 17–22
- synthesising long positions, corporate debt 36
- synthetic CDOs 133–41
 - uses 139–40
 - valuation considerations 140
 - variations 140–1
- synthetic securitization, SPVs 137–41
- third-party asset managers, market participants 23–5
- total return swaps (TRS) 83–90

- market size 20
- mechanisms 83–5
- uses 85–7
- valuation considerations
 - 87–9
- variations 89–90
- transaction costs 5
- TRS *see* total return swaps
- two-asset FTD basket 240–1
- two-asset portfolio, single-name
 - CDS 268
- types, credit derivatives 6–9
- uncertain time of default, credit
 - modeling 190–1
- unconditional loss distribution
 - function 226–8, 237
- unfunded instruments, CDS
 - 35–6
- uses, credit derivatives 29–40
 - credit risk management,
 - banks 29–31
 - hedging by convertible bond
 - investors 38–9
 - hedging vendor-financed
 - deals 38
 - market indicators 39–40
 - portfolio diversification 35–6
 - regulatory capital
 - management, banks
 - 31–5
 - shorting corporate bonds
 - 37–8
 - yield enhancement 35–6
- valuation considerations
 - asset swaps 58–65
 - basket default swaps 101–5
 - CDOs 137
 - CDS 73–81, 181–3, 197–8
 - CLN 125
 - FRNs 45–51
 - PDS 110–15
 - PPNs 119–22
 - spread and bond options
 - 95–6
 - SPVs 130
 - synthetic CDOs 140
 - TRS 87–9
- valuation principles 9–15
- valuing basket default swaps *see*
 - basket default swaps
- valuing bond options 211
- valuing collateralized debt
 - obligations 258–60
- valuing credit default swaptions
 - 208–10
- valuing credit options 205–11
 - forward-starting contracts
 - 205–8
- valuing defaultable bonds
 - 145–56, 191–6
 - alternative recovery
 - assumptions 193–6
 - coupon-paying bonds 150–2
 - nonzero recovery 152–3,
 - 192–3
 - recovery rates 154–6
 - risk-neutral
 - valuation/probability
 - 147–50
 - risky bond spreads 153–4
 - zero-coupon bonds 145–50
- valuing portfolio default swaps
 - 249–60
- valuing principal-protected notes
 - 166–9
- yield enhancement 35–6
- zero-coupon bonds
 - bond math 305, 307–9
 - payout scenarios 147–9
 - PPNs 120–2
 - valuing defaultable bonds
 - 145–50, 191–6